MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, LOW-POWER SCHOTTKY

TTL, COUNTERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, binary and decade counters. Two product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit
01	Decade counter
02	4-bit binary counter
03	Synchronous 4-bit decade counter (asynchronous clear)
04	Synchronous 4-bit binary counter (asynchronous clear)
05	Synchronous 4-bit up/down decade counter
06	Synchronous 4-bit up/down binary counter
07	Synchronous 4-bit up/down decade counter (with clear)
08	Synchronous 4-bit up/down binary counter (with clear)
09	Synchronous 4-bit up/down binary counter (with mode control)
10	Divide-by-twelve counter
11	Synchronous 4-bit decade counter (with synchronous clear)
12	Synchronous 4-bit binary counter (with synchronous clear)
13	Synchronous 4-bit decade counter (with mode control)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
E	D-2 (14-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
2	C-2 (20 terminal, .350" x .350"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center RBE-2, Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 to 7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to 5.5 V dc
Storage temperature range	-65°C to 150°C
Maximum power dissipation, (P_D) 1/	
Device types 05, 06, 07, 08	187 mW
Device types 01, 02, 10	83 mW
Device types 03, 04, 11, 12	176 mW
Device types 09, 13	193 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction to case case (θ_{JC}):	
Cases A, B, D, F	70°C/W
Cases C, E	50°C/W
Case 2	60°C/W
Junction temperature (T_J)	175°C

1.4 Recommended operating conditions. 2/

Maximum low level output current (I_{OL})	4.0 mA
Supply voltage	4.5 V dc minimum to 5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	2.0 V dc
Maximum low-level input voltage (V_{IL})	0.7 V dc
Normalized fanout (each output)	
Types 01, 02, 05, 06, 07, 08, 10	10 maximum
Types 03, 04, 09, 11, 12, 13	
Low-level	10 maximum
High-level	20 maximum
Width of input count pulse, $t_{p(IN)}$	
Types 01, 02, 10	
Input A, reset	15 ns minimum
Input B	30 ns minimum
Types 07, 08	20 ns minimum
Width of reset pulse, $t_{p(reset)}$	
Types 01, 02, 10	25 ns minimum
Count enable time	
Type 09, enable	40 ns minimum
Input clock frequency, f_{clock}	
Types 01, 02, 10	
Input A	0 to 29 MHz
Types 03, 04, 11, 12	0 to 22 MHz
Types 09, 13	0 to 18 MHz
Types 07, 08	0 to 20 MHz
Types 05, 06	0 to 25 MHz
Width of clock pulse, $t_{w(clock)}$	
Types 03, 04, 09, 11, 12, 13	25 ns minimum
Types 05, 06	20 ns minimum
Width of clear pulse, $t_{w(clear)}$	
Types 03, 04, 05, 06, 07, 08, 11, 12	20 ns minimum
Setup time, $t_{(setup)}$	
Types 03, 04, 11, 12	
Enable P	25 ns minimum
Load	25 ns minimum
Clear (types 11 and 12 only)	20 ns minimum
Data inputs	
Types 03, 04, 09, 11, 12, 13	20 ns minimum
Types 07, 08	30 ns minimum
Types 05, 06	
Data, L inputs	15 ns minimum
PE input	20 ns minimum
U/D input	30 ns minimum
EP, ET inputs	15 ns minimum

1/ Must withstand the added P_D due to short circuit (e.g., I_{OS}) test.
2/ A change of states on the U/D input for device types 09 and 13 is not recommended when the clock input is low. This may result in an erroneous count.

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Hold time at any input, $t_{(hold)}$	
Types 09, 13	0 ns minimum
Types 07, 08	10 ns minimum
Types 05, 06	
Data, L, EP, ET inputs	5 ns minimum
PE, U/D inputs	0 ns minimum
Types 03, 04, 11, 12	10 ns minimum
Types 03, 04, 11, 12 t_W (clear)	0 ns minimum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figures 1 and 2.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range unless otherwise specified.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Units
				Min	Max	
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2.0 V V_{IL} = 0.7 V, I_{OL} = 4 mA 1/	All	-	0.4	V
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IH} = 2.0 V V_{IL} = 0.7 V, I_{OH} = -400 µA	All	2.5	-	V
Input clamp voltage	V_{IC}	T_C = 25°C, V_{CC} = 4.5 V I_{IN} = -18 mA	All	-	-1.5	V
Low-level input current at reset inputs	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01, 02, 10	-30	-400	µA
Low-level input current at input A	I_{IL2}		01, 02, 10	-0.5	-2.4	mA
Low-level input current at input B	I_{IL3}		01, 02, 10	-0.4	-3.2	mA
Low-level input current at data, clear, EnP	I_{IL4}		03, 04	-30	-400	µA
Low-level input current at data, EnP	I_{IL4}		01, 12	-30	-400	µA
Low-level input current at clear	I_{IL5}		01, 12	-30	-760	µA
Low-level input current at load	I_{IL5}		03, 04, 11, 12	-30	-800	µA
Low-level input current at EnT	I_{IL6}		03, 04, 11, 12	-30	-860	µA
Low-level input current at clock	I_{IL7}		03, 04 11, 12	0	-.630	mA
Low-level input current at EnG	I_{IL8}		09	-.15	-1.08	mA
Low-level input current at data, clock, down/up	I_{IL8}		13	-.36	-1.08	
Low-level input current at load	I_{IL9}		09, 13	-120	-400	µA
Low-level input current at data	I_{IL10}		09, 13	-100	-400	µA
Low-level input current at load	I_{IL11}		07, 08	-100	-400	µA
Low-level input current at clear, count up, count down	I_{IL12}		07, 08	-100	-400	µA
Low-level input current at data	I_{IL13}		07, 08	-120	-400	µA
Low-level input current at L, clock, down/up	I_{IL14}		05, 06	-3.0	-400	µA
Low-level input current at EP	I_{IL15}		05, 06	-135	-370	µA
Low-level input current at ET			05, 06	-150	-385	µA
			05, 06	-280	-760	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Units
				Min	Max	
High-level input current at reset inputs	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01, 02 10	-	20	µA
High-level input current at reset inputs	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01, 02 10	-	100	µA
High-level input current at input A	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01, 02 10	-	80	µA
High-level input current at input A	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01, 02, 10	-	400	µA
High-level input current at input B	I_{IH5}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01 02, 10	- -	160 80	µA
High-level input current at input B	I_{IH6}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	01 02, 10	- -	800 400	µA
High-level input current at load, clock, EnT	I_{IH9}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	03, 04, 11, 12	-	40	µA
High-level input current at load, clock, EnT	I_{IH10}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	03, 04, 11, 12	-	200	µA
High-level input current at data, EnP	I_{IH11}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	03, 04, 11, 12	-	20	µA
High-level input current at data, EnP	I_{IH12}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	03, 04, 11, 12	-	100	µA
High-level input current at clear	I_{IH13}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	03, 04 11, 12	- -	20 40	µA
High-level input current at clear	I_{IH14}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	03, 04 11, 12	- -	100 + 200	µA
High-level input current at EnG	I_{IH15}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	09, 13	-	60	µA
High-level input current at EnG	I_{IH16}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	09, 13	-	300	µA
High-level input current at data, load, clear, count up, count down, clock, down/up	I_{IH17}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	05, 06 07, 08 09, 13	-	20	µA
High-level input current at data, load, clear, count up, count down, clock, down/up	I_{IH18}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	05, 06 07, 08 09, 13	-	100	µA
High-level input current at ET	I_{IH19}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	05, 06	-	40	µA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Units
				Min	Max	
High-level input current at ET	I_{IH20}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	05, 06	-	200	µA
Short circuit output current	I_{OS}	V_{CC} = 5.5 V 2/	All	-15	-130	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01,02,10 05,06,07,08 09, 13		15 34 35	mA
High-level supply current	I_{CCH}	V_{CC} = 5.5 V, 3/	03, 04, 11, 12	-	31	mA
High-level supply current	I_{CCH}	V_{CC} = 5.5 V, 3/	03, 04, 11, 12	-	31	mA
Low-level supply current	I_{CCL}	V_{CC} = 5.5 V, 4/	03, 04, 11, 12	-	32	mA
Maximum input A, clock, or count up frequency	F_{MAX}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ	05, 06 01,02,10 03, 04, 07, 08, 11, 12 09, 13	25 29 22 18	- - - -	MHz
Propagation delay time, high to low, A to Q_C	t_{PHL1}		01,02,10	3	81	ns
Propagation delay time, low to high, A to Q_C	t_{PLH1}		01, 10 02	3 3	74 74	ns
Propagation delay time, high to low, B to Q_D	t_{PHL2}		01, 10 02	3 3	56 78	ns
Propagation delay time, low to high, B to Q_D	t_{PLH2}		01, 10 02	3 3	52 78	ns
Propagation delay time, low to high, clock to carry	t_{PLH4}		03, 04, 11, 12	3	56	ns
Propagation delay time, high to low, clock to carry	t_{PHL4}		03, 04, 11, 12	3	56	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Units
				Min	Max	
Propagation delay time, low to high, clock to Q	t_{PLH5}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ	03, 04, 11, 12	3	41	ns
Propagation delay time, high to low, clock to Q	t_{PHL5}		03, 04, 11, 12	3	45	ns
Propagation delay time, low to high, clock to Q	t_{PLH5}		05, 06	3	26	ns
Propagation delay time, high to low, clock to Q	t_{PHL5}		05, 06	3	26	ns
Propagation delay time, low to high, clock (data) to Q	t_{PLH6}		03, 04, 11, 12	3	42	ns
Propagation delay time, high to low, clock (data) to Q	t_{PHL6}		03, 04, 11, 12	3	48	ns
Propagation delay time, low to high, EnT to carry	t_{PLH7}		03, 04, 11, 12	3	28	ns
Propagation delay time, high to low, EnT to carry	t_{PHL7}		03, 04, 11, 12	3	28	ns
Propagation delay time, low to high, ET to RC	t_{PLH7}		05, 06	3	18	ns
Propagation delay time, high to low, ET to RC	t_{PHL7}		05, 06	3	28	ns
Propagation delay time, high to low, clear to Q	t_{PHL8}		03, 04, 11, 12	3	46	ns
Propagation delay time, low to high, load to Q	t_{PLH8}		07, 08	3	63	ns
Propagation delay time, high to low, load to Q	t_{PHL10}		07, 08	3	63	ns
Propagation delay time, low to high, counts up and down to Q, U/$\overline{D}$ to RC	t_{PLH9}		07, 08	3	60	ns
			05, 06	3	26	ns

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Units
				Min	Max	
Propagation delay time, high to low, counts up and down to Q, U/$\overline{D}$ to RC	t_{PHL11}	V_{CC} = 5.0 V, C_L = 50 pF, R_L = 2 kΩ	07, 08	3	73	ns
			05, 06	3	33	ns
Propagation delay time, high to low, clear to Q	t_{PHL12}		07, 08	3	56	ns
Propagation delay time, low to high, load to Q	t_{PLH10}		09, 13	3	53	ns
Propagation delay time, high to low, load to Q	t_{PHL13}		09, 13	3	77	ns
Propagation delay time, low to high, clock to Q	t_{PLH11}		09, 13	3	41	ns
Propagation delay time, high to low, clock to Q	t_{PHL14}		09, 13	3	57	ns
Propagation delay time, low to high, clock to $\frac{\text{Max}}{\text{Min}}$	t_{PLH12}		09, 13	3	66	ns
Propagation delay time, high to low, clock to $\frac{\text{Max}}{\text{Min}}$	t_{PHL15}		09, 13	3	80	ns

1/ Use I_{OL} + $I_{IL3(MAX)}$ for V_{OL} test on Q_A.

2/ Not more than one output should be shorted at a time.

3/ I_{CCH} is measured: (a) With the load input high; and (b) Then again with the load input low with all other inputs high and all outputs open.

4/ I_{CCL} is measured: (a) With the clock input high; and (b) Then again with the clock input low with all other inputs low and all outputs open.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,8,9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1, 2, 3
Additional electrical subgroups for group C periodic inspections	N/A	None
Group D end-point electrical parameters (method 5005)	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1 (see 4.2c).

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

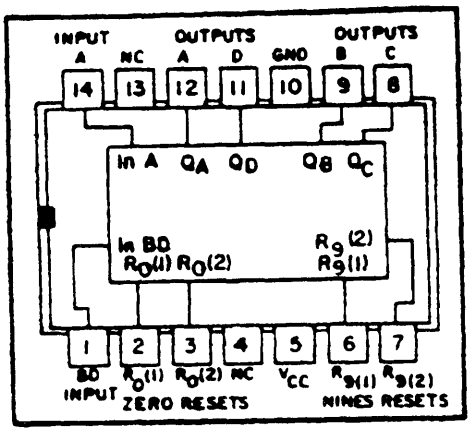
a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D using the circuit shown on figure 13, or equivalent.

(2) T_A = +125°C minimum.

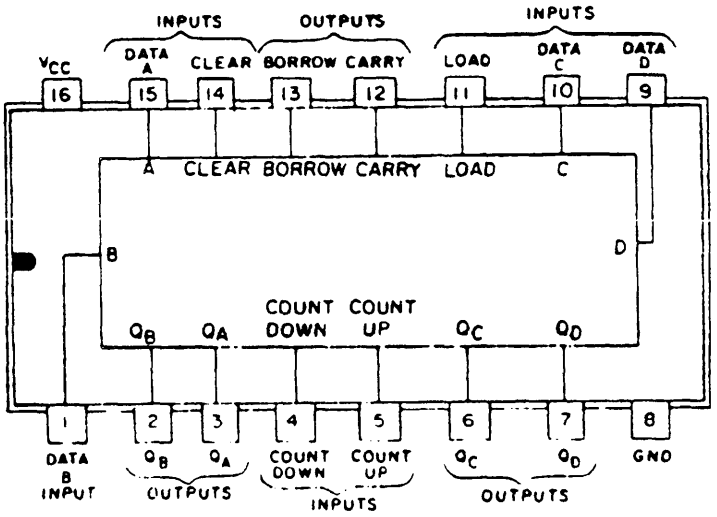
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CASE A, B, C, & D



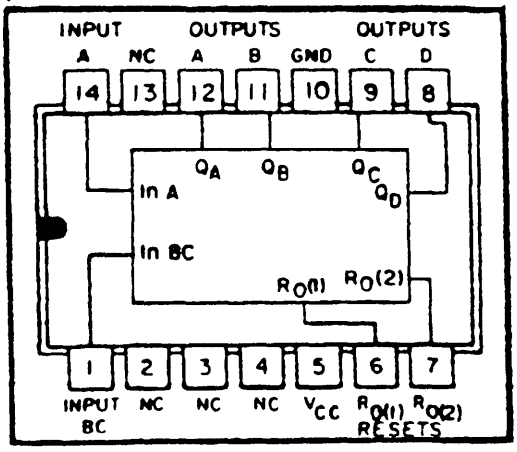
DEVICE TYPE 01

CASES E AND F



DEVICE TYPES 07 & 08

CASE A, B, C & D

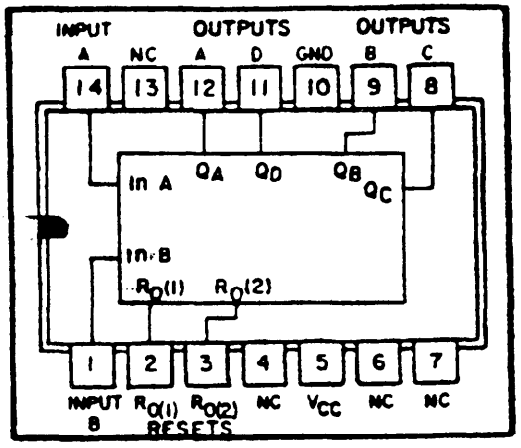


DEVICE TYPE 10

FIGURE 1. Terminal connections (top view).

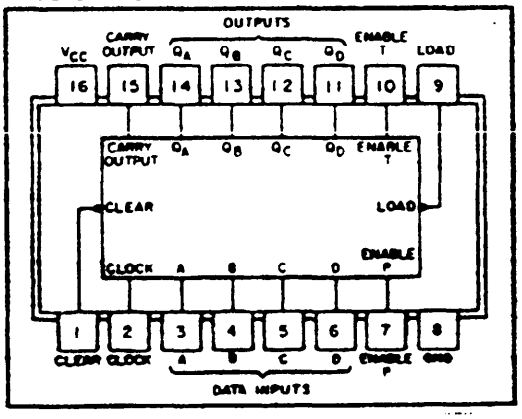
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CASE A, B, C & D



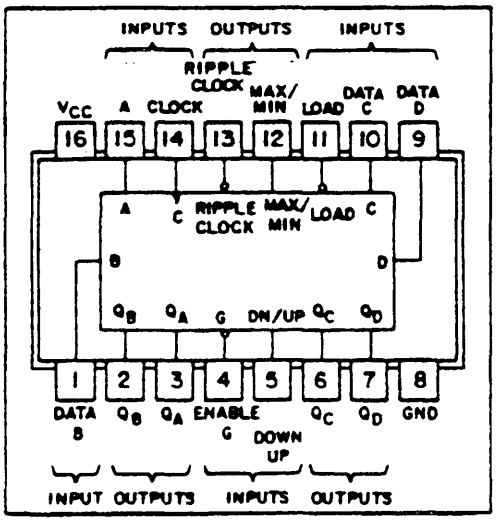
DEVICE TYPE 02

CASE E AND F



DEVICE TYPES 03, 04, 11 & 12

CASE E AND F



DEVICE TYPES 09 & 13

FIGURE 1. Terminal connections (top view) - Continued.

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Device types 05 and 06

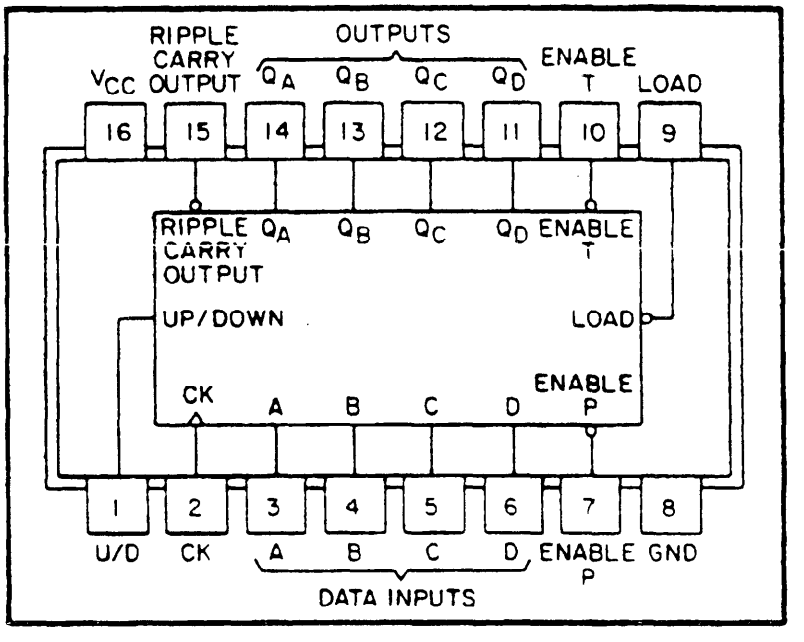
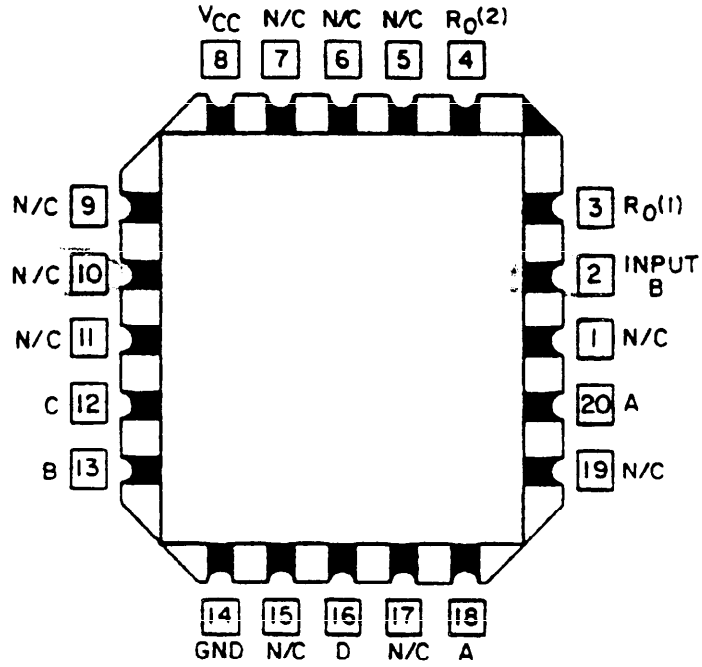


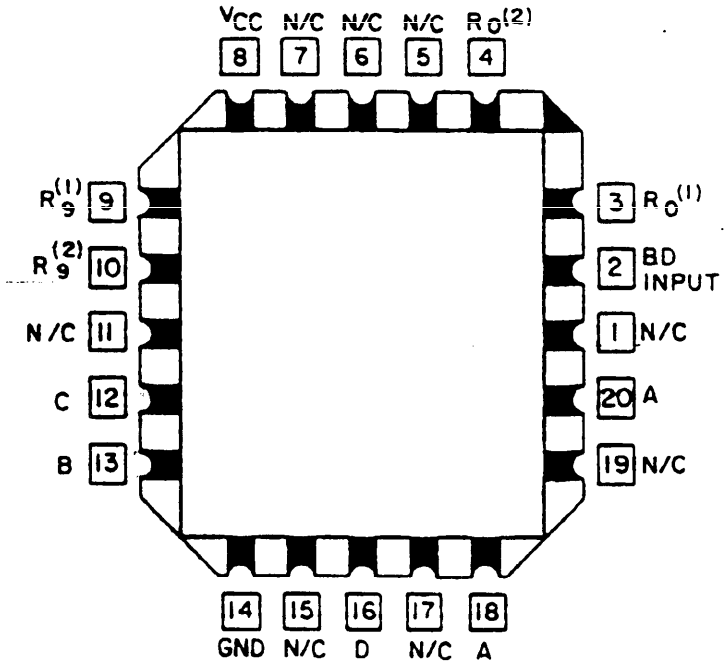
FIGURE 1. Terminal connections (top view) - Continued.

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Device type 01

Case 2

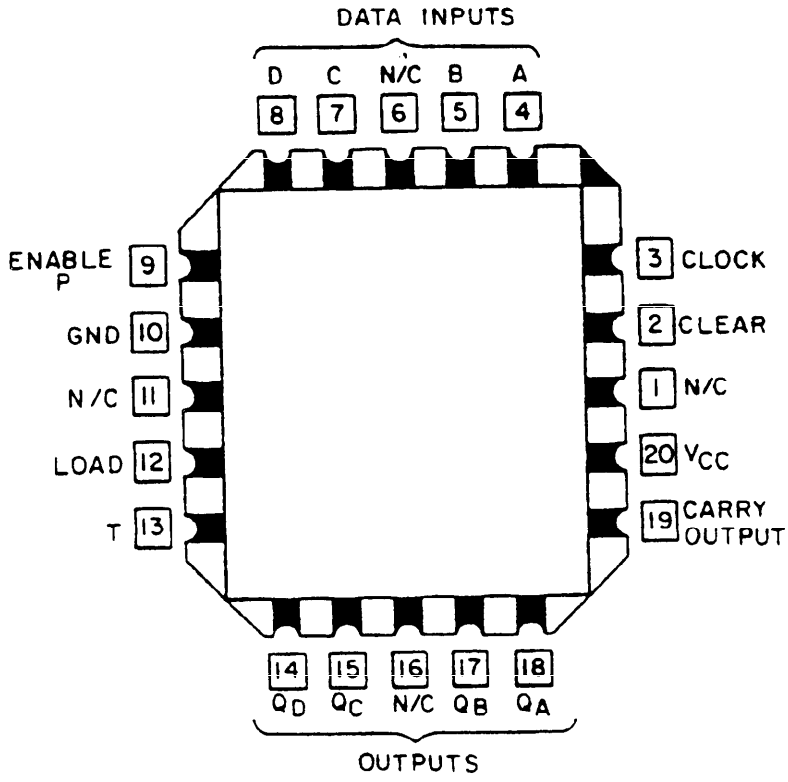


Device type 02

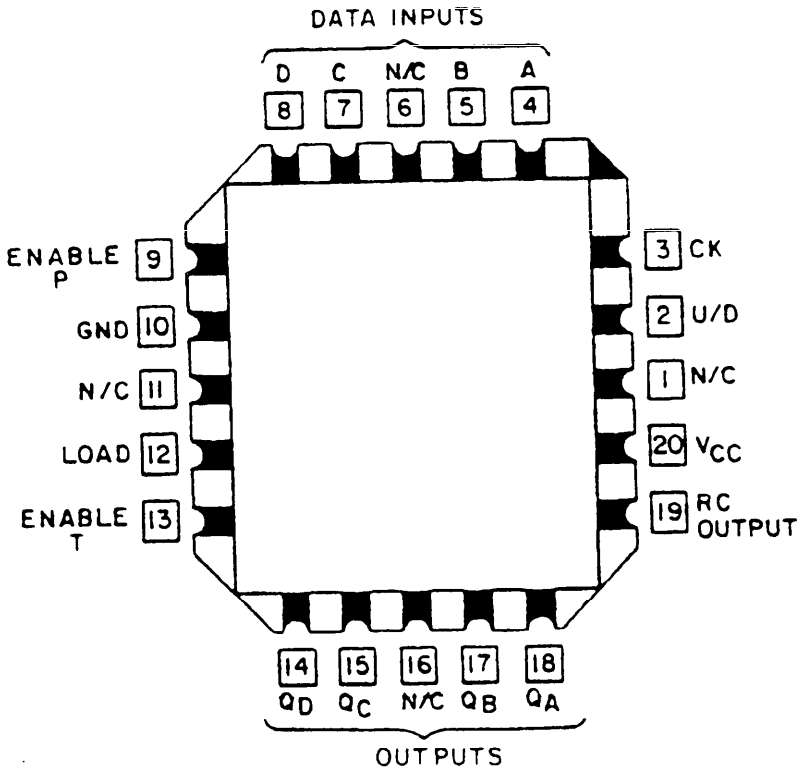
Case 2

FIGURE 1. Terminal connections (top view) - Continued.

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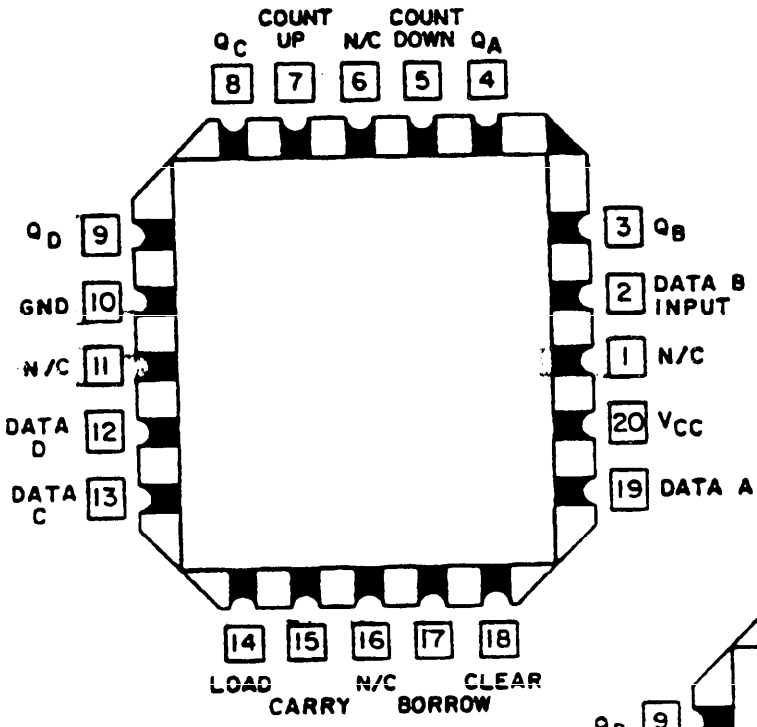
Device types 03, 04, 11 and 12
Case 2



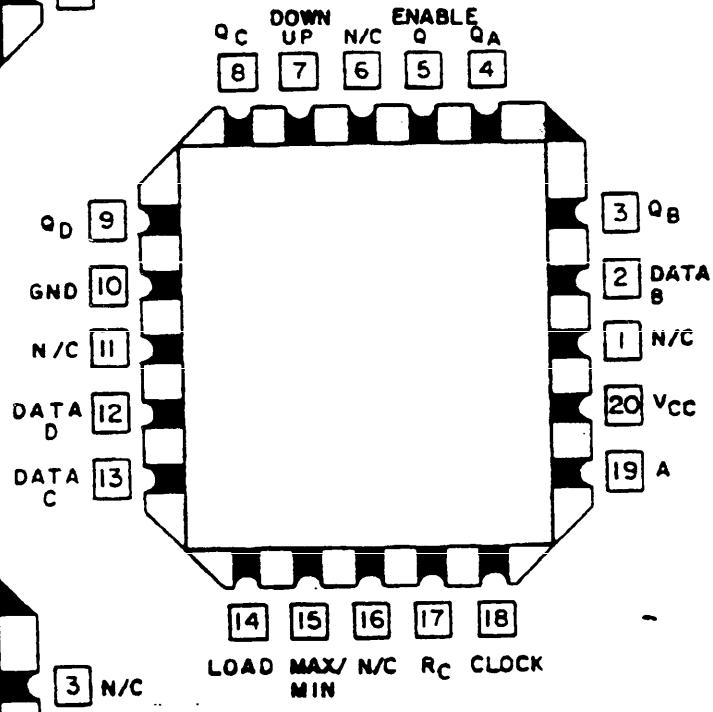
Device types 05 and 06
Case 2

FIGURE 1. Terminal connections (top view) - Continued.

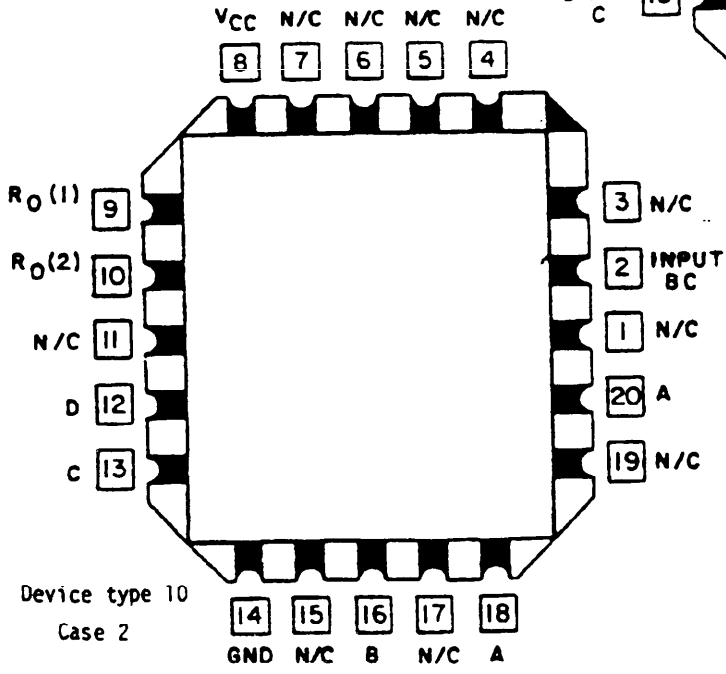
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
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Device types 07 and 08
Case 2



Device types 09 and 13
Case 2



Device type 10
Case 2

FIGURE 1. Terminal connections (top view) - Continued.

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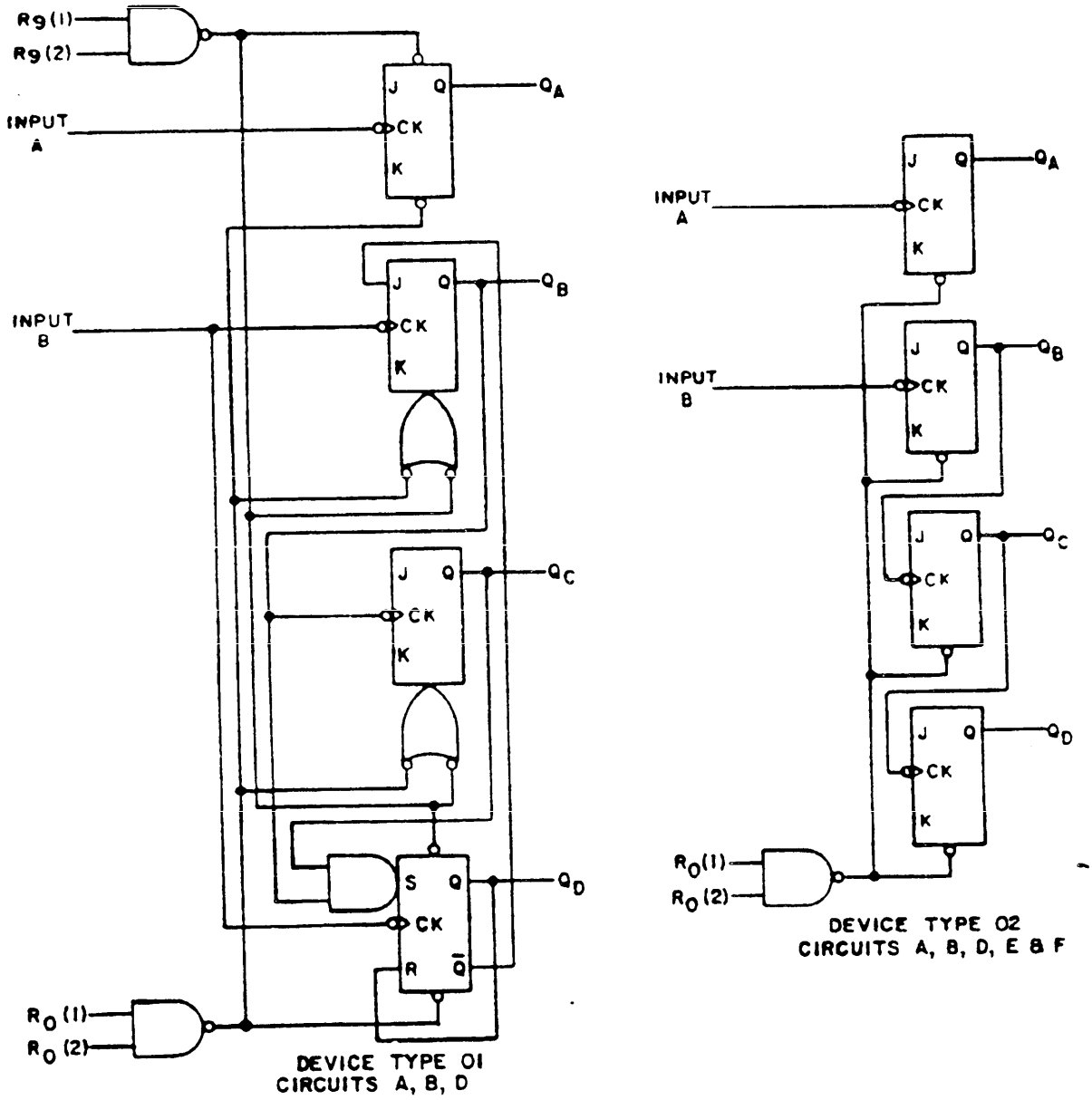
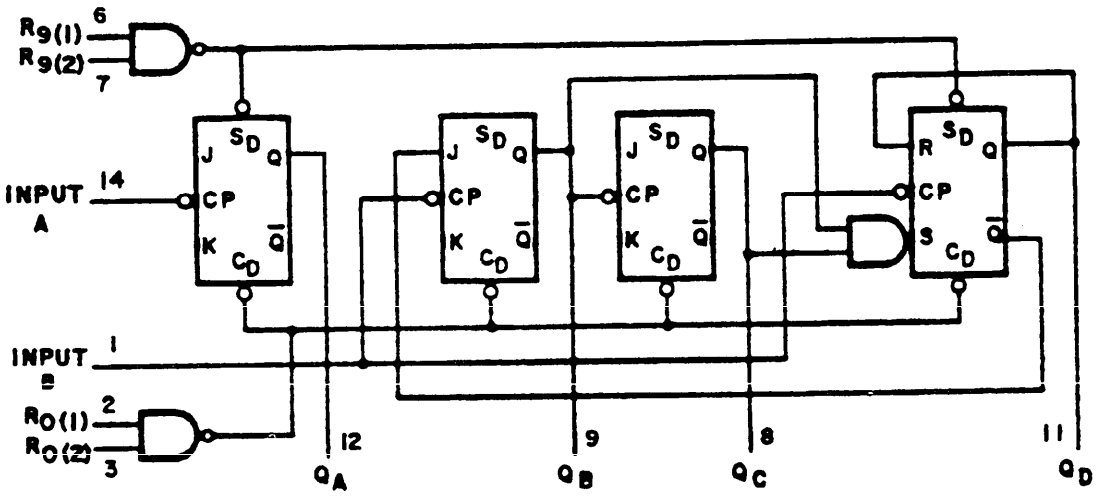


FIGURE 2. Logic diagrams.

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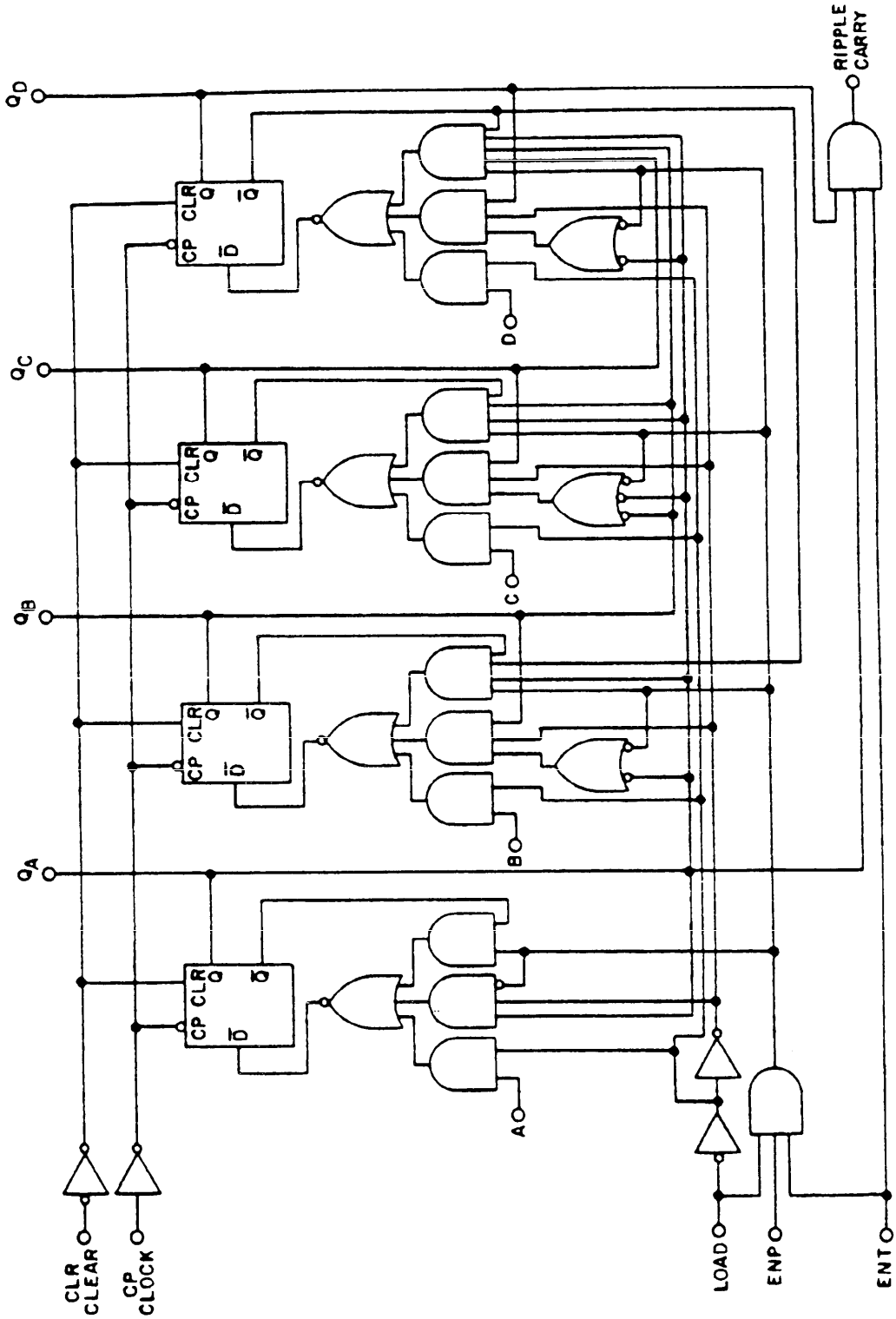


DEVICE TYPE 01

CIRCUITS E AND F

FIGURE 2. Logic diagrams - Continued.

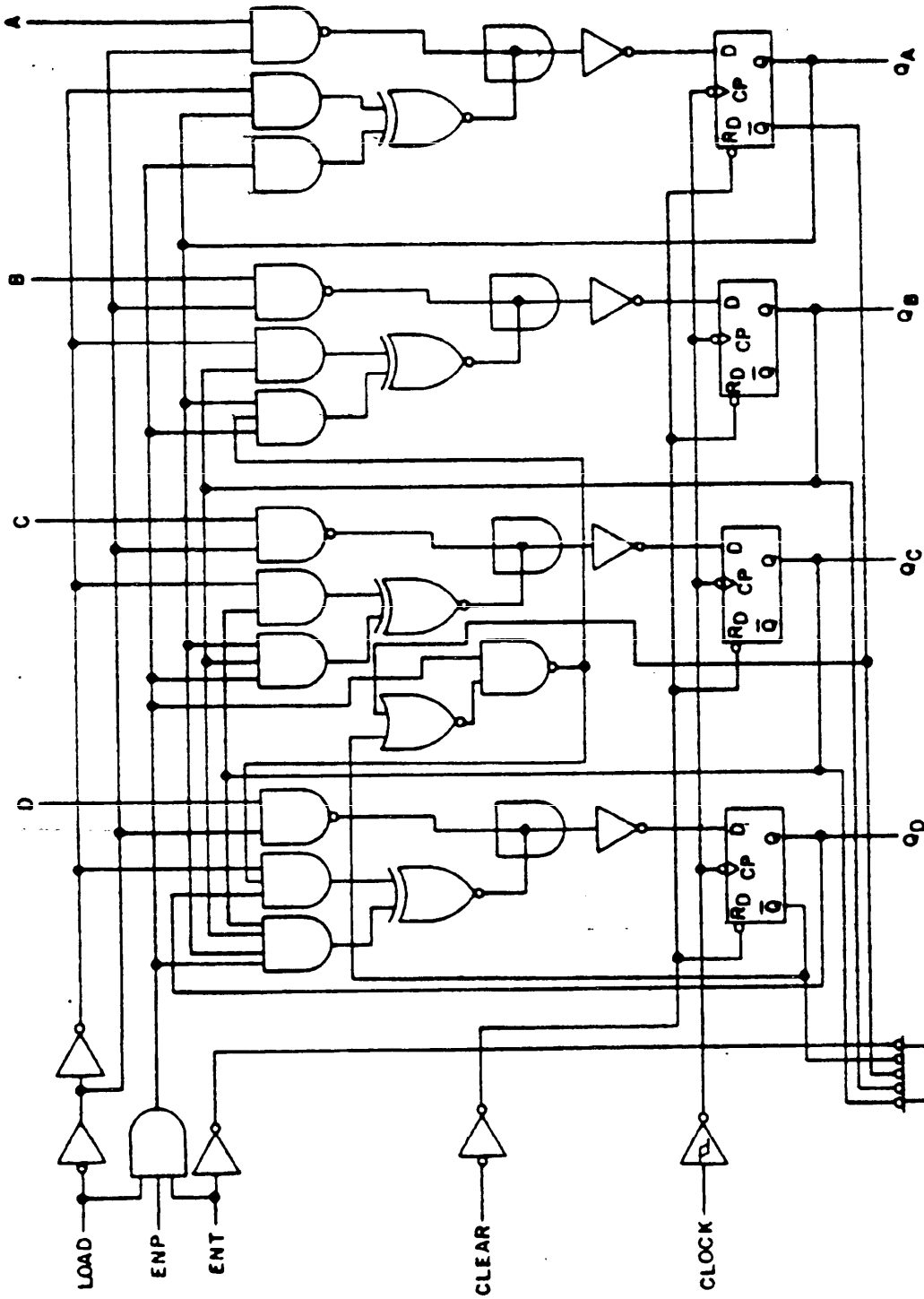
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
Check the source to verify that this is the current version before use.



DEVICE TYPE 03
CIRCUITS A, C, D & G

FIGURE 2. Logic diagrams - Continued.

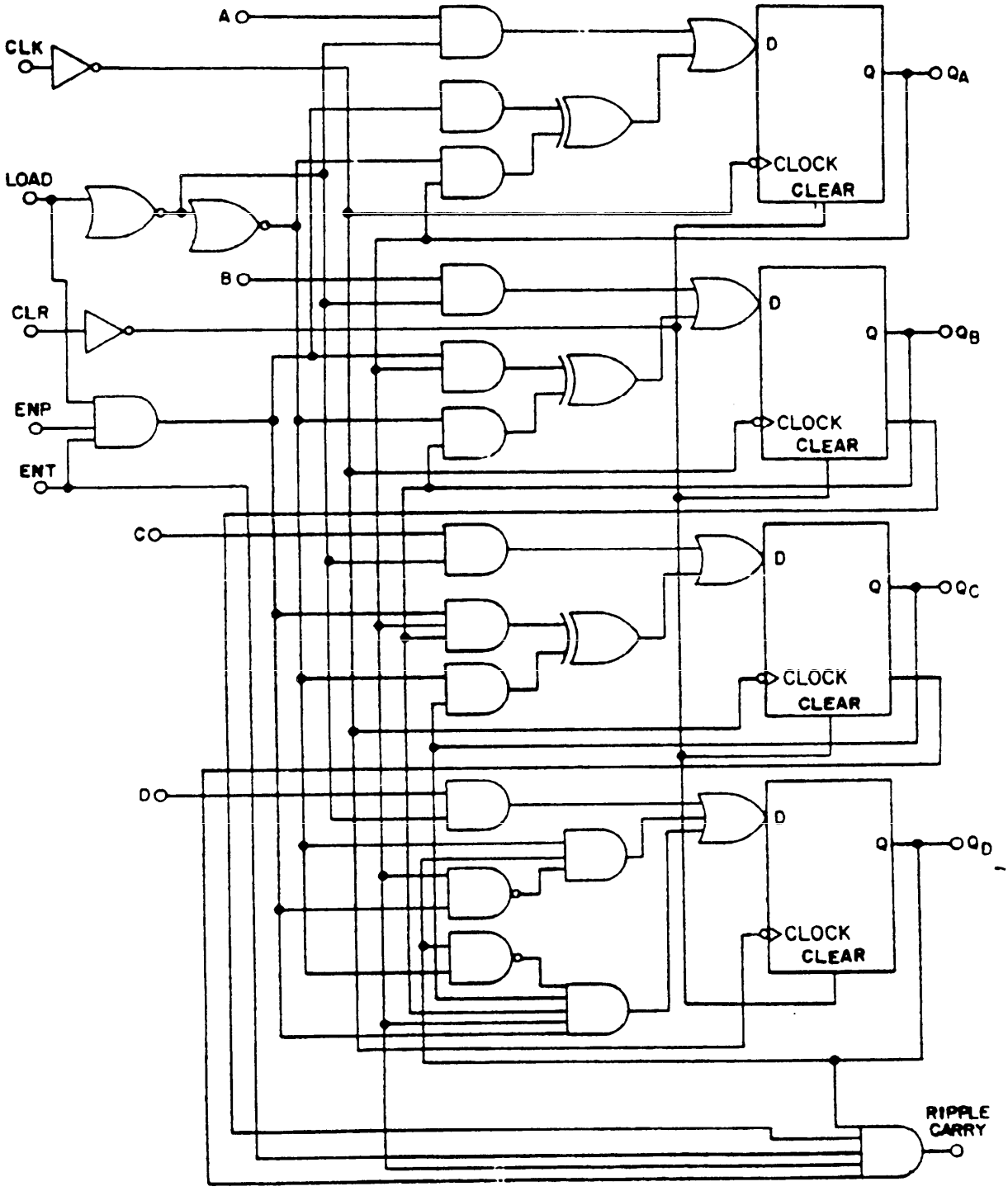
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DEVICE TYPE 03
CIRCUIT B

FIGURE 2. Logic diagram - Continued.

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DEVICE TYPE 03
CIRCUIT E

FIGURE 2. Logic diagrams - Continued.

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Device type 03

Circuit F

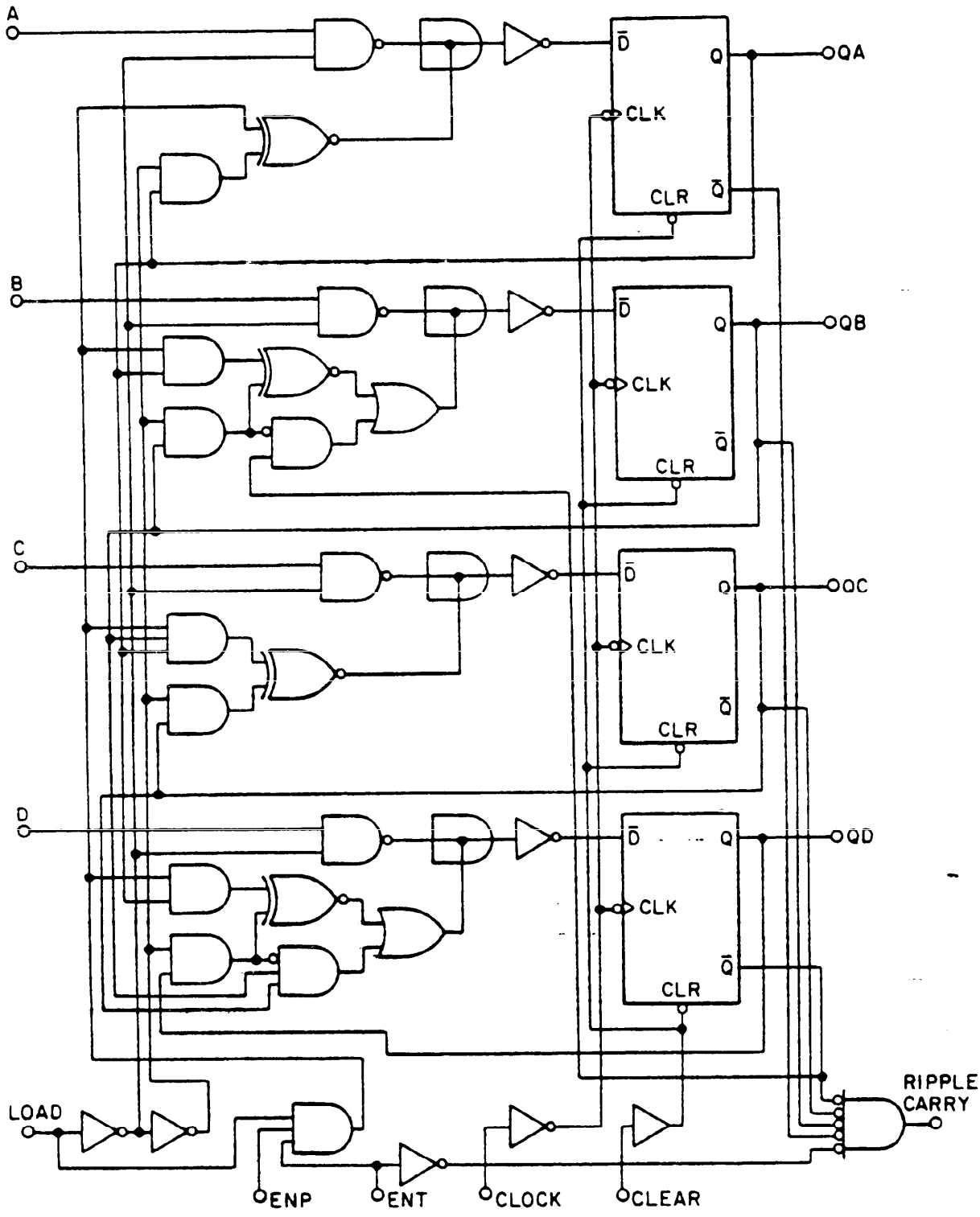


FIGURE 2. Logic diagrams - Continued.

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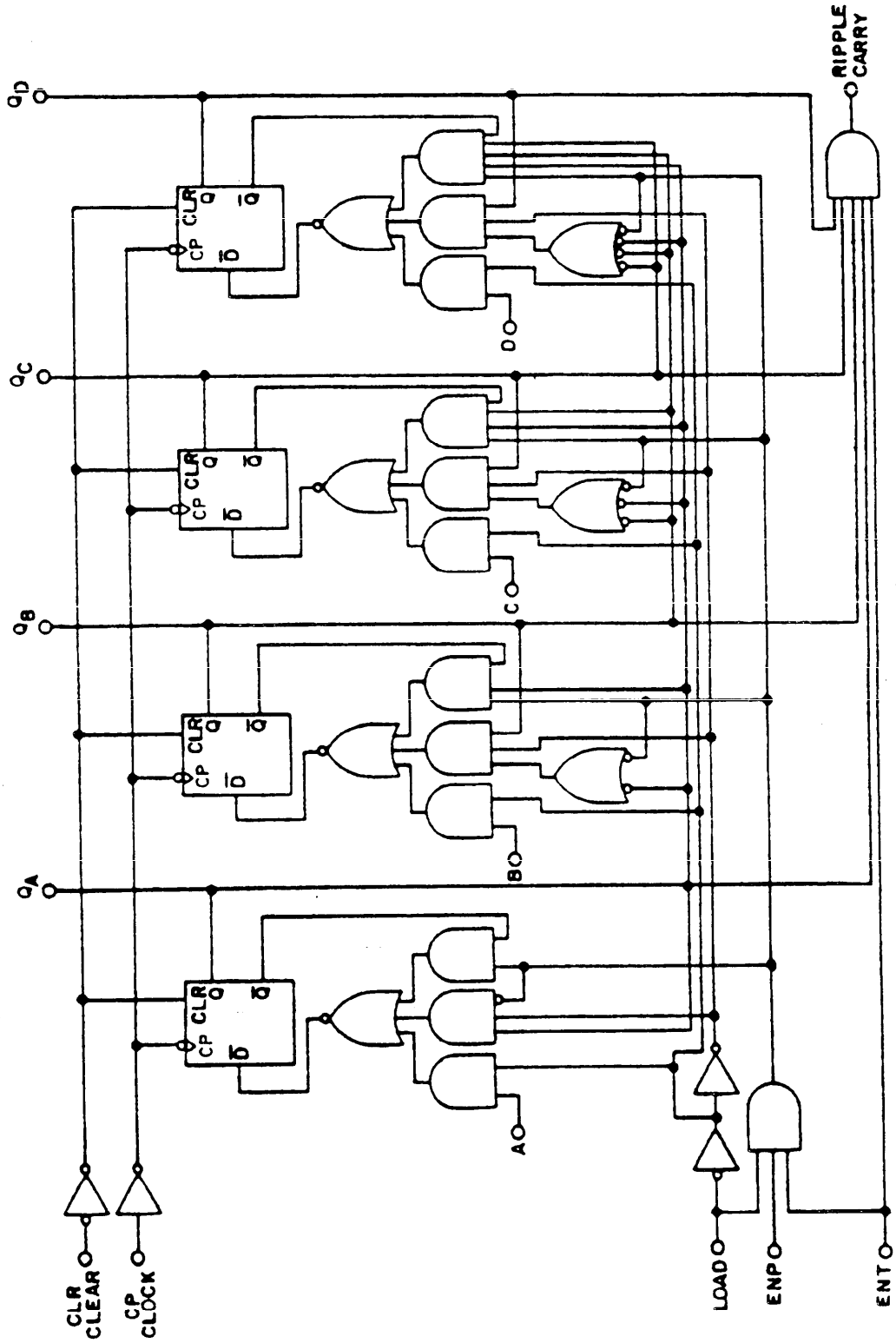


FIGURE 2. Logic diagrams - Continued.

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Check the source to verify that this is the current version before use.

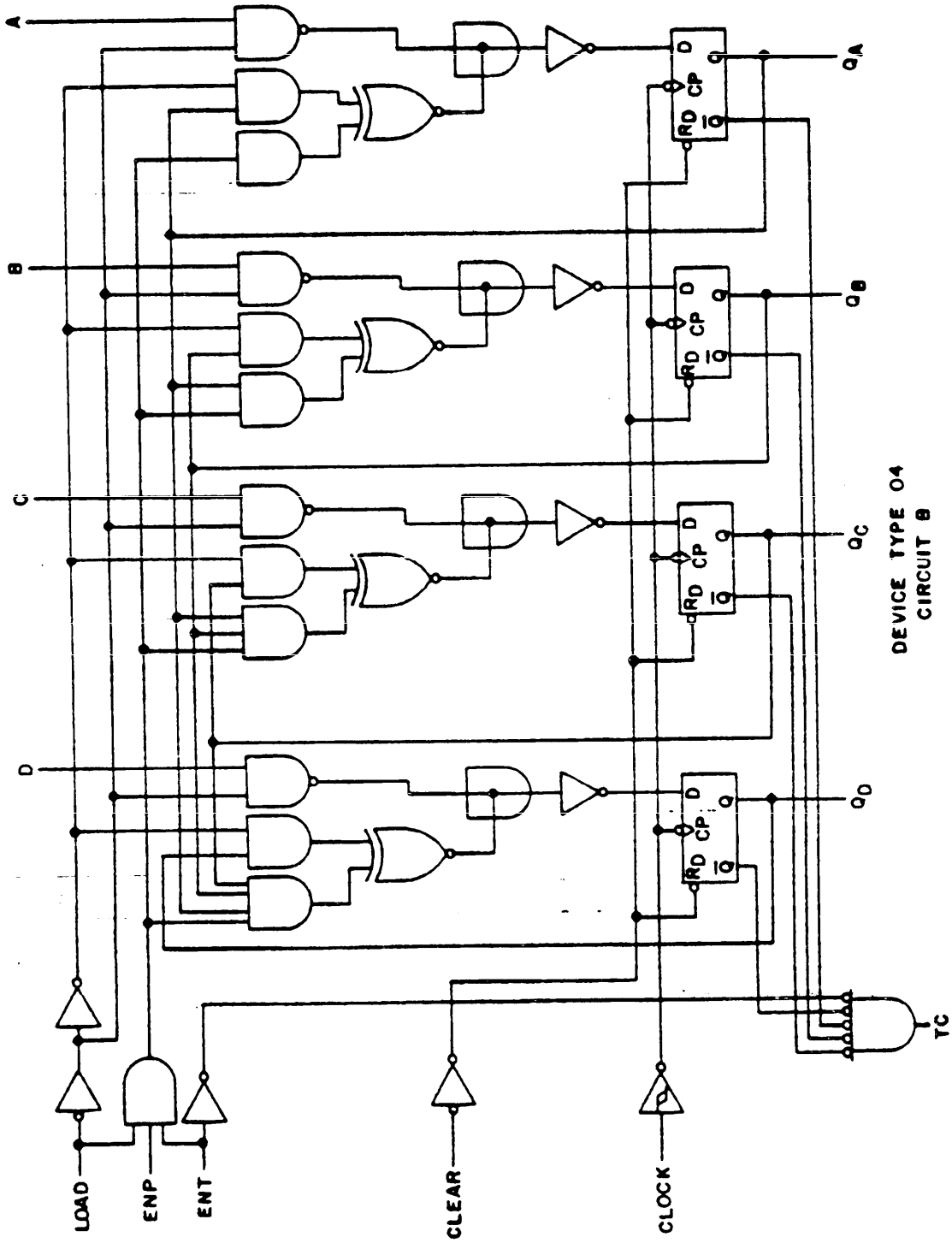
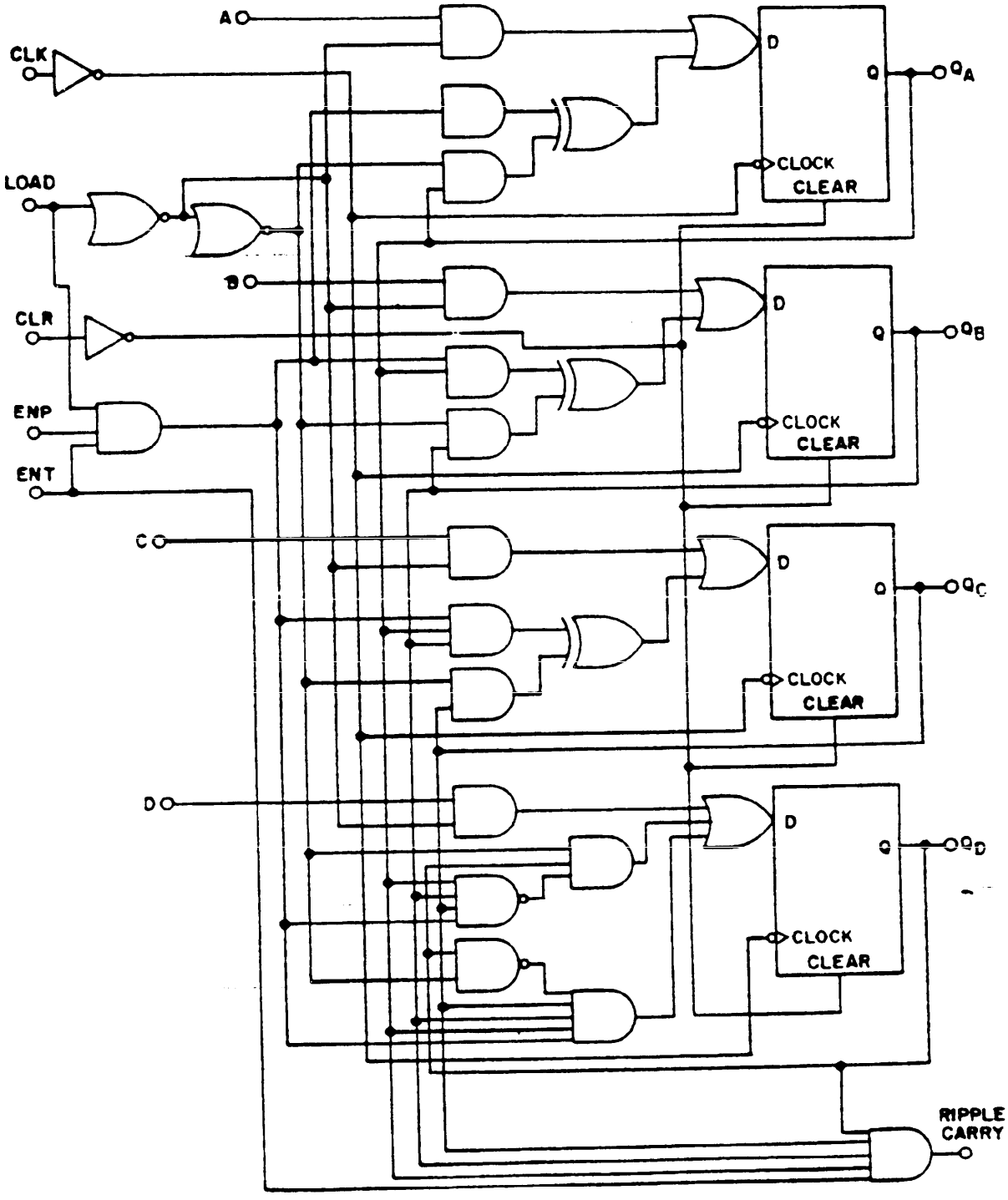


FIGURE 2. Logic diagram - Continued.

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DEVICE TYPE 04
CIRCUIT E

FIGURE 2. Logic diagrams - Continued.

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Device type 04

Circuit F

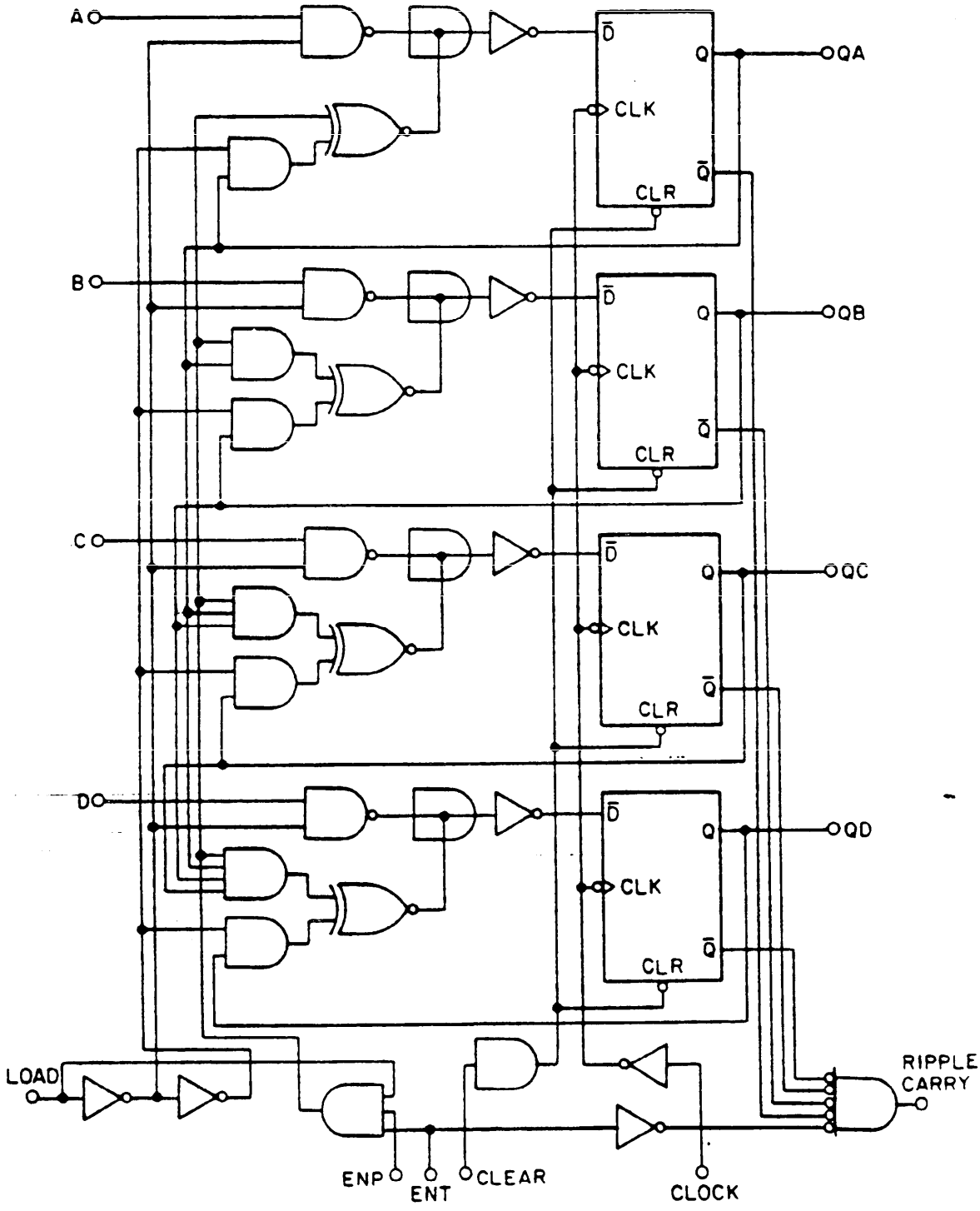


FIGURE 2. Logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
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Device type 05
Circuit E

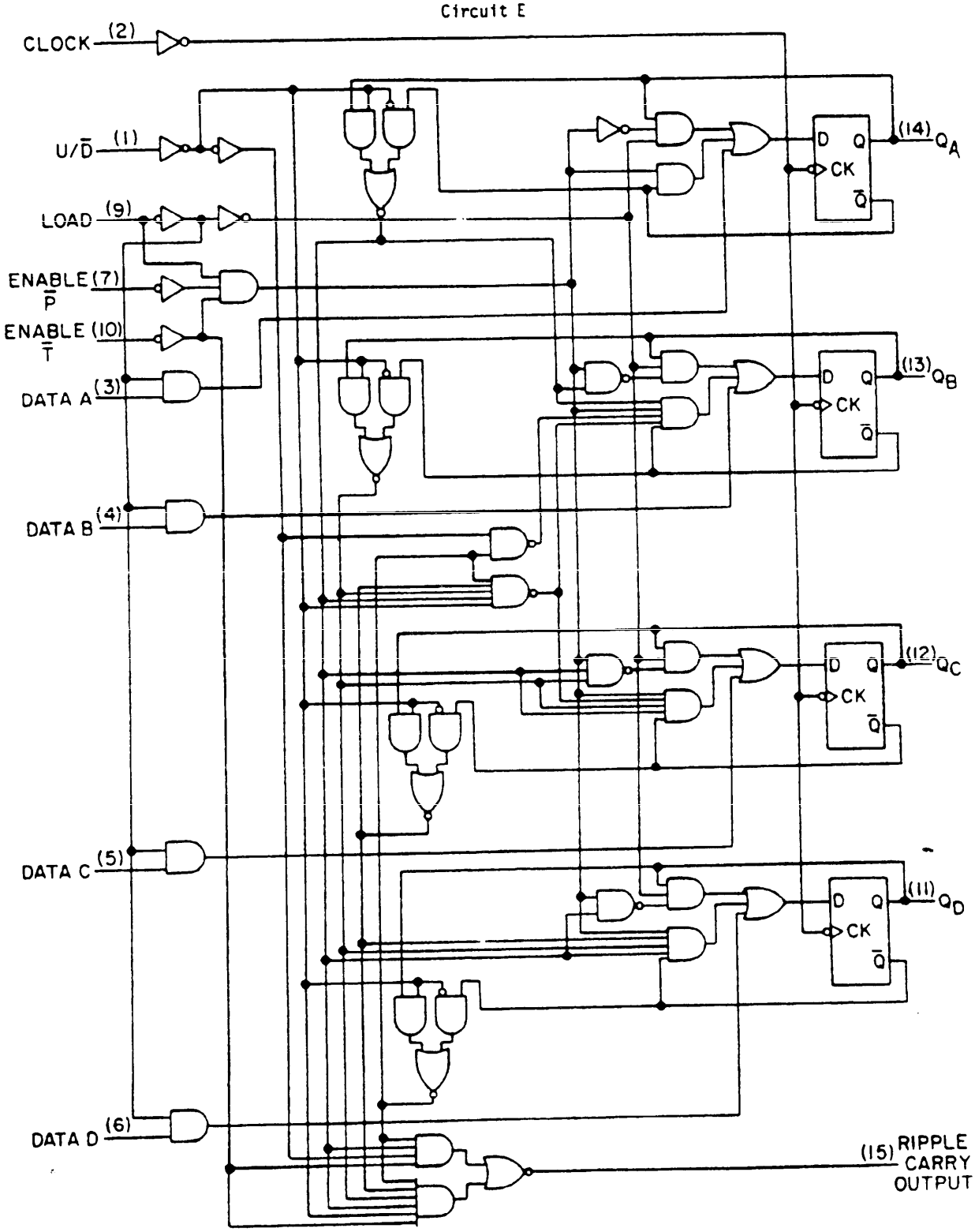
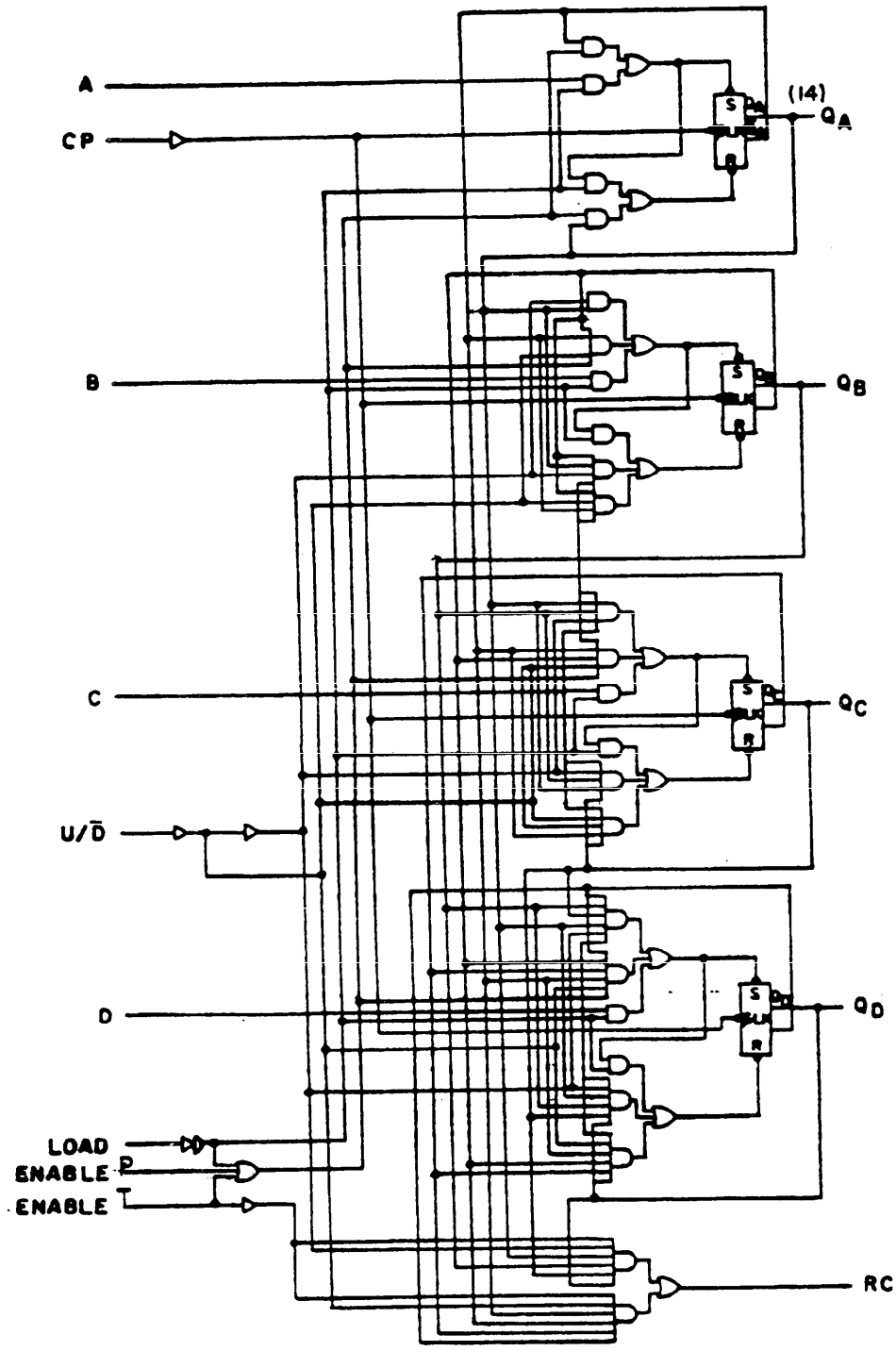


FIGURE 2. Logic diagrams - Continued.

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Device type 06
Circuit C

FIGURE 2. Logic diagram - Continued.

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Device type 06

Circuit E

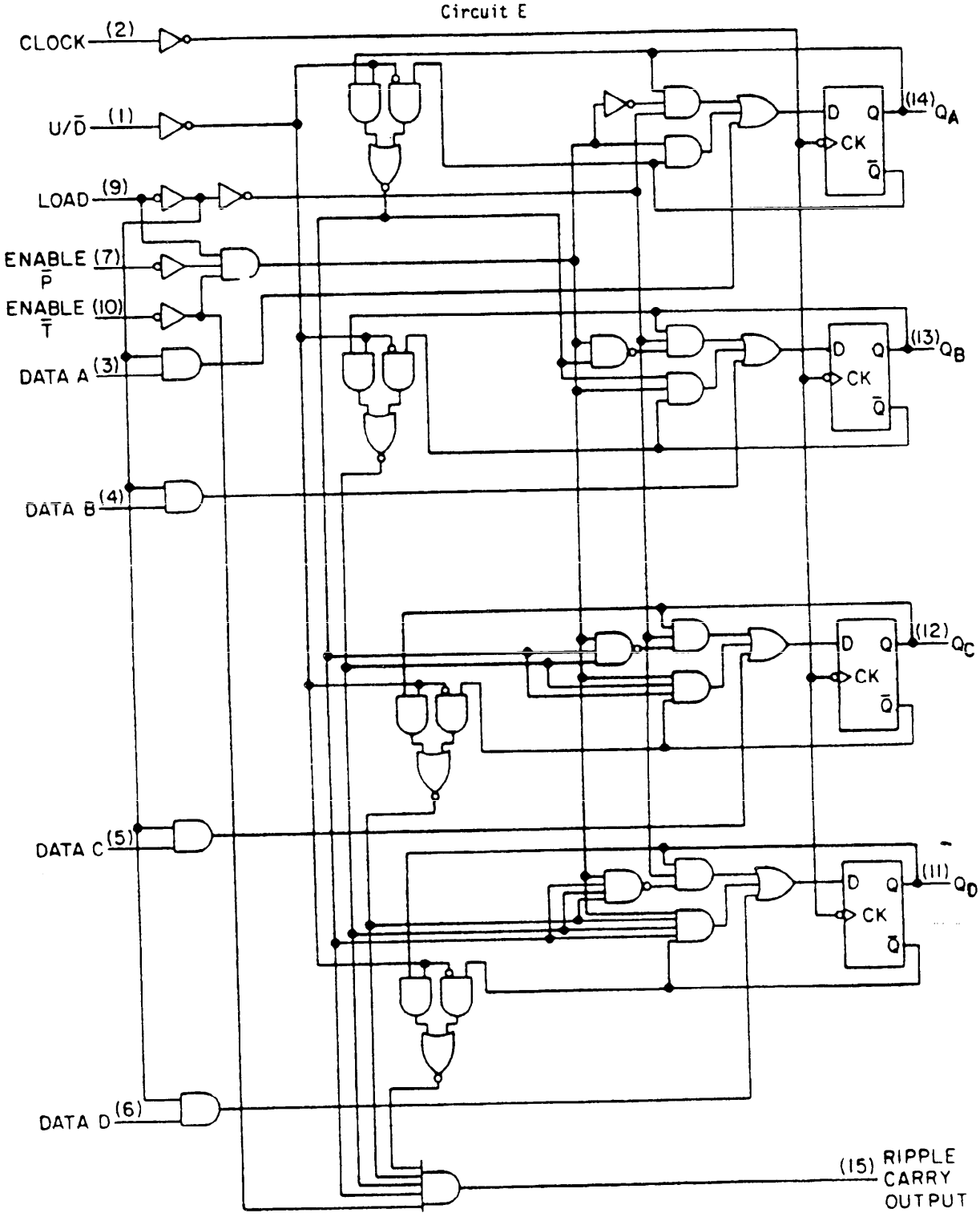


FIGURE 2. Logic diagrams - Continued.

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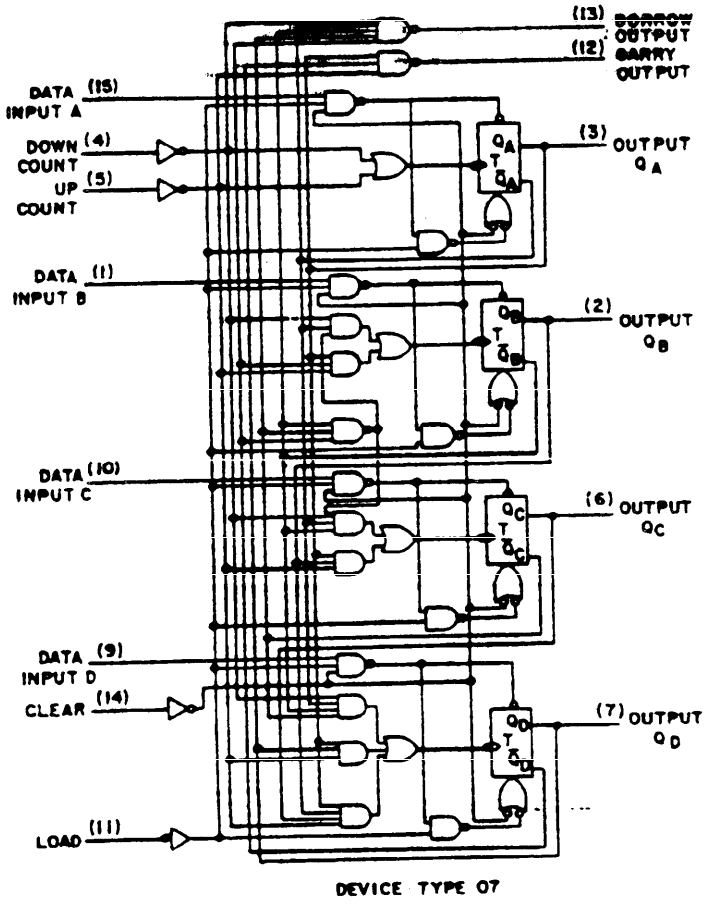


FIGURE 2. Logic diagrams - Continued.

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Check the source to verify that this is the current version before use.

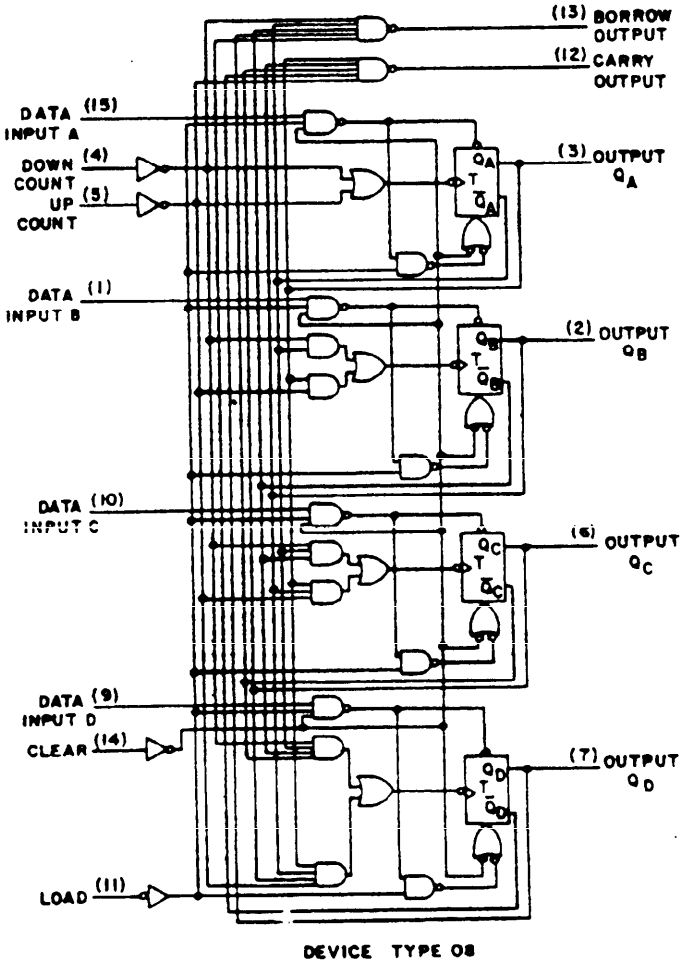
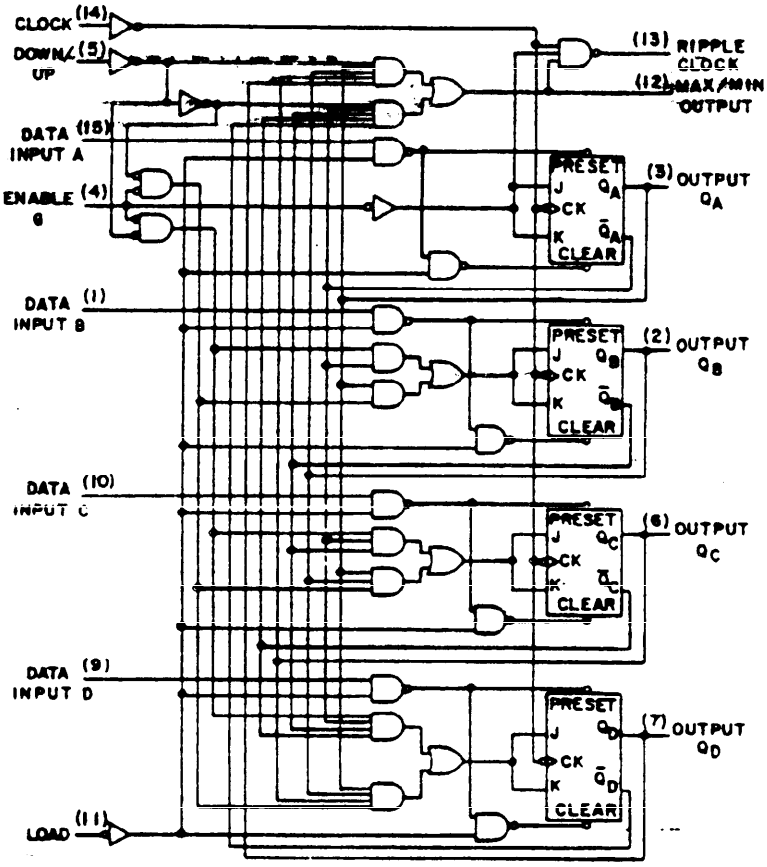


FIGURE 2. Logic diagrams - Continued.

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DEVICE TYPE 09

FIGURE 2. Logic diagrams - Continued.

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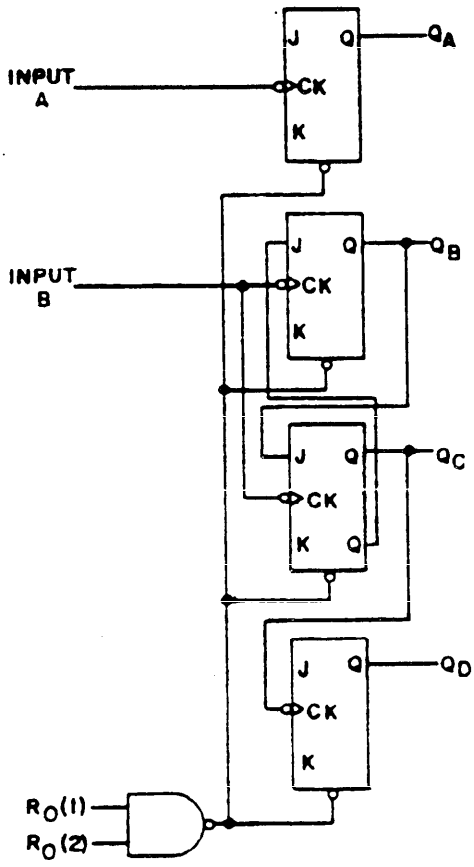
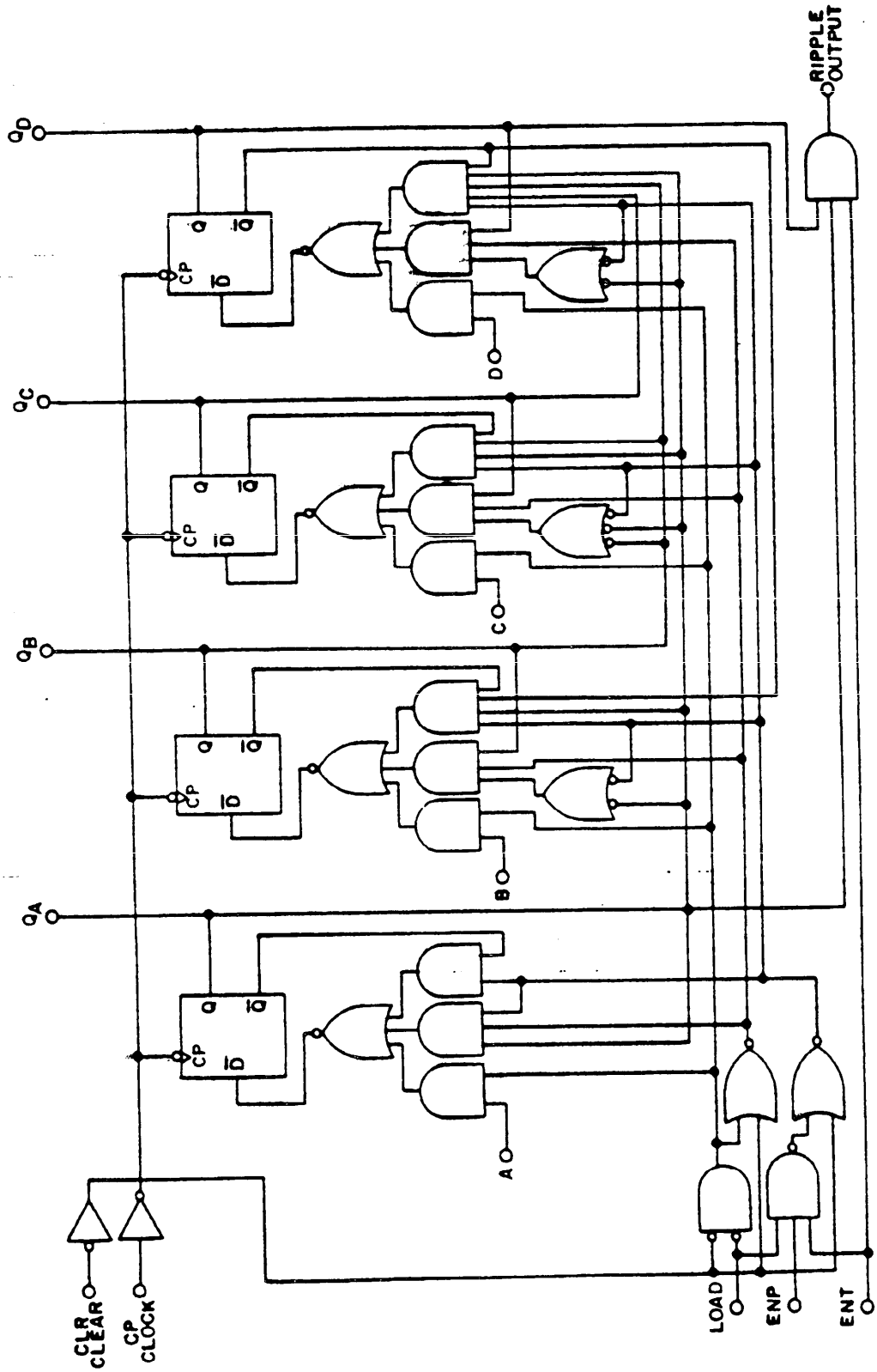


FIGURE 2. Logic diagrams - Continued.

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DEVICE TYPE 11
CIRCUITS A, C, D & G

FIGURE 2. Logic diagrams - Continued.

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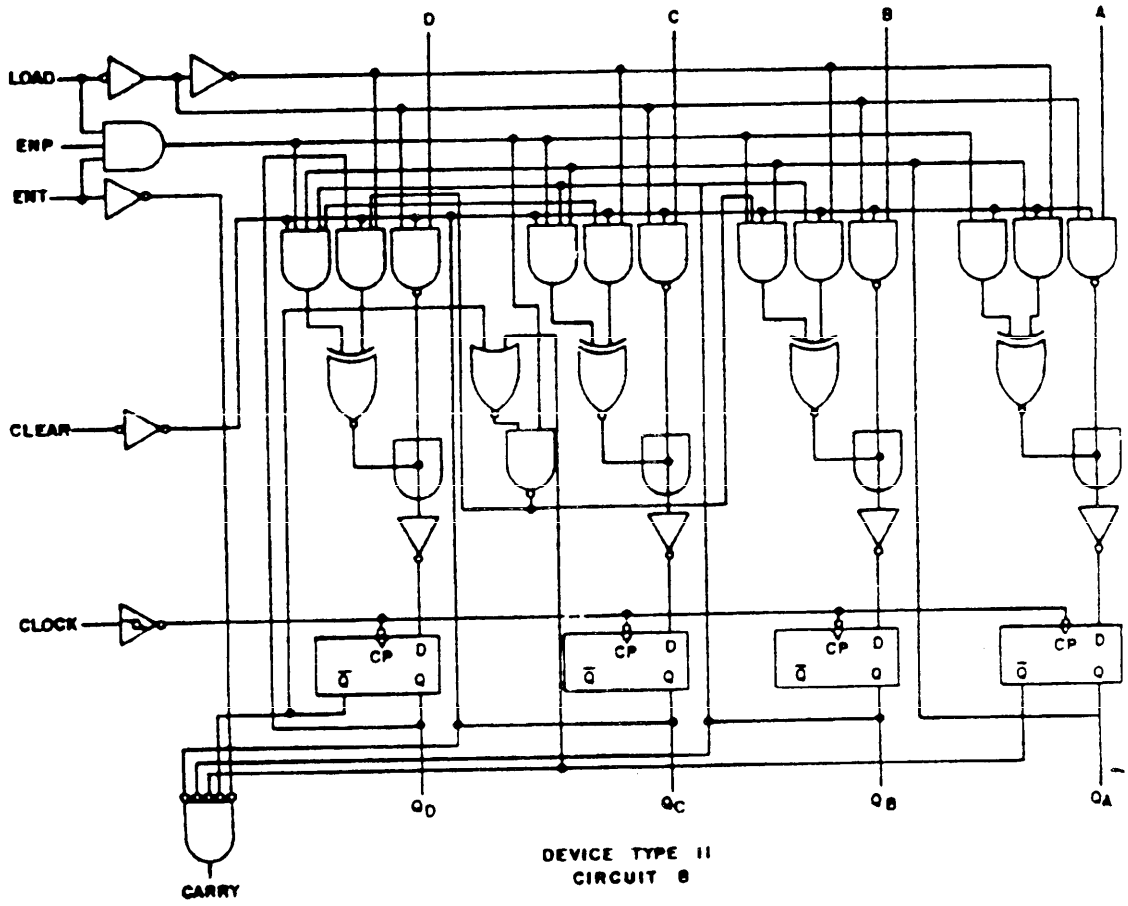
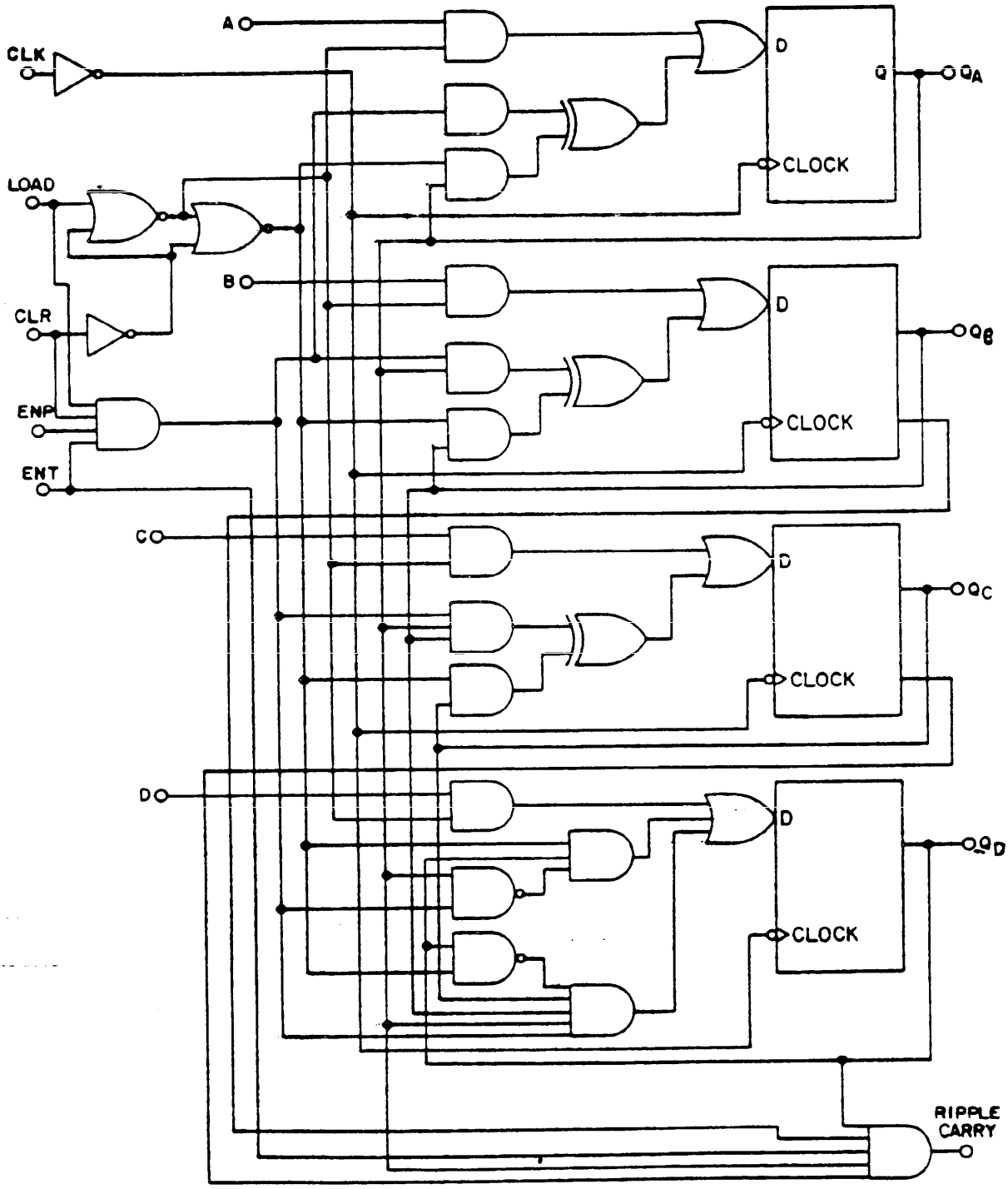


FIGURE 2. Logic diagram - Continued.

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DEVICE TYPE 11
CIRCUIT E

FIGURE 2. Logic diagrams - Continued.

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Device type 11
Circuit F

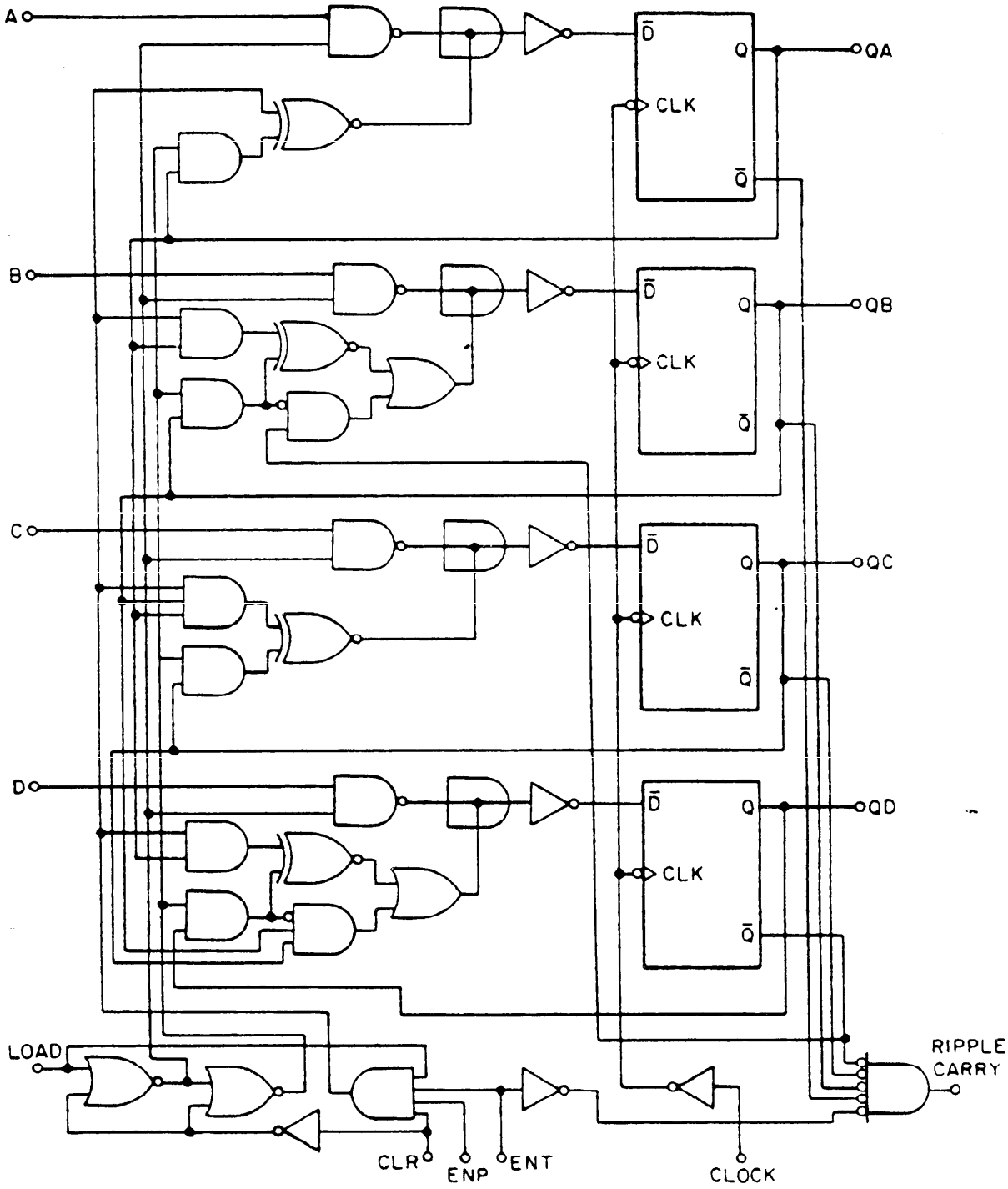


FIGURE 2. Logic diagrams - Continued.

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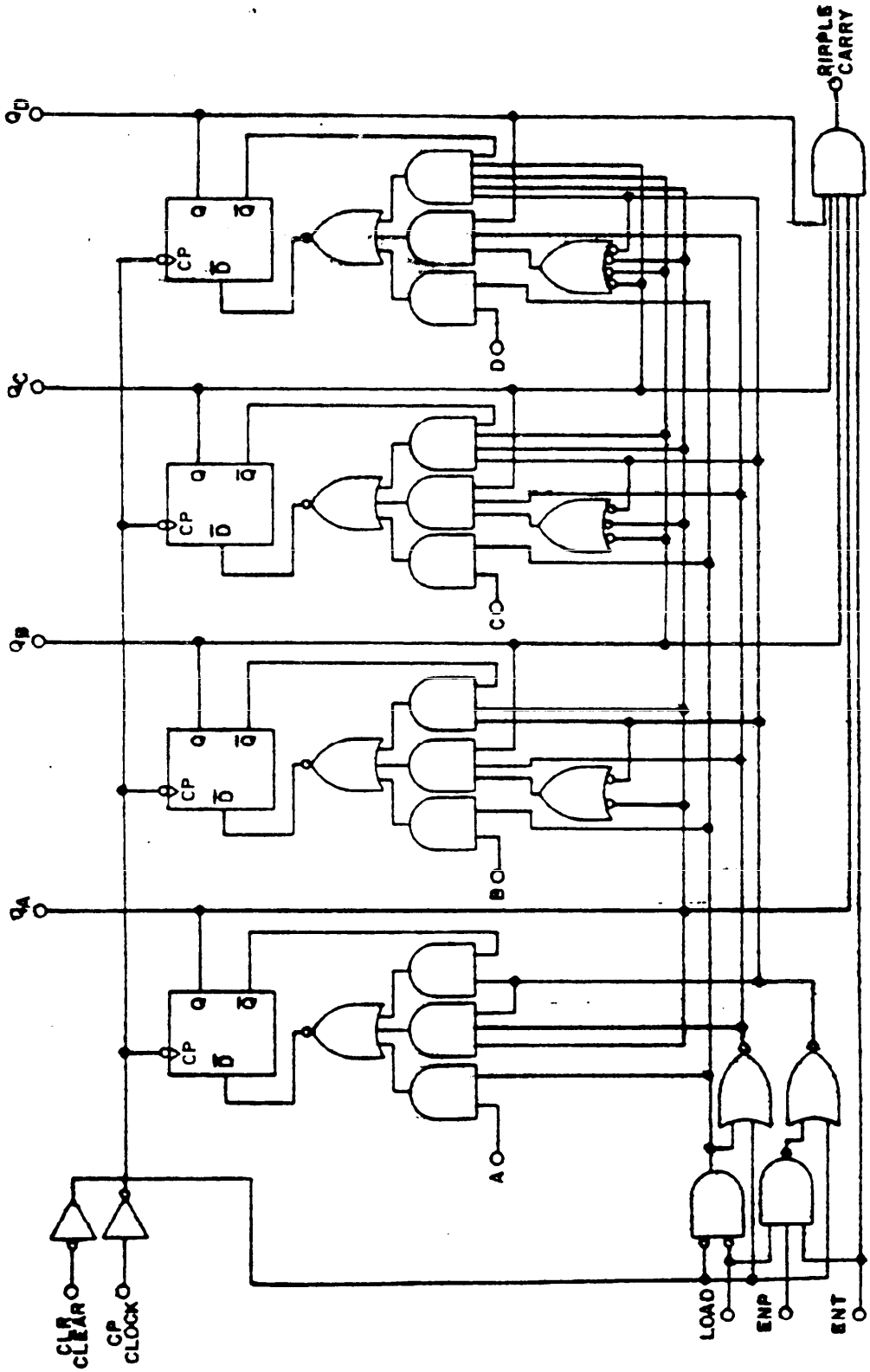


FIGURE 2. Logic diagrams - Continued.

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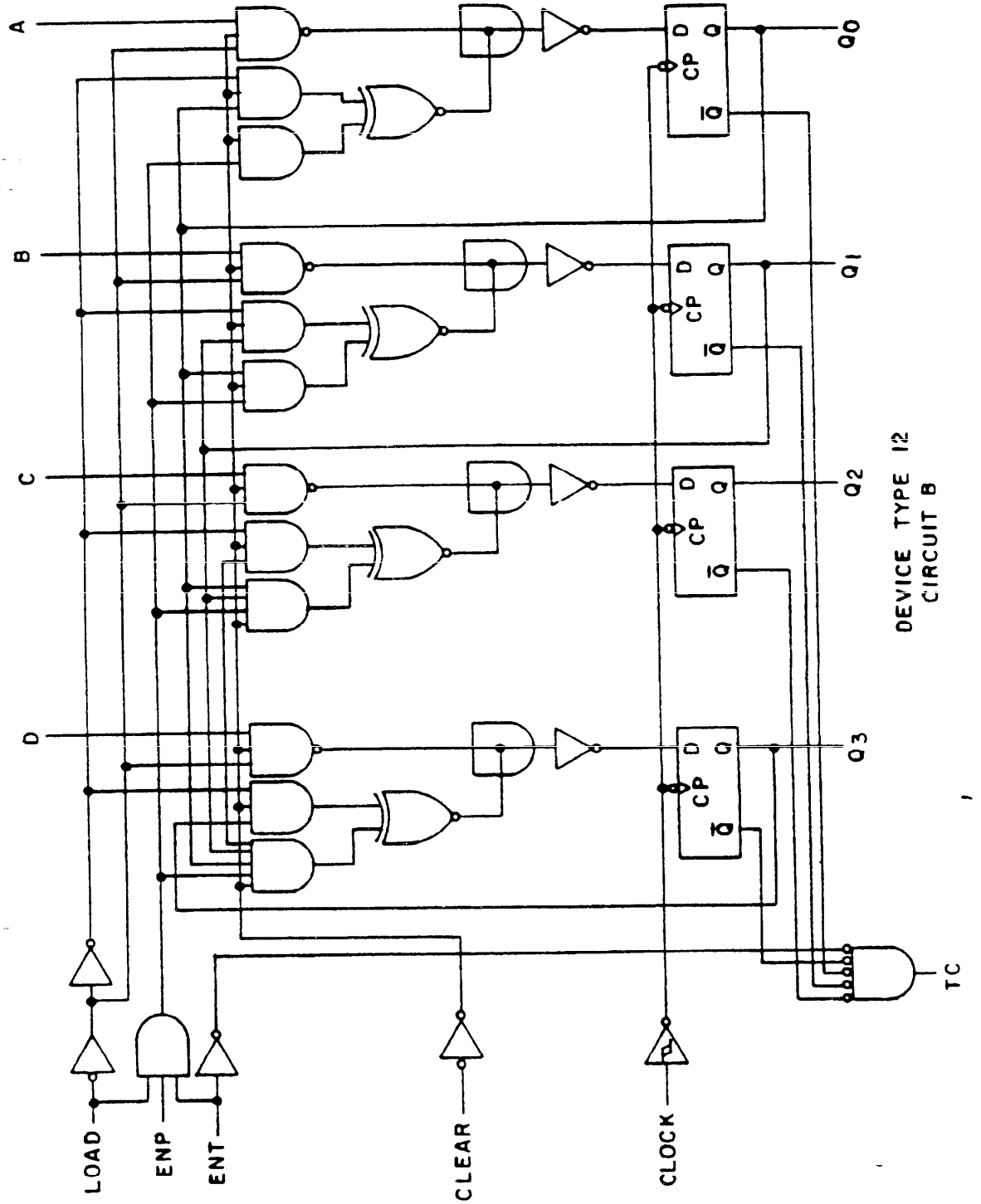
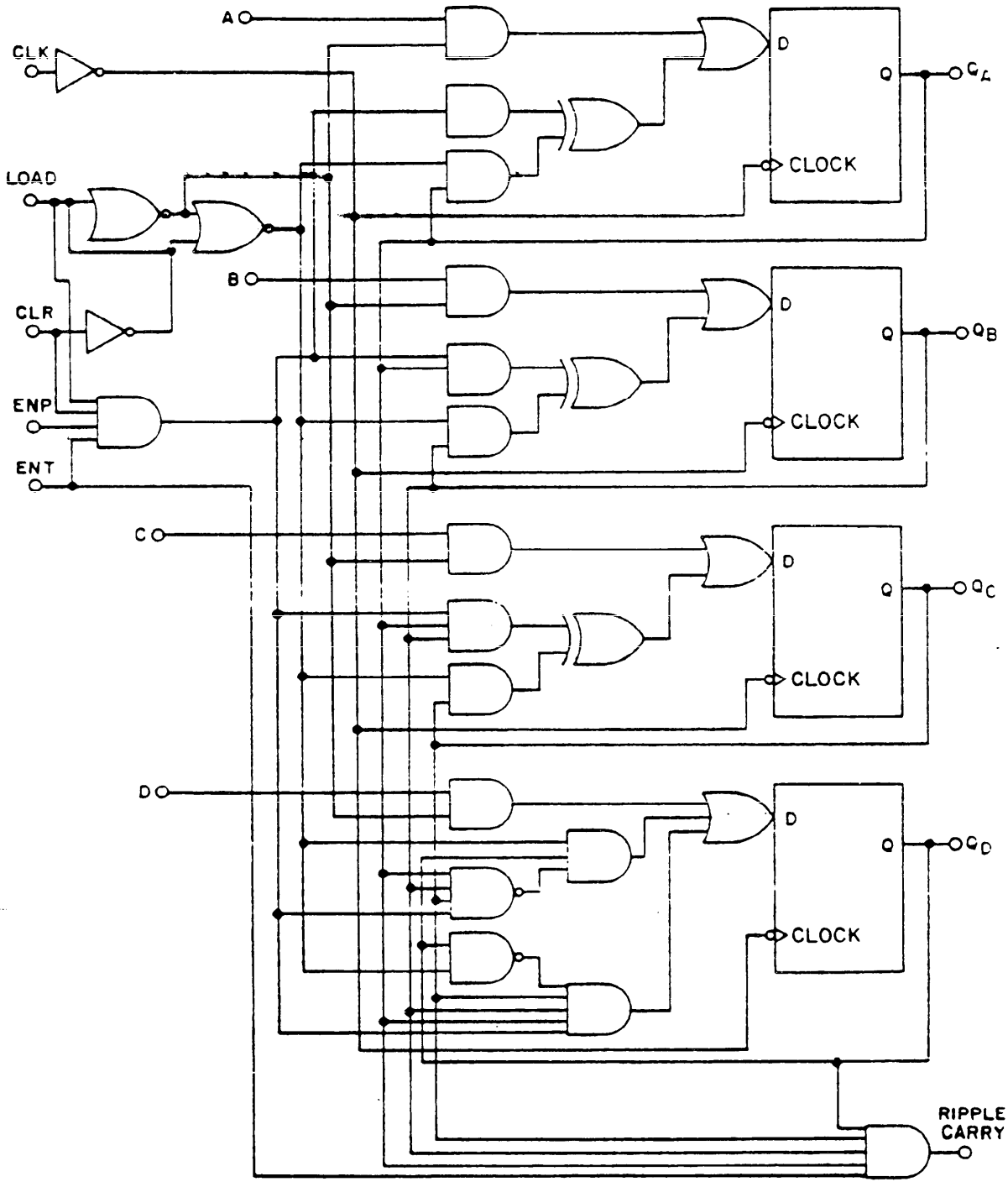


FIGURE 2. Logic diagram - Continued.

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DEVICE TYPE 12
CIRCUIT E

FIGURE 2. Logic diagrams - Continued.

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Device type 12
Circuit F

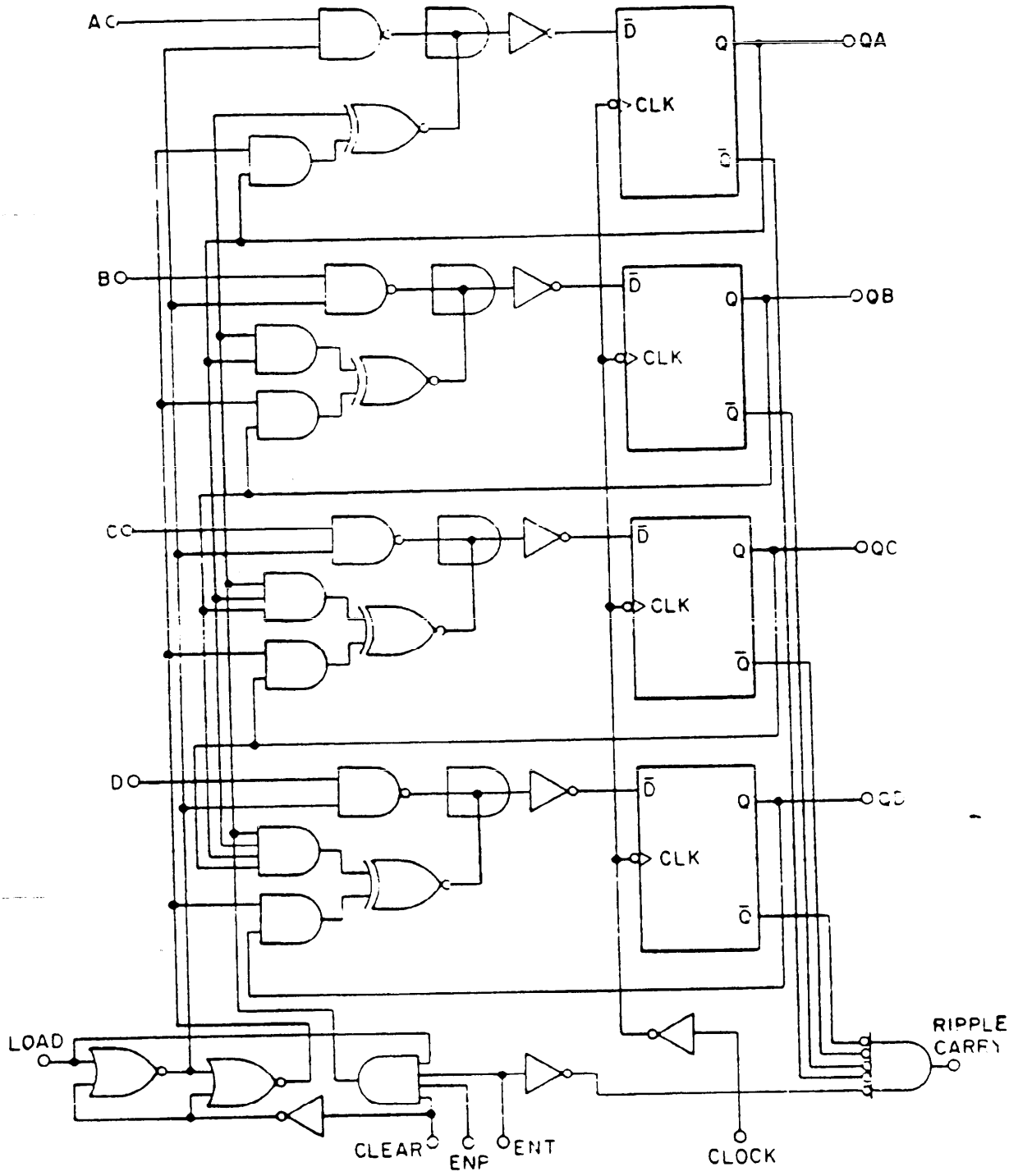
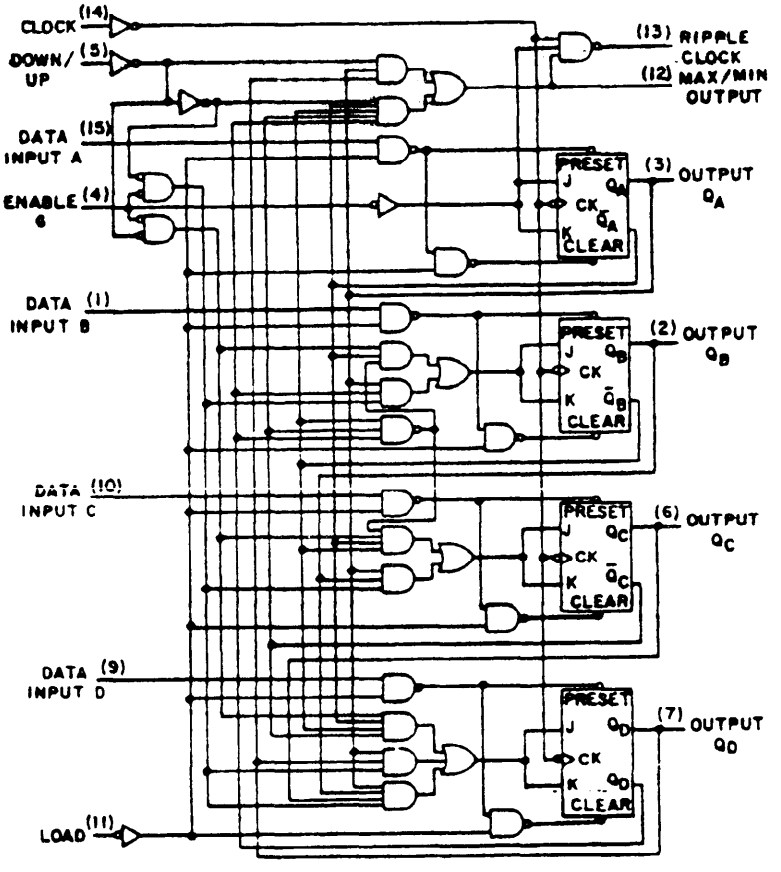


FIGURE 2. Logic diagrams - Continued.

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DEVICE TYPE 13

FIGURE 2. Logic diagrams - Continued.

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DEVICE TYPE 01

BCD COUNT SEQUENCE
(See Note A)

COUNT	OUTPUT			
	Q_D	Q_C	Q_B	Q_A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H

BI-QUINARY (5-?)
(See Note B)

COUNT	OUTPUT			
	Q_A	Q_D	Q_C	Q_B
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	H	L	L	L
6	H	L	L	H
7	H	L	H	L
8	H	L	H	H
9	H	H	L	L

RESET/COUNT FUNCTION TABLE

RESET INPUTS				OUTPUT			
$R_{0(1)}$	$R_{0(2)}$	$R_{9(1)}$	$R_{9(2)}$	Q_D	Q_C	Q_B	Q_A
H	H	L	X	L	L	L	L
H	H	X	L	L	L	L	L
X	X	H	H	H	L	L	H
X	L	X	L	COUNT			
L	X	L	X	COUNT			
L	X	X	L	COUNT			
X	L	L	X	COUNT			

NOTES: A. Output Q_A is connected to input B for BCD count.
B. Output Q_D is connected to input A for bi-quinary count.

DEVICE TYPE 02

COUNT SEQUENCE
(See Note)

COUNT	OUTPUT			
	Q_D	Q_C	Q_B	Q_A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	L	H	H	L
7	L	H	H	H
8	H	L	L	L
9	H	L	L	H
10	H	L	H	L
11	H	L	H	H
12	H	H	L	L
13	H	H	L	H
14	H	H	H	L
15	H	H	H	H

RESET/COUNT FUNCTION TABLE

RESET INPUTS		OUTPUT			
$R_{0(1)}$	$R_{0(2)}$	Q_D	Q_C	Q_B	Q_A
H	H	L	L	L	L
L	X	COUNT			
X	L	COUNT			

NOTE: Output Q_A is connected to input B.

DEVICE TYPE 10

COUNT SEQUENCE
(See Note)

COUNT	OUTPUT			
	Q_D	Q_C	Q_B	Q_A
0	L	L	L	L
1	L	L	L	H
2	L	L	H	L
3	L	L	H	H
4	L	H	L	L
5	L	H	L	H
6	H	L	L	L
7	H	L	L	H
8	H	L	H	L
9	H	L	H	H
10	H	H	L	L
11	H	H	L	H

DEVICE TYPE 10
RESET/COUNT FUNCTION TABLE

RESET INPUTS		OUTPUT			
$R_{0(1)}$	$R_{0(2)}$	Q_D	Q_C	Q_B	Q_A
H	H	L	L	L	L
L	X	COUNT			
X	L	COUNT			

NOTE: Output Q_A is connected to input B.

FIGURE 3. Truth tables.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
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SYNCHRONOUS TRUTH TABLE, DEVICE TYPES 3 AND 11

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry output
CP	L	X	H	X	X	X	X	H	NC	NC	NC	NC	NC
CP	X	L	H	X	X	X	X	H	NC	NC	NC	NC	L
CP	H	H	H	X	X	X	X	H	Previous count plus 1 (note 1)				H if count = 9 L if count < 9
CP	X	H	L	X	X	X	X	H	A	B	C	D	H if count = 9 L if count < 9
CP	X	L	L	X	X	X	X	H	A	B	C	D	L
CP	X	X	X	X	X	X	X	L	L	L	L	L	L

ASYNCHRONOUS TRUTH TABLE, DEVICE TYPE 3

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable $\overline{T}$	Load	$\overline{A}$	$\overline{B}$	$\overline{C}$	$\overline{D}$	Clear	Q_A	Q_B	Q_C	Q_D	Carry output
X	X	X	X	X	X	X	X	L	L	L	L	L	L

NOTES:
1. See up count sequence table.
2. L = V_{IL} for inputs, V_{OL} for outputs.
3. H = V_{IH} for inputs, V_{OH} for outputs.
4. X = V_{IH} or V_{IL}.
5. CP = Clock pulse.
6. NC = No change.

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H

FIGURE 3. Truth tables - Continued.

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SYNCHRONOUS TRUTH TABLE, DEVICE TYPES 4 AND 12

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry output
CP	L	X	H	X	X	X	X	H	NC	NC	NC	NC	NC
CP	X	L	H	X	X	X	X	H	NC	NC	NC	NC	L
CP	H	H	H	X	X	X	X	H	Previous count plus 1 (note 1)				H if count = 15 L if count < 15
CP	X	H	L	X	X	X	X	H	A	B	C	D	H if count = 15 L if count < 15
CP	X	L	L	X	X	X	X	H	A	B	C	D	L
CP	X	X	X	X	X	X	X	L	L	L	L	L	L

ASYNCHRONOUS TRUTH TABLE, DEVICE TYPE 4

Inputs at time t_n									Outputs at time t_{n+1}				
Clock	Enable P	Enable T	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry output
X	X	X	X	X	X	X	X	L	L	L	L	L	L

NOTES:
1. See up count sequence table.
2. L = V_{IL} for inputs, V_{OL} for outputs.
3. H = V_{IH} for inputs, V_{OH} for outputs.
4. X = V_{IH} or V_{IL}.
5. CP = Clock pulse.
6. NC = No change.

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	H

FIGURE 3. Truth tables - Continued.

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Device type 05

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H

Device type 06

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)
L	L	L	L
H	L	L	L
L	H	L	L
H	H	L	L
L	L	H	L
H	L	H	L
L	H	H	L
H	H	H	L
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	H

Device types 05 and 06

MODE SELECT TABLE

L	EP	ET	U/$\overline{D}$	Action on Rising Clock Edge
L	X	X	X	Load ($D_n \rightarrow Q_n$)
H	L	L	H	Count Up (increment)
H	L	L	L	Count Down (decrement)
H	H	X	X	No Change (Hold)
H	X	H	X	No Change (Hold)

H = High voltage level
L = Low voltage level
X = Don't care

FIGURE 3. Truth tables - Continued.

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DEVICE TYPE 7 TRUTH TABLE

Inputs at time t_n								Outputs at time t_{n+1}					
Count up	Count down	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry	Borrow
H	H	H	X	X	X	X	L	NC	NC	NC	NC	H	H
H	H	H	X	X	X	X	H	L	L	L	L	H	H
H	H	L	X	X	X	X	L	A	B	C	D	H	H
P	H	H	X	X	X	X	L	Previous count plus 1 (note 1)				H	H
H	P	H	X	X	X	X	L	Previous count minus 1 (note 2)				H	H
N	H	H	X	X	X	X	L	NC	NC	NC	NC	N if count = 9 H if count ≠ 9	H
H	N	H	X	X	X	X	L	NC	NC	NC	NC	H	N if count = 0 H if count ≠ 0

NOTES:
1. See up count sequence table.
2. See down count sequence table.
3. L = V_{IL} for inputs, V_{OL} for outputs.
4. H = V_{IH} for inputs, V_{OH} for outputs.
5. X = V_{IH} or V_{IL}.
6. NC = No change.
7. NA = Not applicable.
8. P = Positive going pulse.
9. N = Negative going pulse.

FIGURE 3. Truth tables - Continued.

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DEVICE TYPE 07

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)	Carry
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	L

DOWN COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)	Borrow
H	L	L	H	H
L	L	L	H	H
H	H	H	L	H
L	H	H	L	H
H	L	H	L	H
L	L	H	L	H
H	H	L	L	H
L	H	L	L	H
H	L	L	L	H
L	L	L	L	L

DEVICE TYPE 8 TRUTH TABLE

Inputs at time t_n								Outputs at time t_{n+1}					
Count up	Count down	Load	A	B	C	D	Clear	Q_A	Q_B	Q_C	Q_D	Carry	Borrow
H	H	H	X	X	X	X	L	NC	NC	NC	NC	H	H
H	H	H	X	X	X	X	H	L	L	L	L	H	H
H	H	L	X	X	X	X	L	A	B	C	D	H	H
P	H	H	X	X	X	X	L	Previous count plus 1 (note 1)				H	H
H	P	H	X	X	X	X	L	Previous count minus 1 (note 2)				H	H
N	H	H	X	X	X	X	L	NC	NC	NC	NC	N if count = 15 H if count ≠ 15	H
H	N	H	X	X	X	X	L	NC	NC	NC	NC	H	N if count = 0 H if count ≠ 0

NOTES:
1. See up count sequence table.
2. See down count sequence table.
3. L = V_{IL} for inputs, V_{OL} for outputs.
4. H = V_{IH} for inputs, V_{OH} for outputs.
5. X = V_{IH} or V_{IL}.
6. NC = No change.
7. NA = Not applicable.
8. P = Positive going pulse.
9. N = Negative going pulse.

FIGURE 3. Truth tables - Continued.

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DEVICE TYPE 08

UP COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)	Carry
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

DOWN COUNT SEQUENCE TABLE

Q_A (LSB)	Q_B	Q_C	Q_D (MSB)	Borrow
H	H	H	H	H
L	H	H	H	H
H	L	H	H	H
L	L	H	H	H
H	H	L	H	H
L	H	L	H	H
H	L	L	H	H
L	L	L	H	H
H	H	H	L	H
L	H	H	L	H
H	L	H	L	H
L	L	H	L	H
H	H	L	L	H
L	H	L	L	H
H	L	L	L	H
L	L	L	L	L

DEVICE TYPES 09 AND 13

Mode select table

Inputs				Mode
Load	Enable G	U/D	CLK	
H	L	L	⌐	Count up
H	L	H	⌐	Count down
L	X	X	X	Preset (Asyn)
H	H	X	X	No change (Hold)

Ripple carry truth table

Inputs		Outputs	
Enable G	CLK	Max/Min	RC output
L	⌐⌙	H	⌐⌙
H	X	X	H
X	X	L	H

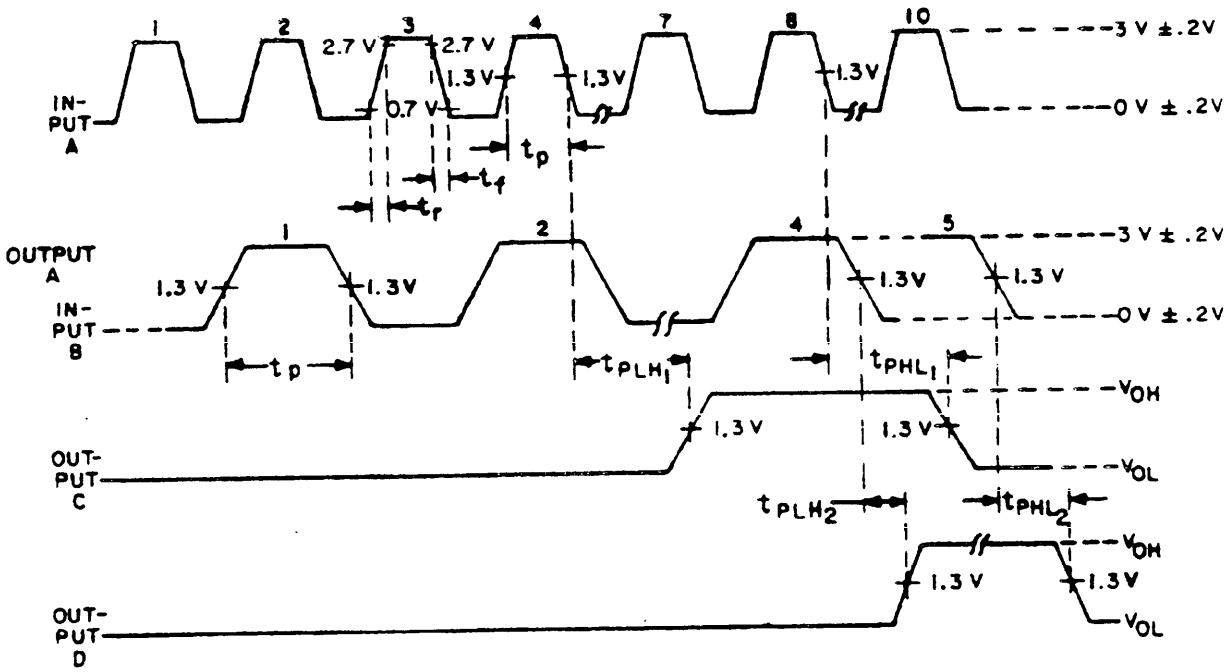
L = Low voltage level
H = High voltage level
X = Don't care
⌐ = Low-to-high clock transition
⌐⌙ = Negative going clock pulse

NOTE: The up count and down count sequence for device type 09 is identical as that for device type 08.

The up count and down count sequence for device type 13 is identical as that for device type 07.

FIGURE 3. Truth tables - Continued.

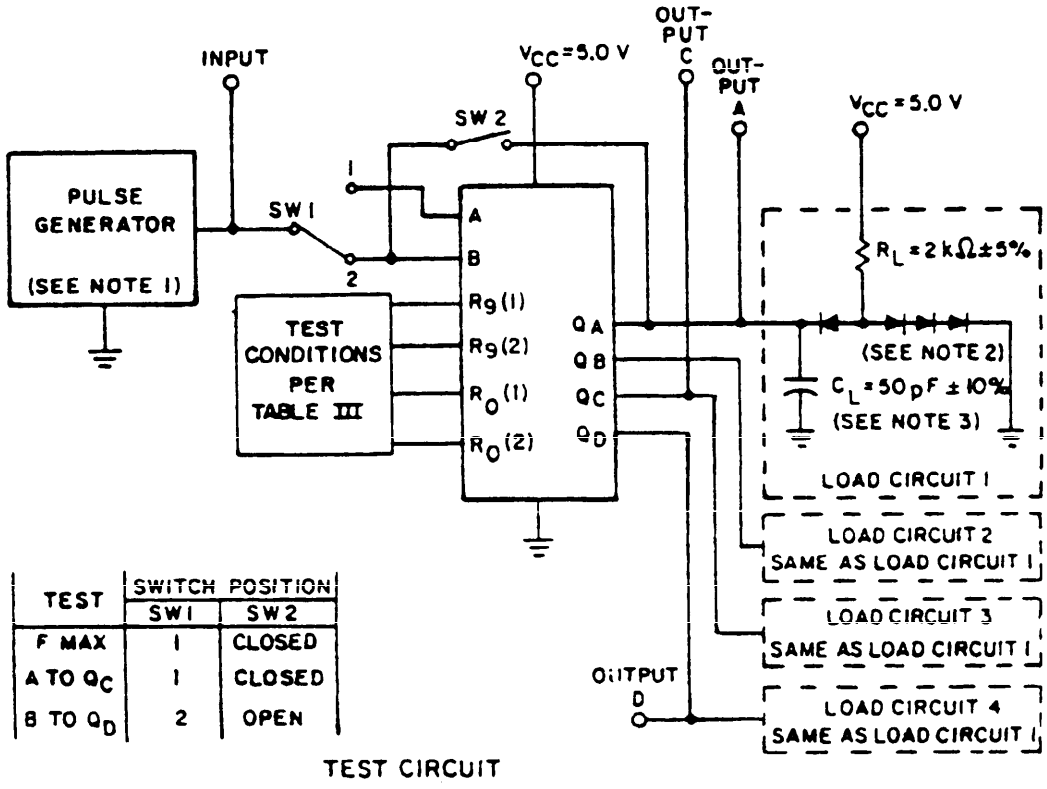
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VOLTAGE WAVEFORMS

FIGURE 4. Switching time test circuit and waveforms for device type 01.

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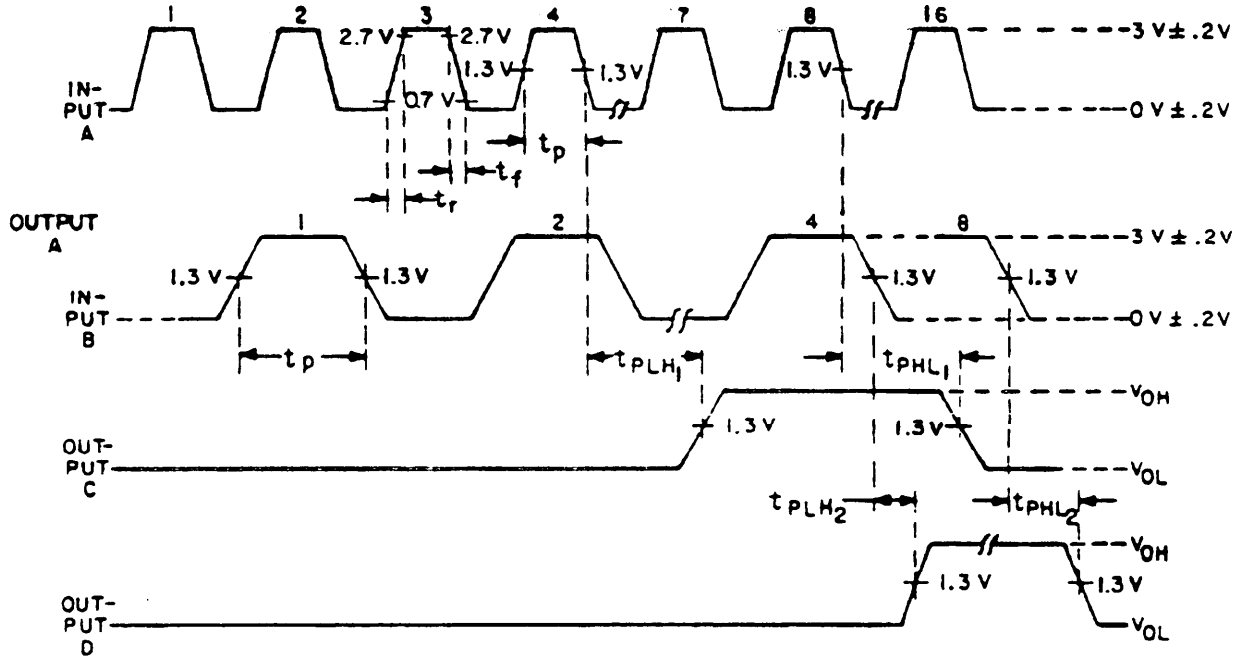
TEST	SWITCH POSITION	
	SW1	SW2
F MAX	1	CLOSED
A TO Q_C	1	CLOSED
B TO Q_D	2	OPEN

TEST CIRCUIT

NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, t_p = .5 µs, PRR ≤ 1 MHz, $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX}: $t_r = t_f \leq 6$ ns.

FIGURE 4. Switching time test circuit and waveforms for device type 01 - Continued.

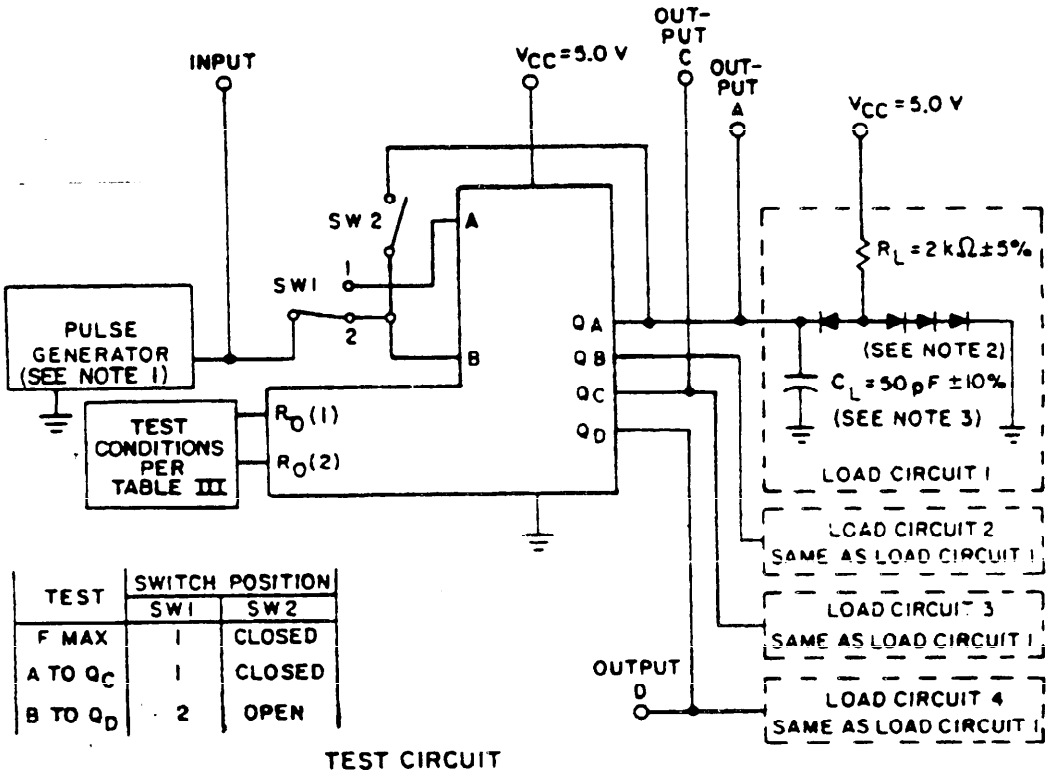
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VOLTAGE WAVEFORMS

FIGURE 5. Switching time test circuit and waveforms for device type 02.

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TEST	SWITCH POSITION	
	SW1	SW2
F MAX	1	CLOSED
A TO Q_C	1	CLOSED
B TO Q_D	2	OPEN

TEST CIRCUIT

NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, t_p = .5 µs, PRR ≤ 1 MHz, $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX}: $t_r = t_f \leq 6$ ns.

FIGURE 5. Switching time test circuit and waveforms for device type 02 - Continued.

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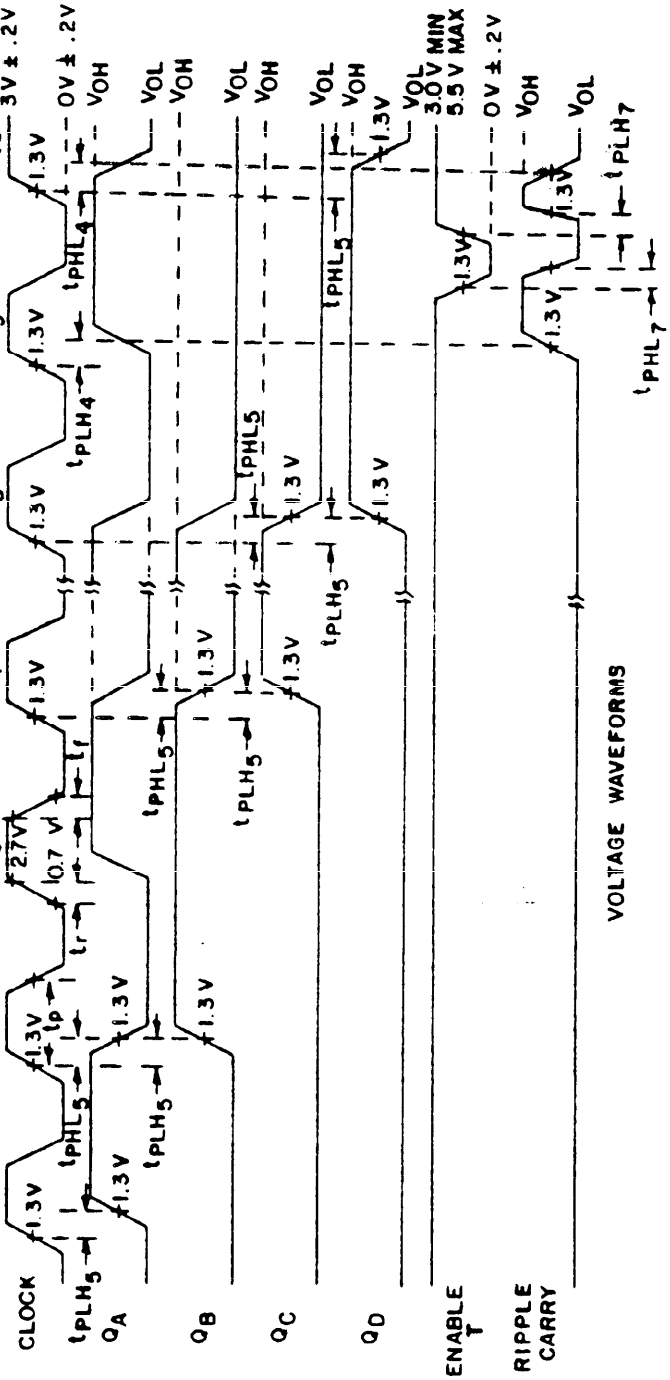


FIGURE 6. Switching time test circuit and waveforms for device types 03, 04, 11, and 12.

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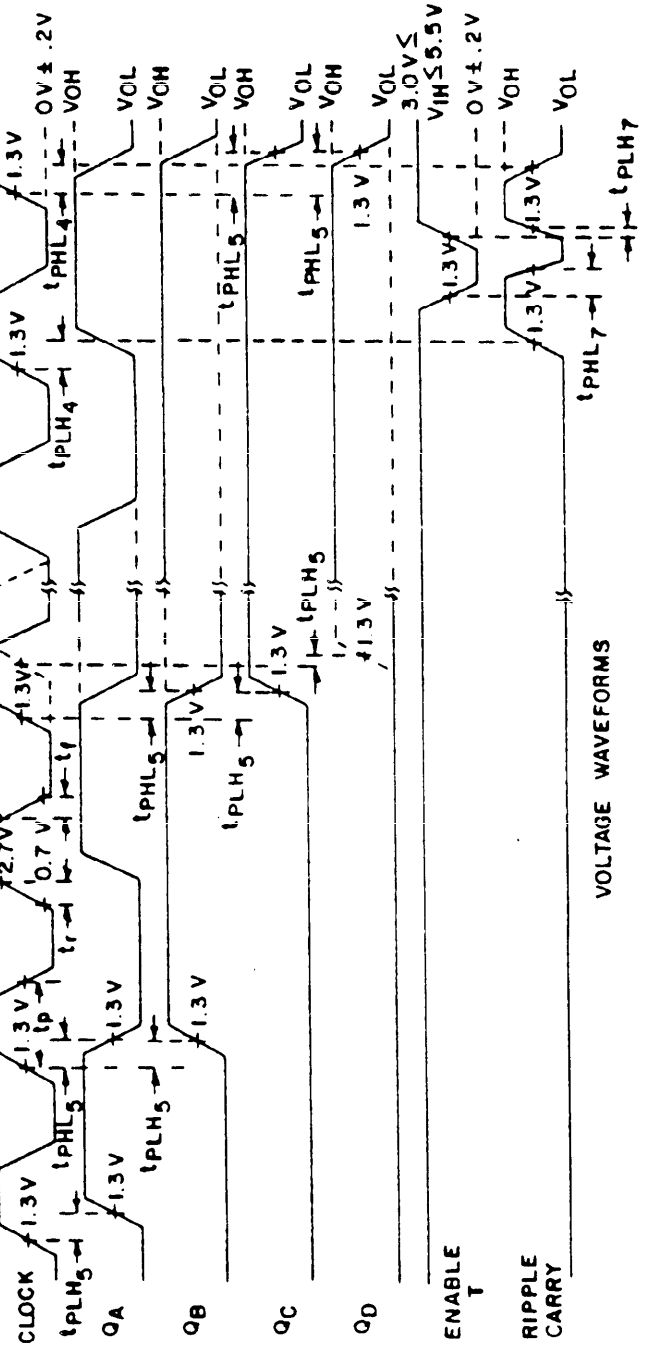
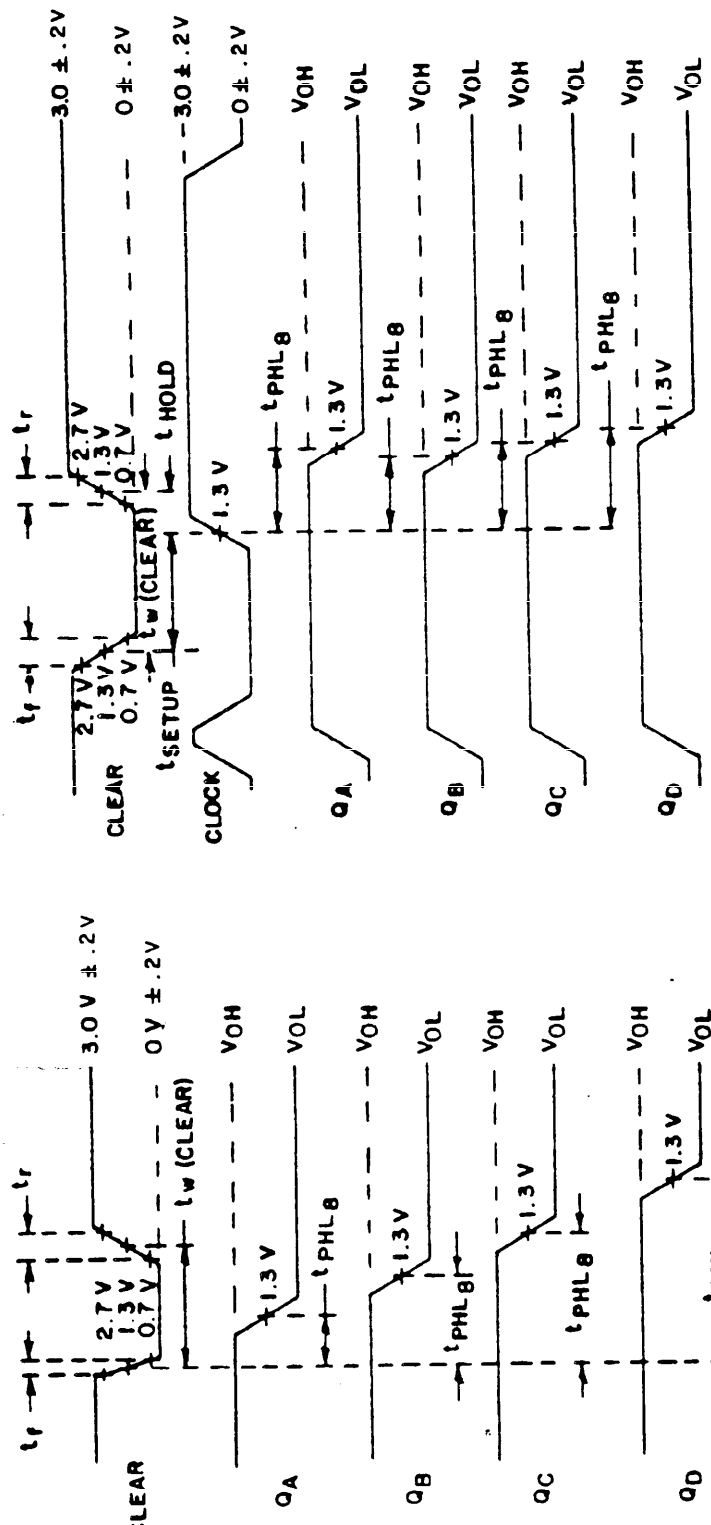


FIGURE 6. Switching time test circuit and waveforms for device types 03, 04, 11, and 12 - Continued.

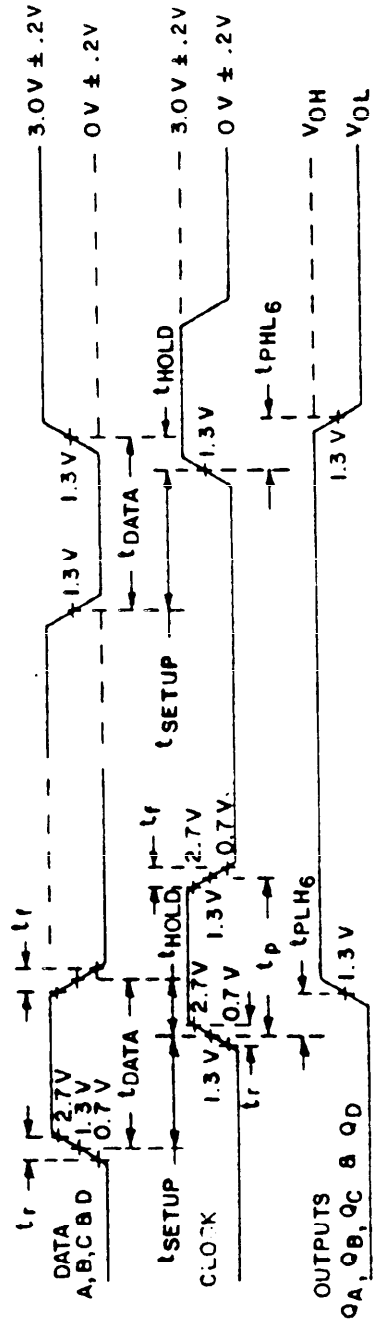
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NOTE: The clear pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_r \leq 6$ ns, $t_f \leq 15$ ns, $t_{w(clear)}$ = 20 ns; for types 11 and 12, t_{setup} = 20 ns, t_{hold} = 0 ns.

FIGURE 6. Switching time test circuit and waveforms for device types 03, 04, 11, and 12 - Continued.

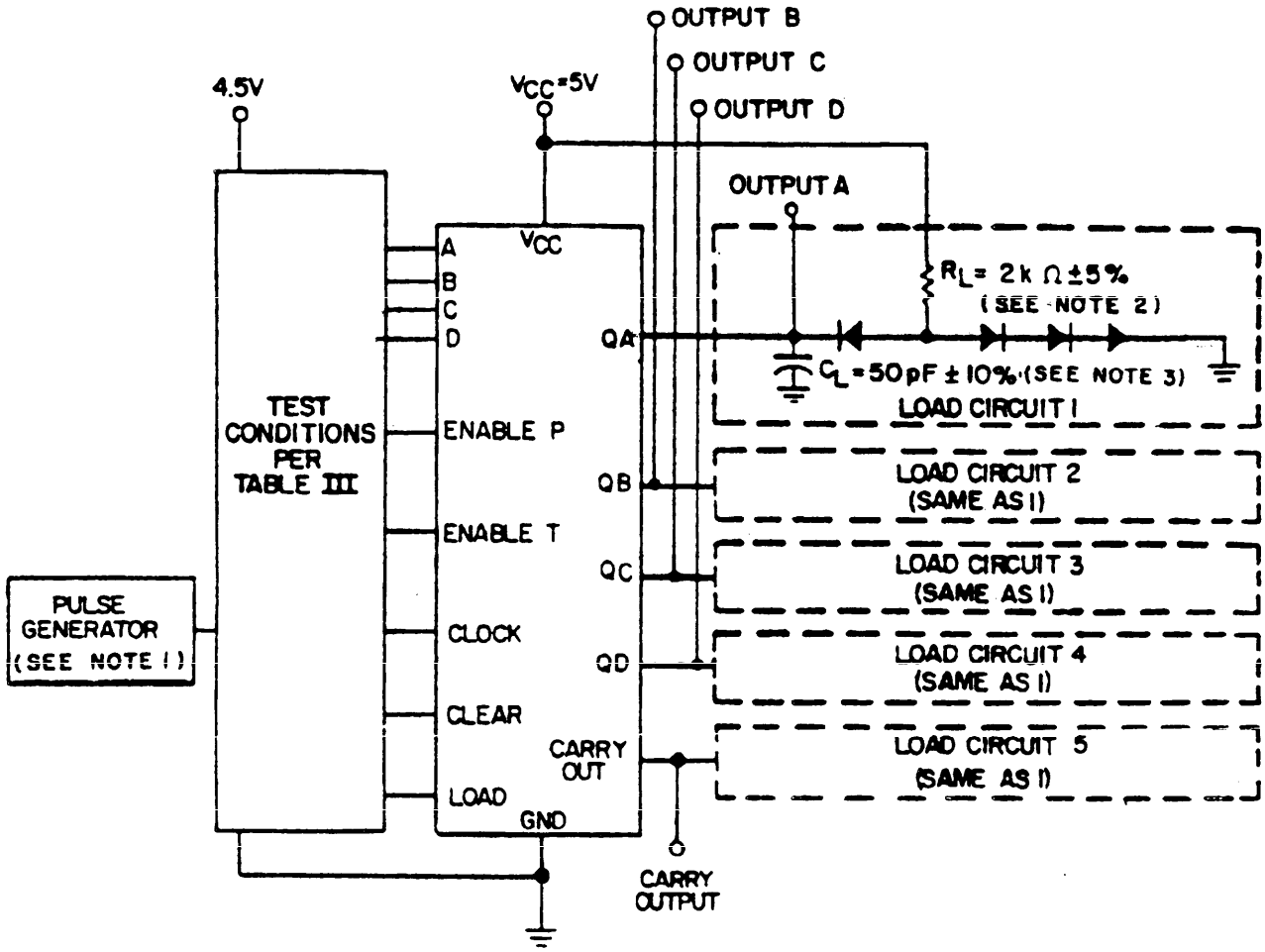
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NOTE: The data pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, t_{DATA} = 30 ns, t_{SETUP} = 20 ns, t_{HOLD} = 10 ns.

FIGURE 6. Switching time test circuit and waveforms for device types 03, 04, 11, and 12 - Continued.

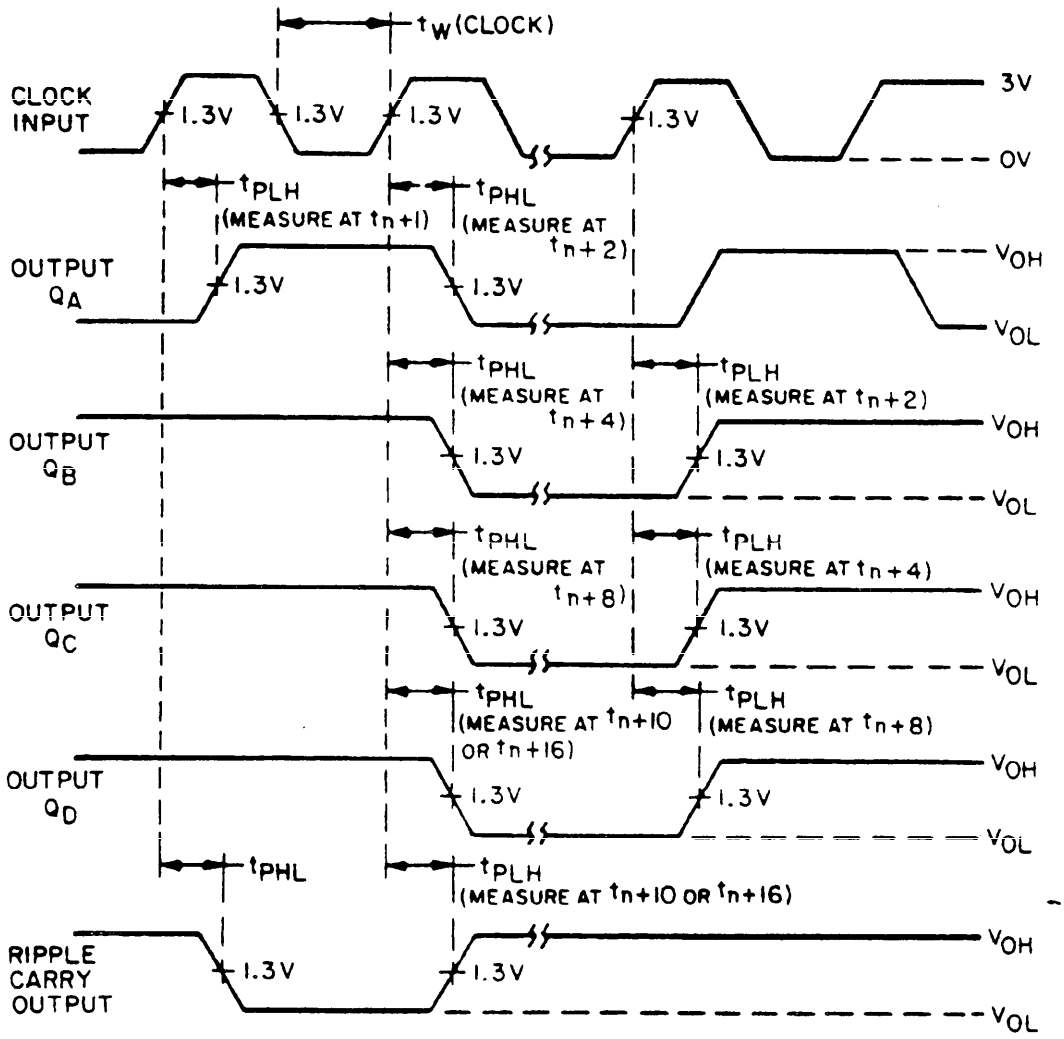
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NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, t_p = .5 µs, PRR ≤ 1 MHz, $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground termina].
5. F_{MAX}: $t_r = t_f \leq 6$ ns.

FIGURE 6. Switching time test circuit and waveforms for device types 03, 04, 11, and 12 - Continued.

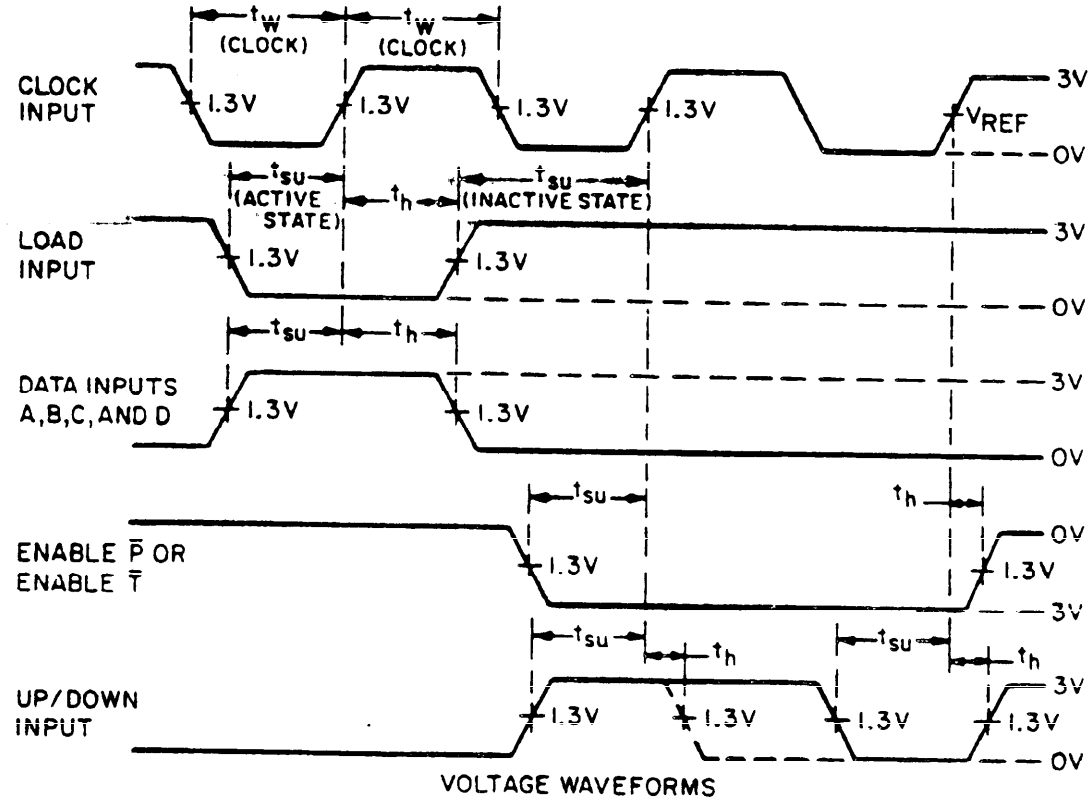
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UP-COUNT VOLTAGE WAVEFORMS

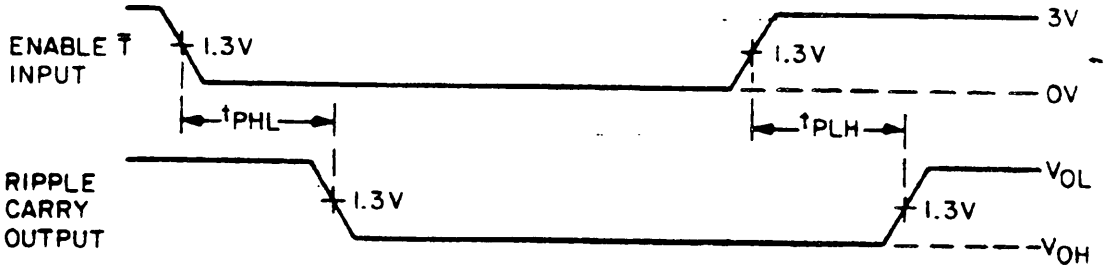
FIGURE 7. Switching time waveforms for device types 05 and 06.

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VOLTAGE WAVEFORMS

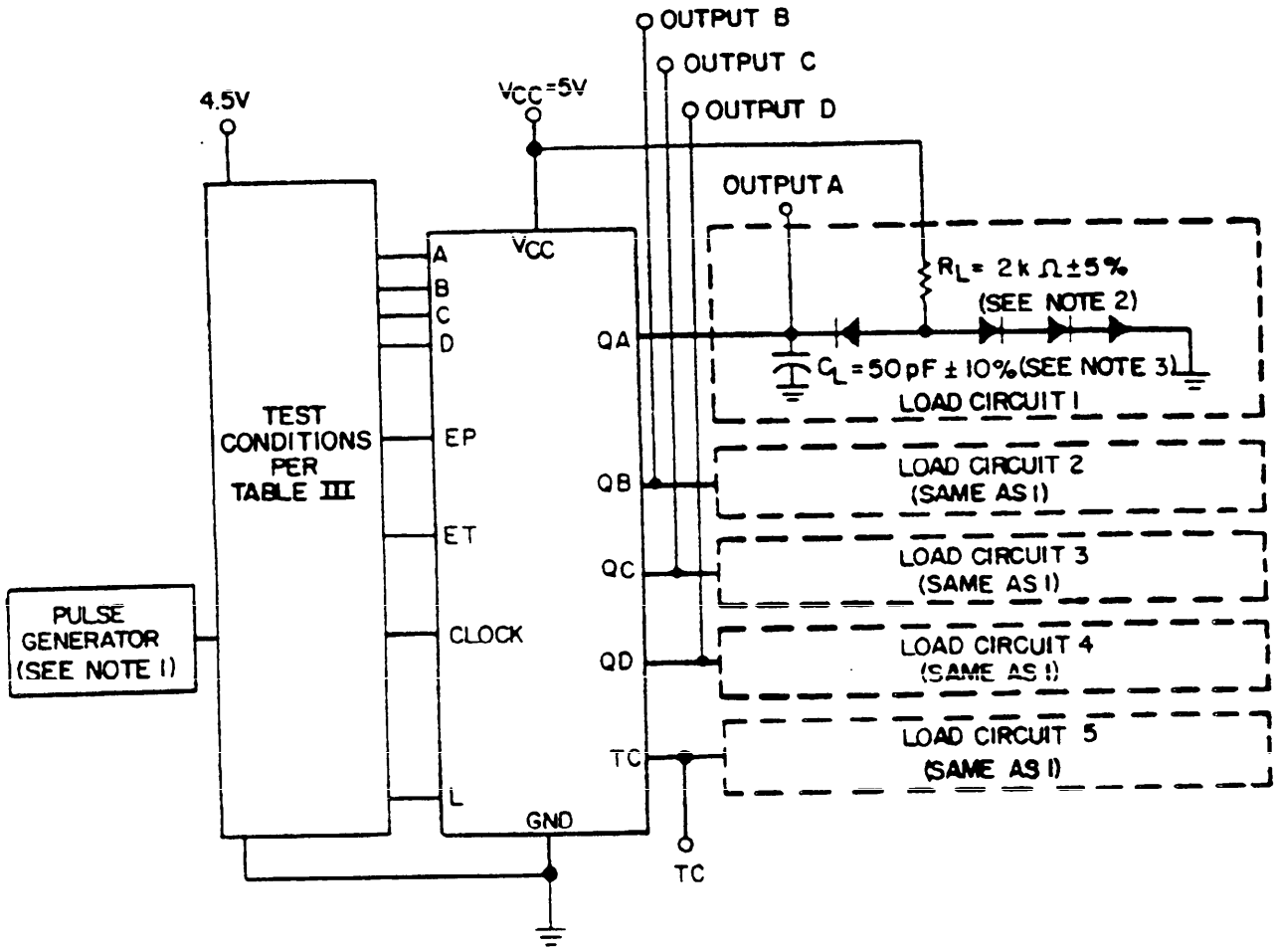
PULSE WIDTHS, SETUP TIMES, HOLD TIMES



VOLTAGE WAVEFORMS

FIGURE 7. Switching time waveforms for device types 05 and 06 - Continued.

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NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, t_p = .5 us, PRR ≤ 1 MHz, $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX}: $t_r = t_f \leq 6$ ns.
6. The clear pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, $t_{w(clear)}$ = 20 ns.

FIGURE 7. Switching time waveforms for device types 05 and 06 - Continued.

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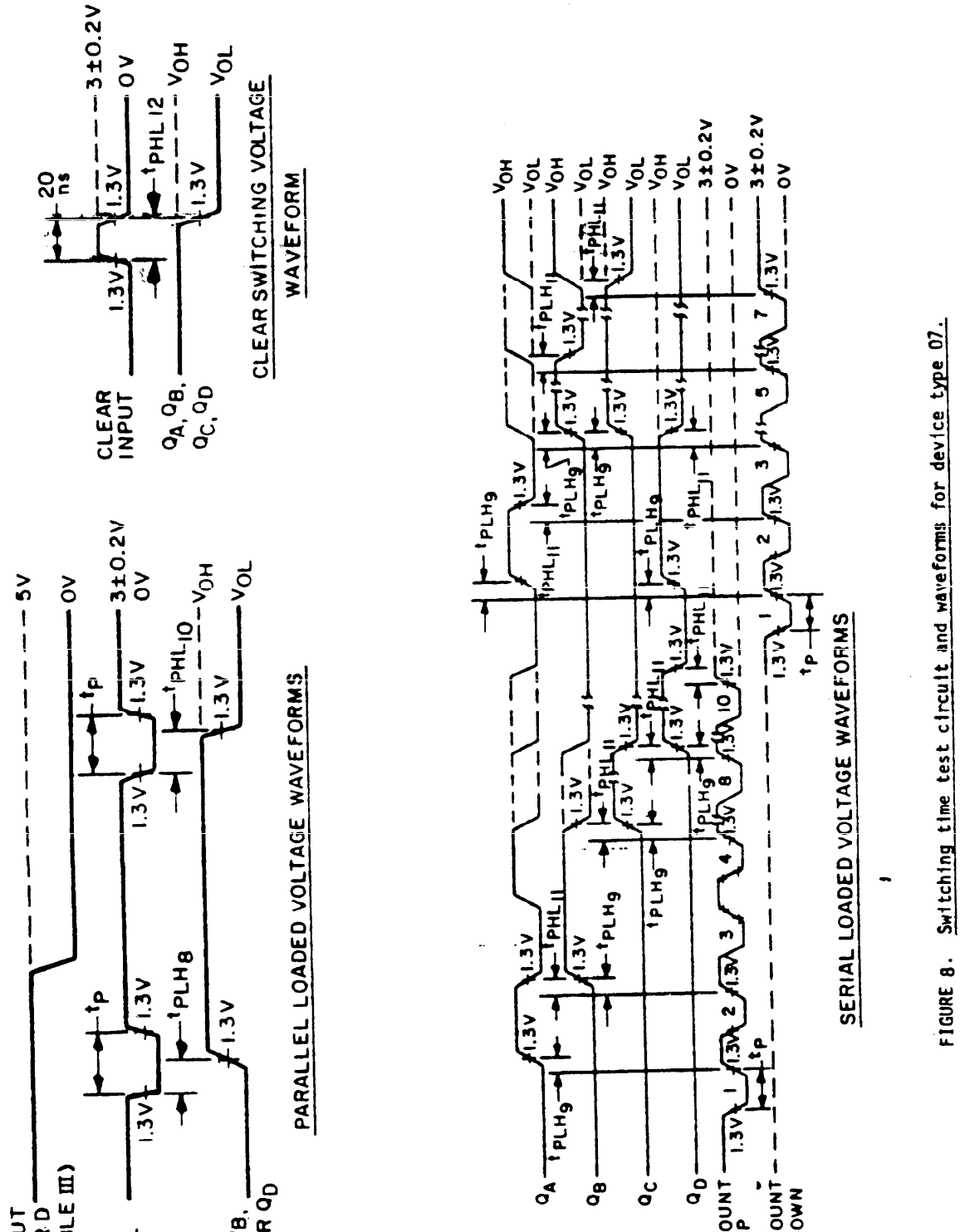
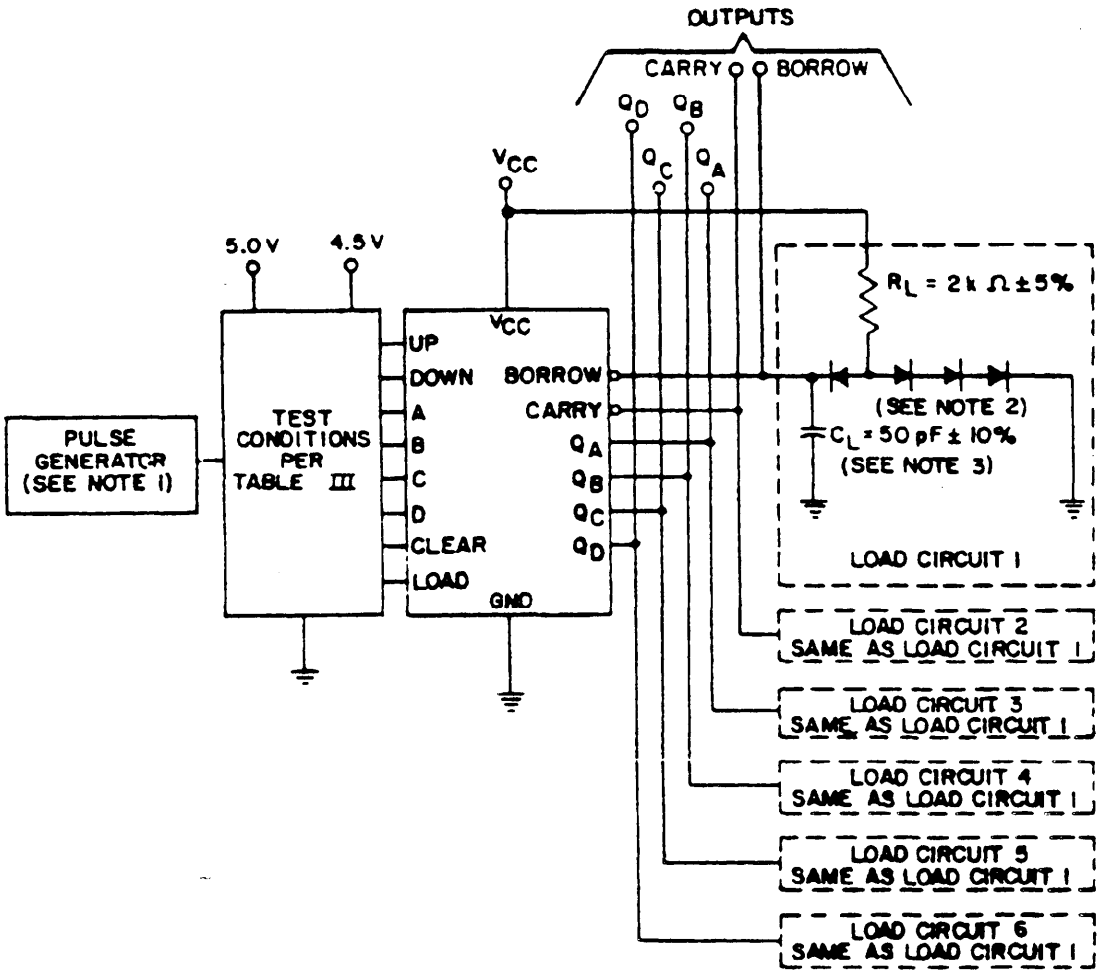


FIGURE 8. Switching time test circuit and waveforms for device type 07.

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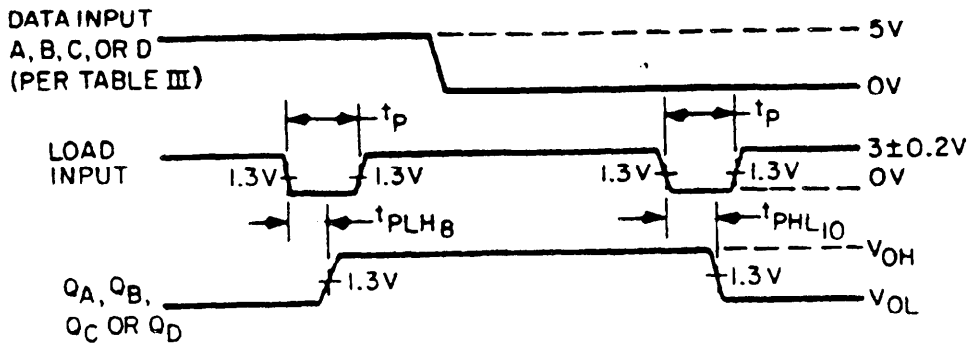


TEST CIRCUIT

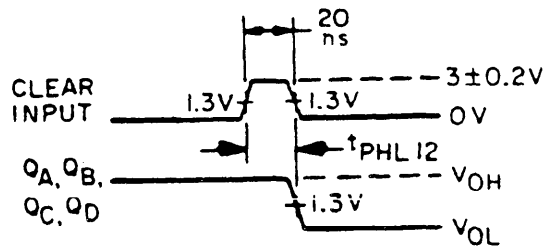
NOTES:
1. The load and count pulse generators have the following characteristics: V_{gen} = 3 V, t_p = .5 µs, PRR ≤ 1 MHz, $Z_{out} \approx 50\Omega$, $t_r \leq 15$ ns, $t_f \leq 6$ ns between 0.7 V and 2.7 V.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX}: $t_r = t_f \leq 6$ ns.
6. The clear pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns between 0.7 V and 2.7 V, $t_{w(clear)}$ = 20 ns.

FIGURE 8. Switching time test circuit and waveforms for device type 07 - Continued.

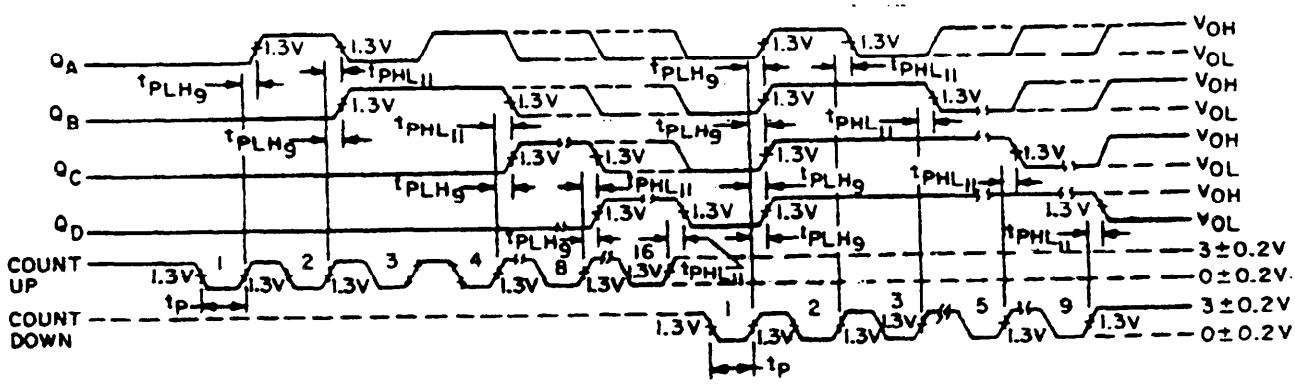
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PARALLEL LOADED VOLTAGE WAVEFORMS



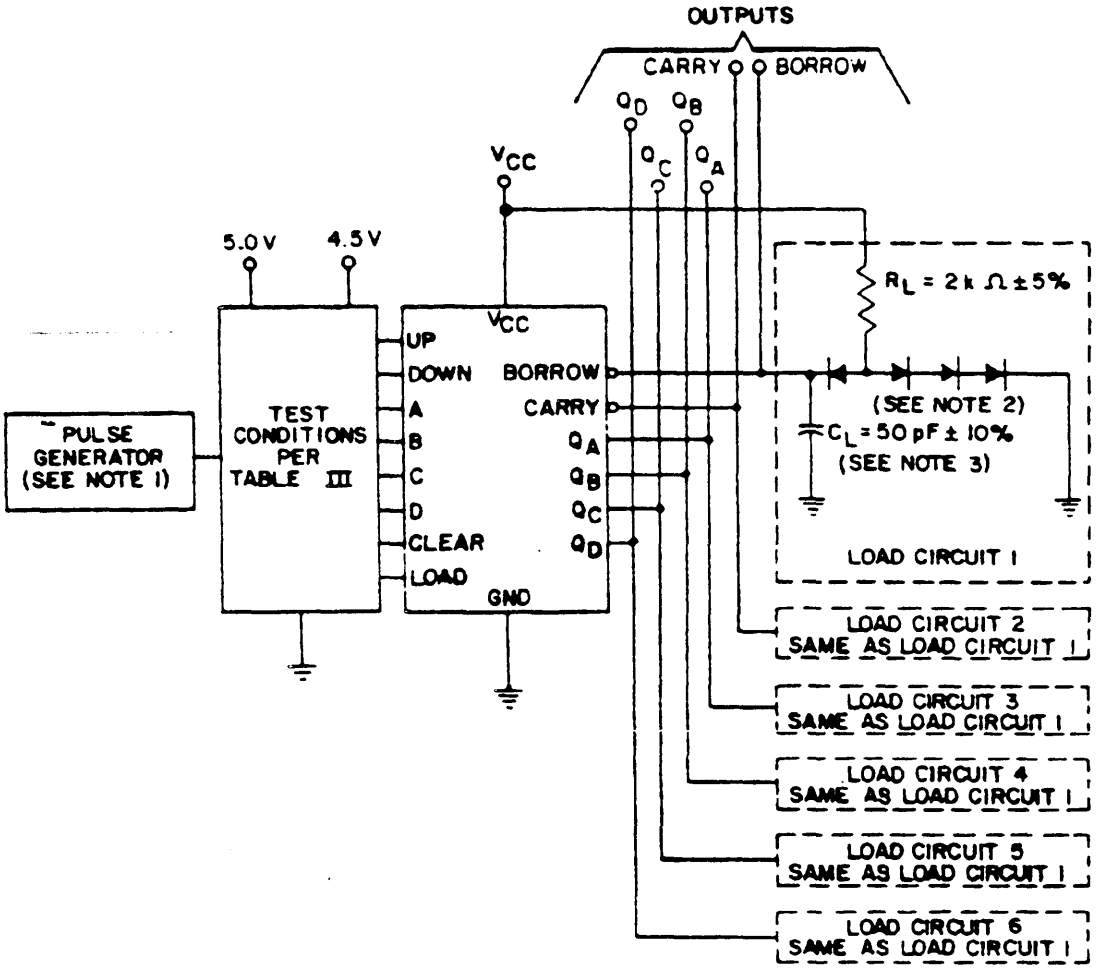
CLEAR SWITCHING VOLTAGE WAVEFORM



SERIAL LOADED VOLTAGE WAVEFORMS

FIGURE 9. Switching time test circuit and waveforms for device type 08.

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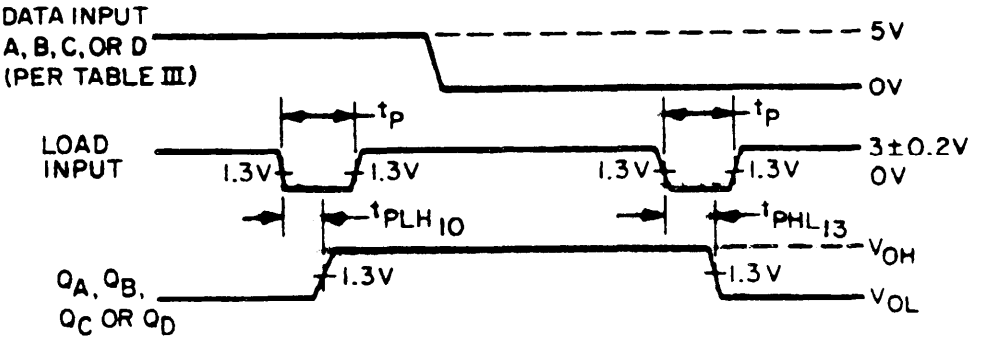


TEST CIRCUIT

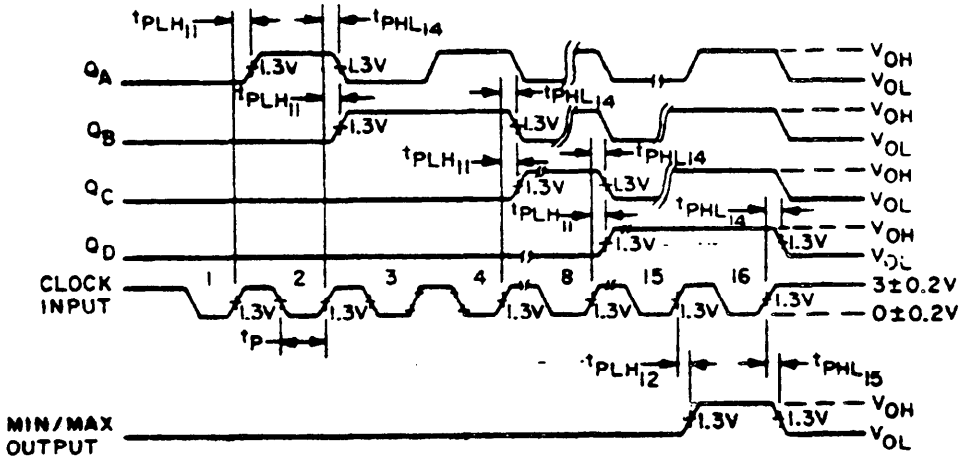
NOTES:
1. The load and count pulse generators have the following characteristics: V_{gen} = 3 V, t_p = .5 µs, PRR ≤ 1 MHz, $Z_{out} \approx 50\Omega$, $t_r \leq 15$ ns, $t_f \leq 6$ ns between 0.7 V and 2.7 V.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX}: $t_r = t_f \leq 6$ ns.
6. The clear pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns between 0.7 V and 2.7 V, $t_{w(clear)}$ = 20 ns.

FIGURE 9. Switching time test circuit and waveforms for device type 08 - Continued.

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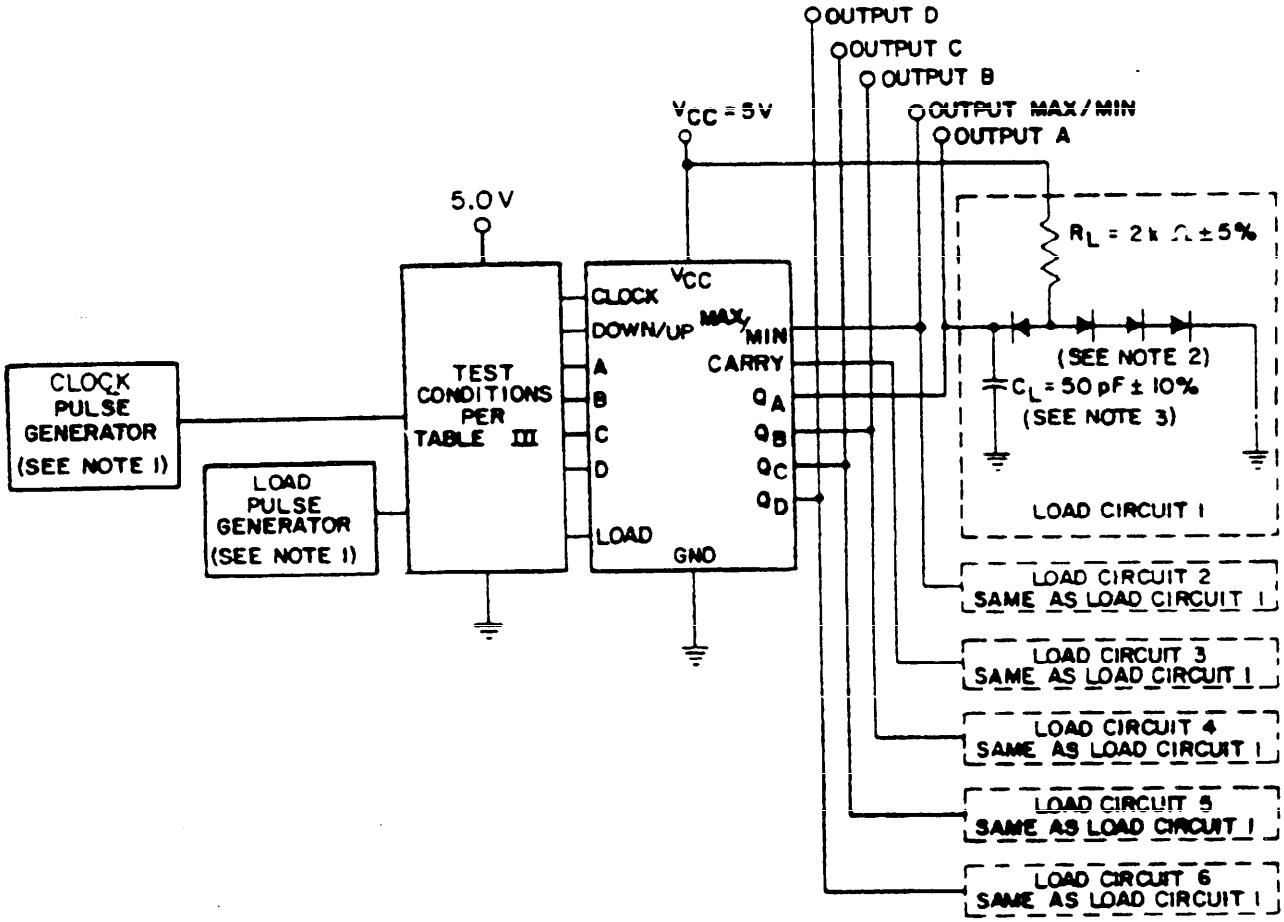
PARALLEL LOADED VOLTAGE WAVEFORMS



SERIAL LOADED VOLTAGE WAVEFORMS
(COUNT UP MODE)

FIGURE 10. Switching time test circuit and waveforms for device type 09.

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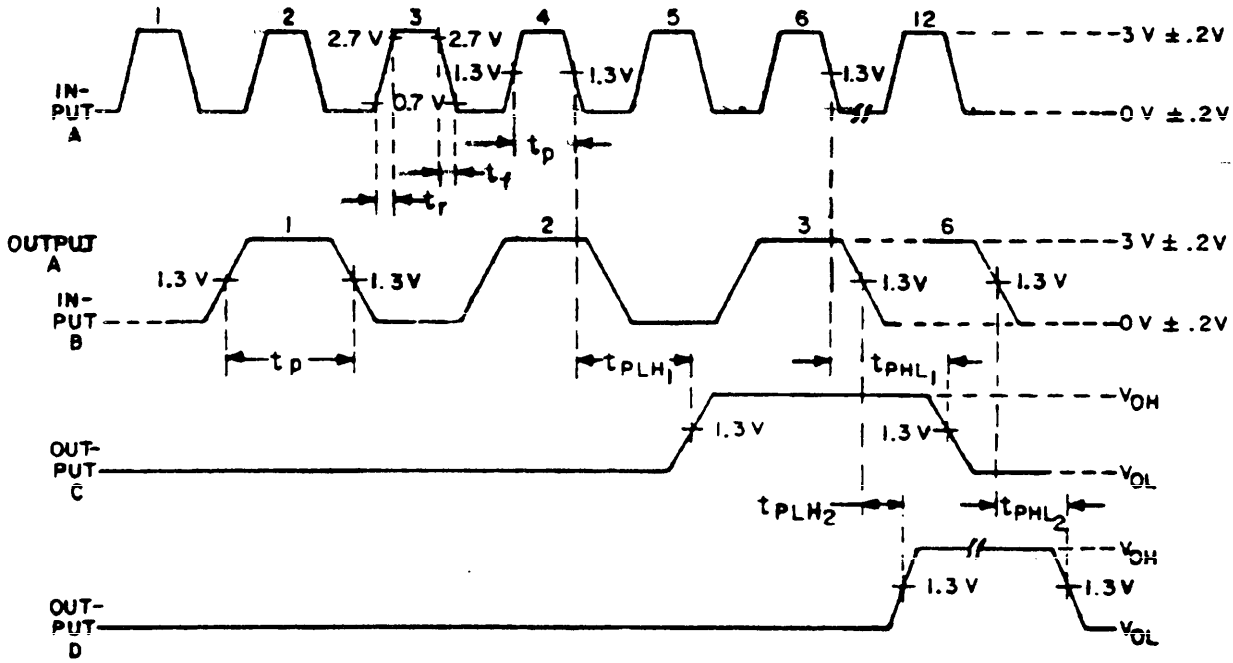


TEST CIRCUIT

NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, t_p = .5 µs, PRR ≤ 1 MHz, $Z_{out} \cong 50$, $t_r \leq 15$ ns, $t_f \leq 6$ ns, between 0.7 V and 2.7 V.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX}: $t_r = t_f \leq 6$ ns.

FIGURE 10. Switching time test circuit and waveforms for device type 09 - Continued.

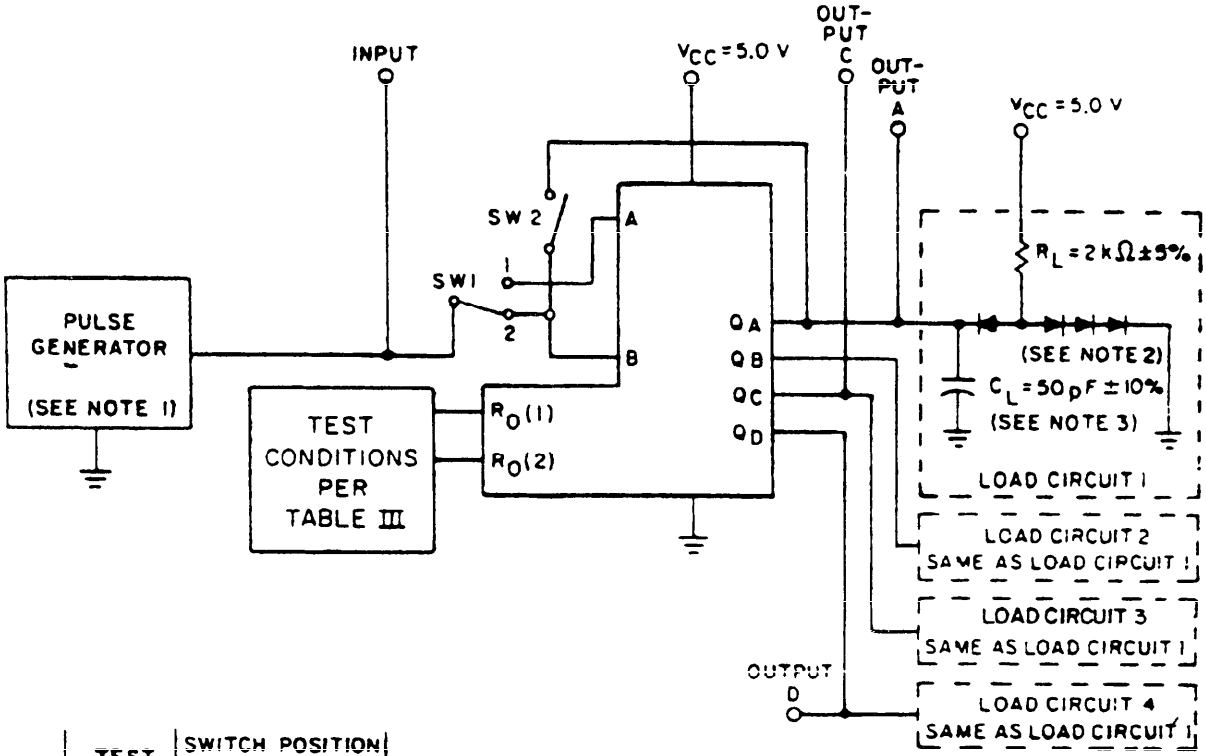
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VOLTAGE WAVEFORMS

FIGURE 11. Switching time test circuit and waveforms for device type 10.

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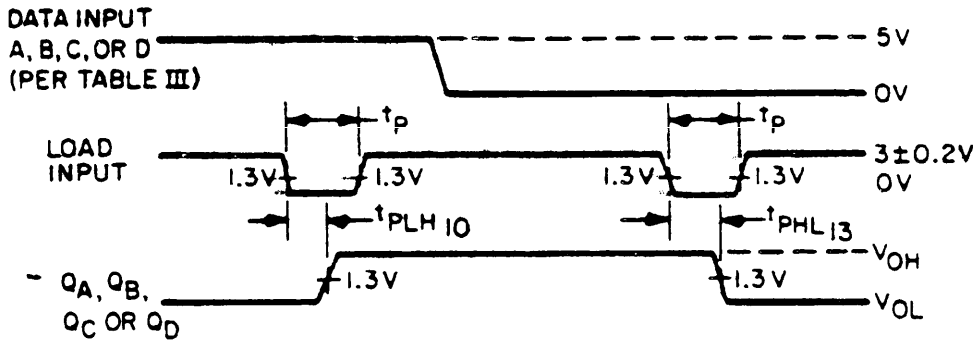
TEST	SWITCH POSITION	
	SW1	SW2
F MAX	1	CLOSED
A TO Q_C	1	CLOSED
B TO Q_D	2	OPEN

TEST CIRCUIT

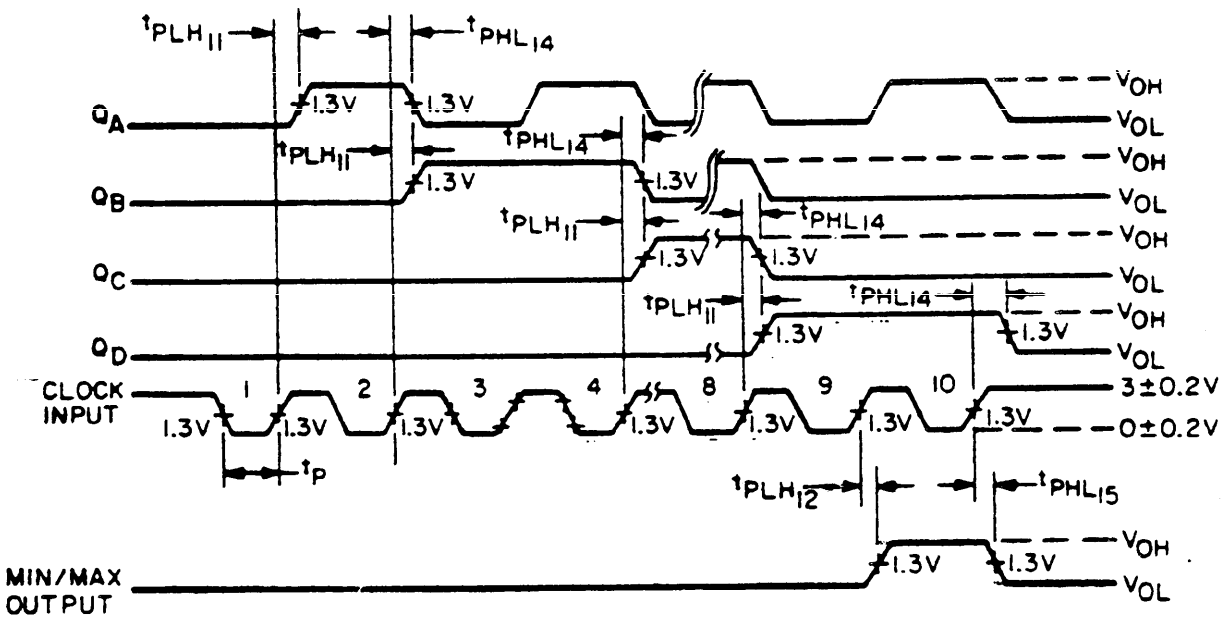
NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, $t_r \leq 15$ ns, $t_f \leq 6$ ns, t_p = .5 µs, PRR ≤ 1 MHz, $Z_{out} \approx 50\Omega$.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX}: $t_r = t_f \leq 6$ ns.

FIGURE 11. Switching time test circuit and waveforms for device type 10 - Continued.

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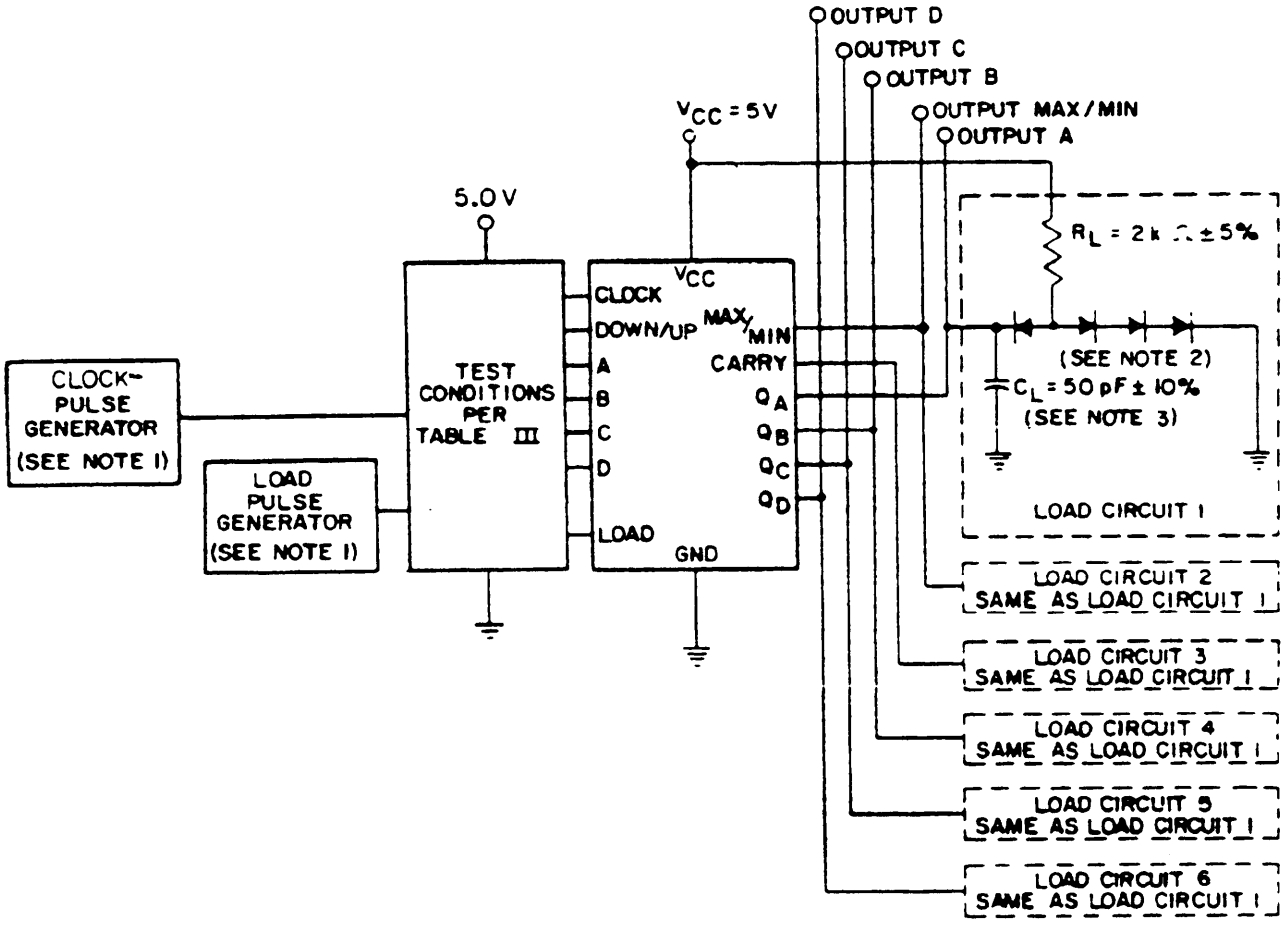
PARALLEL LOADED VOLTAGE WAVEFORMS



SERIAL LOADED VOLTAGE WAVEFORMS
(COUNT UP MODE)

FIGURE 12. Switching time test circuit and waveforms for device type 13.

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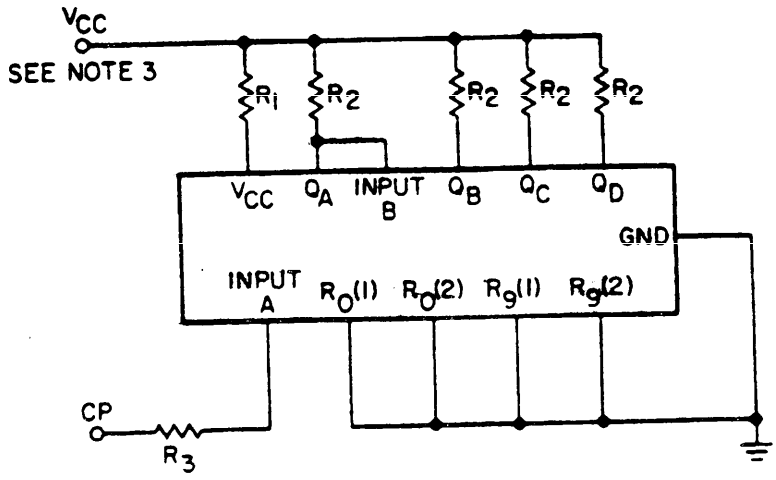
TEST CIRCUIT

NOTES:
1. The pulse generator has the following characteristics: V_{gen} = 3 V, t_p = .5 µs, PRR ≤ 1 MHz, $Z_{out} \approx 50$, $t_r \leq 15$ ns, $t_f \leq 6$ ns, between 0.7 V and 2.7 V.
2. All diodes are 1N3064 or equivalent.
3. C_L includes probe and jig capacitance.
4. Voltage values are with respect to ground terminal.
5. F_{MAX}: $t_r = t_f \leq 6$ ns.

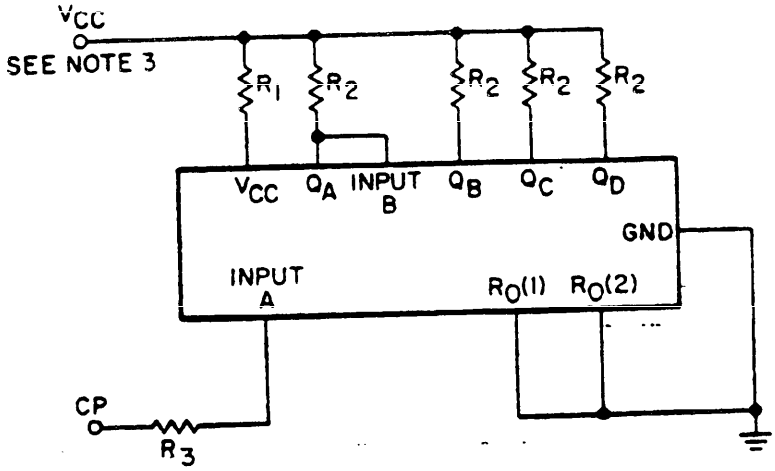
FIGURE 12. Switching time test circuit and waveforms for device type 13 - Continued.

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Device type 01



Device types 02 and 10

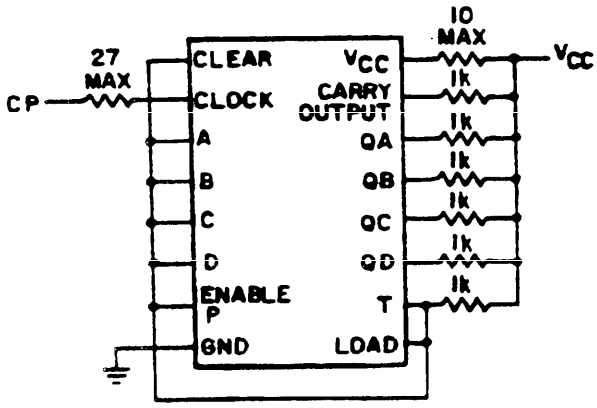


NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; -0.5 V ≤ V_{IL} ≤ 0.7 V; 2.0 V ≤ V_{IH} ≤ 5.5 V.
2. R1 = 10Ω maximum, R2 = 1.0 kΩ ±5%, R3 = 27Ω maximum.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 13. Burn-in and life test circuit.

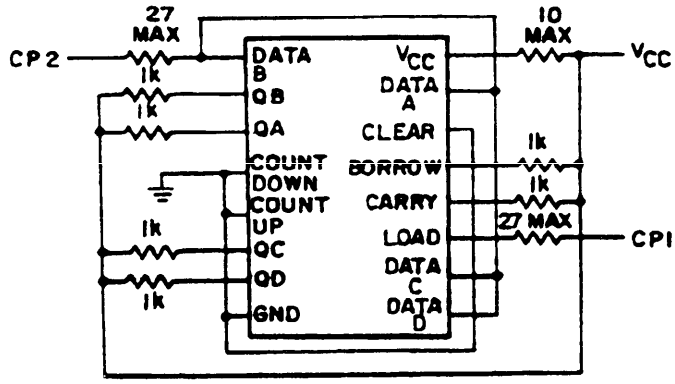
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:08Z
Check the source to verify that this is the current version before use.

Device types 03, 04, 11 and 12



NOTES:
1. CP; 100 kHz ±50% square wave; duty cycle = 50% ±15%; 2.0 V $\leq V_{IH} \leq$ 5.5 V, -0.5 V $\leq V_{IL} \leq$ 0.7 V.
2. Resistances in ohms, ±5%.
3. Voltage at $V_{CC} \geq$ 5.0 V.

Device types 07 and 08

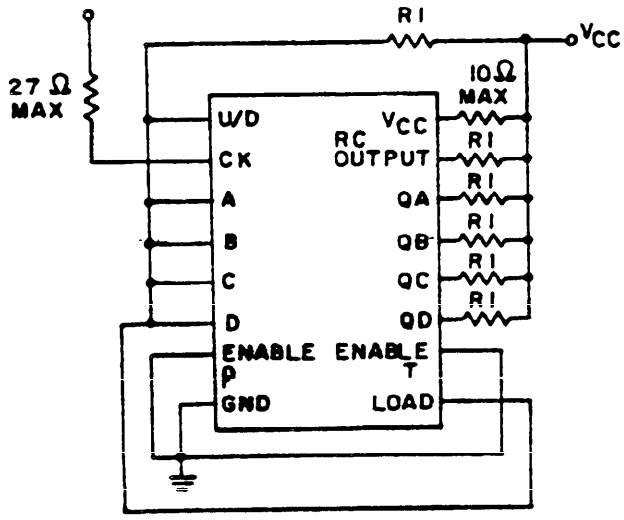


NOTES:
1. CP1 = 100 kHz ±50% square wave; CP2 = 50 kHz ±50% square wave; duty cycle = 50% ±15%; 2.0 V $\leq V_{IH} \leq$ 5.5 V, -0.5 V $\leq V_{IL} \leq$ 0.7 V.
2. Resistance in ohms, ±5%.
3. Voltage at $V_{CC} \geq$ 5.0 V.

FIGURE 13. Burn-in and life test circuit - Continued.

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Device types 05 and 06

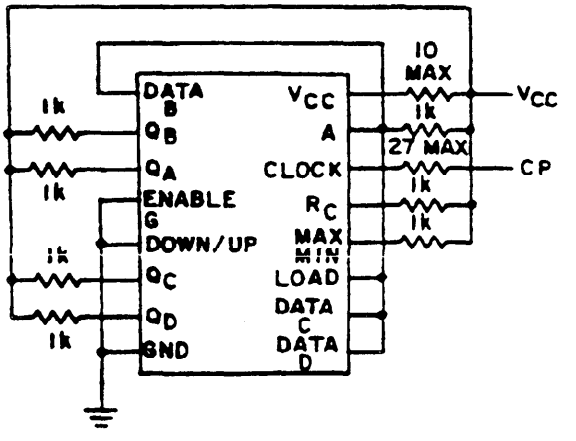


NOTES:
1. R1 = 1,000Ω. All resistances ±5%.
2. CP = 100 kHz ±50% square wave; duty cycle = 50% ±15%; 2.0 V ≤ V_{IH} ≤ 5.5 V, -0.5 V ≤ V_{IL} ≤ 0.7 V.
3. Voltage at V_{CC} ≥ 5.0 V.

FIGURE 13. Burn-in and life test circuit - Continued.

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Device types 09 and 13



NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50% ±15%; 2.0 V $\leq V_{IH} \leq$ 5.5 V, -0.5 V $\leq V_{IL} \leq$ 0.7 V.
2. Resistances in ohms, ±5%.
3. Voltage at $V_{CC} \geq$ 5.0 V.

FIGURE 13. Burn-in and life test circuit - Continued.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 B	2 3 Ro(1)	3 4 Ro(2)	4 15 NC	5 8 V_{CC}	6 9 R9(1)	7 10 R9(2)	8 12 Q_C	9 13 Q_B	10 14 GND	11 16 Q_D	12 18 Q_A	13 19 NC	14 20 A	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1	2.0 V	2.0 V	2.0 V		4.5 V	GND	GND	4 mA	4 mA	GND	4 mA	2/		GND	Q_C		0.4	V
			2	"	"	"		"	GND	GND			"				"	Q_B		"	"
			3	"	"	"		"	2.0 V	0.7 V			"				"	Q_D		"	"
			4	GND	"	"		"	0.7 V	2.0 V			"				2.0 V	Q_A		"	"
			5	2.0 V	0.7 V	0.7 V		"	2.0 V	"	4 mA	4 mA	"				GND	Q_C		"	"
			6	2.0 V	"	2.0 V		"	"	"			"				"	Q_B		"	"
	V_{OH}	3006	7	2.0 V	2.0 V	0.7 V		"	"	"			"	-0.4 mA			"	Q_D	2.5		"
		"	8	GND	0.7 V	2.0 V		"	"	"			"	-0.4 mA	-0.4 mA		2.0 V	Q_A	"		"
		"	9	2.0 V	"	0.7 V		"	"	"			"		-0.4 mA		GND	Q_B	"		"
		"	10	GND	"	0.7 V		"	"	"			"				2.0 V	Q_A	"		"
		"	11	3/ 4/	3/	3/		"	0.7 V	0.7 V	-0.4 mA	-0.4 mA	"				GND	Q_C	"		"
		"	12	3/ 5/	3/	3/		"	0.7 V	0.7 V			"				GND	Q_B	"		"
	I_{IL1}	3009	13		0.4 V	5.5 V		5.5 V					"					Ro(1)	6/	6/	mA
		"	14		5.5 V	0.4 V		"	0.4 V	5.5 V			"					Ro(2)	"	"	"
		"	15					"	5.5 V	0.4 V			"					R9(1)	"	"	"
		"	16					"					"					R9(2)	"	"	"
	I_{IL2}		17		GND	GND		"	3/	3/			"				0.4 V	A	"	"	"
	I_{IL3}		18	0.4 V	GND	GND		"	3/	3/			"					B	"	"	"
	V_{IC}		19					4.5 V	-18 mA	-18 mA			"				-18 mA	R9(1)		-1.5	V
			20					"					"					R9(2)		"	"
			21					"					"					A		"	"
			22	-18 mA				"					"					B		"	"
			23		-18 mA			"					"					Ro(1)		"	"
			24			-18 mA		"					"					Ro(2)		"	"
	I_{IH1}	3010	25					5.5 V	2.7 V	2.7 V			"					R9(1)		20	uA
		"	26					"					"					R9(2)		"	"
		"	27		2.7 V			"					"					Ro(1)		"	"
		"	28			2.7 V		"					"					Ro(2)		"	"
	I_{IH2}	"	29					"	5.5 V	5.5 V			"					R9(1)		100	"
		"	30					"					"					R9(2)		"	"
		"	31		5.5 V			"					"					Ro(1)		"	"
		"	32			5.5 V		"					"					Ro(2)		"	"
	I_{IH3}	"	33					"					"				2.7 V	A		80	"
	I_{IH4}	"	34					"					"				5.5 V	A		400	"
	I_{IH5}	"	35	2.7 V				"					"					B		160	"
	I_{IH6}	"	36	5.5 V				"					"					B		800	"

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	B	Ro(1)	Ro(2)	NC	V_{CC}	R9(1)	R9(2)	Q_C	Q_B	GND	Q_D	Q_A	NC	A				
1 T_C = +25°C	I_{OS}	3011	37		GND	GND		5.5 V	5.5 V	5.5 V			GND	GND	GND		GND	Q_D Q_A Q_B Q_C	-15	-100	mA
			38		GND	GND			5.5 V	GND											
			39	3/ 5/	3/	3/			GND	GND	GND	GND									
			40	3/ 4/																	
	I_{CC}		41	GND														V_{CC}		15	
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Functional tests 7/	3014	42	A 8/	A 8/	A 8/		4.5 V	A	A	L	L	GND	H	H		B 8/	See 9/			
			43	B					A	A				H	H						
			44	B					B	B	L			L	L						
			45	A	B	B			A	A	H			H	H						
			46	B	B	B			A	A	H			H	H						
			47	B	A	A			B	B	L			L	L						
			48	A	A					B											
			49	B	A					A											
			50	A	A																
			51	B	A							H									
			52	A	B							L									
			53	B	B							H									
			54	A								L									
			55	B							H										
			56	A								L									
			57	B								H									
			58	A								L									
			59	B							L			H							
			60	A								H									
			61	B								L									
			62	B		B				B											
			63	A					A			H									
			64	B	A							L									
			65	A								H									
			66	B							H										
			67	A								L									
			68	B								H									
			69	A								H									
			70	B							L			H							
			71	A								L									
			72	B								H									
			73	B	B				B												
			74	A	B							L									
			75	B	A					A		H									
			76	A								L									
			77	B							H										
			78	A								L									
			79	B								H									
			80	B		A				B	L										
			81	B	B				A			H									
			82	A								L									
			83	B								H									
			84	A								L									
			85	B							H										

See footnotes at end of device type 01.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 B	2 3 Ro(1)	3 4 Ro(2)	4 6 NC	5 8 V_{CC}	6 9 R9(1)	7 10 R9(2)	8 12 Q_C	9 13 Q_B	10 14 GND	11 16 Q_D	12 18 Q_A	13 19 NC	14 20 A	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 7/	3014	86	A 8/	B 8/	A		4.5 V	A	B	H	L	GND	L	L		B	See 9/			
			87	B		A				B	H	H		L	L						
			88	A		B				A	L	L		H	H						
			89			B				B				H	L						
			90		A	A			B					L							
			91	B	A	A															
			92	A	B	B						H									
			93	B								H									
			94	A																	
			95	B							H	L									
			96	A								L									
			97	B								H									
			98	A								H									
			99	B	A	A						L									
			100	B	B	B															
			101	A								H									
			102	B								H									
			103	A								L									
			104	B							H										
			105	A								L									
			106	B								H									
			107	A								H									
			108	A					A	A	L			H	H						
			109	B					A	B											
			110	B					B												
			111	A										L	L						
			112	A	A	A															
			113	B	A	A											A				
			114		B	B									H		B				
			115												H		A				
			116														B				
			117												L		A				
			118			A											B				
			119		A	A											A				
			120		A	B									H		B				
			121		B					A							A				
			122														B				
			123						A					H			A				
			124		A					B							B				
			125		A	A				A											
8	Same tests, terminal conditions, and limits as subgroup 7, except T_C = +125°C and -55°C.																				
9 T_C = +25°C	F_{MAX}	3003 (Fig. 4)	126		GND			5.0 V	GND				GND		OUT		IN 10/	A to Q_A	29		MHz
	t_{PLH1}		127		11/	A 8/					OUT						IN	A to Q_C	3	53	ns
	t_{PHL1}		128		GND						OUT						IN	A to Q_C		58	
	t_{PLH2}		129	IN	11/	A								OUT				B to Q_D		37	
	t_{PHL2}		130	IN	GND									OUT				B to Q_D		40	
10 T_C = +125°C	F_{MAX}		131		GND										OUT		IN 10/	A to Q_A	29		MHz
	t_{PLH1}		132		11/	A					OUT						IN	A to Q_C	3	74	ns
	t_{PHL1}		133		GND						OUT						IN	A to Q_C		81	
	t_{PLH2}		134	IN	11/	A								OUT				B to Q_D		52	
	t_{PHL2}		135	IN	GND									OUT				B to Q_D		56	
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

See footnotes at end of device type 01.

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1/ Case 2 pins not referenced are N/C.

2/ Test 4, Pin 12; 4 mA + $I_{IL3(MAX)}$.

3/ Apply 4.5 V pulse then ground prior to taking measurements to set device in the desired state.

4/ Apply two pulses after Ro (reset) pulse.

5/ Apply one pulse after Ro (reset) pulse.

6/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Terminals	Circuits						
		A	B	C	D	E	F	G
I_{IL1}	Ro(1) Ro(2) R9(1) R9(2)	-.12/-.36 " " "	-.03/-.40 " " "		-.03/-.40 " " "	-.12/-.36 " " "	-.12/-.36 " " "	
I_{IL2}	A	-0.5/-2.0	-1.0/-2.4		-1.0/-2.4	-1.0/-2.4	-0.5/-2.0	
I_{IL3}	B	-0.4/-1.6	-1.3/-3.2		-1.3/-3.2	-1.3/-3.2	-1.0/-2.4	

7/ Only a summary of attributes data is required.

8/ A = 3.0 V minimum, B = 0.0 V or GND.

9/ H > 1.5 V; L < 1.5 V.

10/ F_{MAX} minimum limit specified is the frequency of the input pulse. The output pulse shall be one-half of the input frequency.

11/ Momentary 3.0 V (min), then ground. Maintain ground for measurement.

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 B	2 3 Ro(1)	3 4 Ro(2)	4 6 NC	5 8 V_{CC}	6 9 NC	7 10 NC	8 12 Q_C	9 13 Q_B	10 14 GND	11 16 Q_D	12 18 Q_A	13 19 NC	14 20 A	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1	GND	2.0 V	2.0 V		4.5 V				4 mA	GND		2/		2.0 V	Q_A		0.4	V
		"	2	2.0 V	"	"		"			4 mA		"				GND	Q_B		"	"
		"	3	"	"	"		"					"				"	Q_C		"	"
		"	4	"	"	"		"					"	4 mA			"	Q_D		"	"
	V_{OH}	3006	5	GND		3/		"					"		-0.4 mA		3/ 4/	Q_A	2.5		"
		"	6	3/ 4/	3/	"		"				-0.4 mA	"				GND	Q_B	"		"
		"	7	3/ 5/	"	"		"			-0.4 mA		"				"	Q_C	"		"
		"	8	3/ 6/	"	"		"					"	-0.4 mA			"	Q_D	"		"
	V_{IC}		9	-18 mA				"					"				-18 mA	A		-1.5	"
			10		-18 mA			"					"					B		"	"
			11			-18 mA		"					"					Ro(1)		"	"
			12					"					"					Ro(2)		"	"
	I_{IL1}	3009	13		0.4 V	5.5 V		5.5 V					"					Ro(1)	7/	7/	mA
		"	14		5.5 V	0.4 V		"					"					Ro(2)	"	"	"
	I_{IL2}	"	15		3/	3/		"					"				0.4 V	A	"	"	"
	I_{IL3}	"	16	0.4 V	3/	3/		"					"					B	"	"	"
	I_{IH1}	3010	17		2.7 V	GND		"					"					Ro(1)		20	μA
	I_{IH1}	"	18		GND	2.7 V		"					"					Ro(2)		20	"
	I_{IH2}	"	19		5.5 V	GND		"					"					Ro(1)		100	"
	I_{IH2}	"	20		GND	5.5 V		"					"					Ro(2)		100	"
	I_{IH3}	"	21		5.5 V	"		"					"				2.7 V	A		80	"
	I_{IH4}	"	22		"	"		"					"				5.5 V	A		400	"
	I_{IH5}	"	23	2.7 V	"	"		"					"					B		80	"
	I_{IH6}	"	24	5.5 V	"	"		"					"					B		400	"
	I_{OS}	3011	25	GND	3/	3/		"					"		GND		3/ 4/	Q_A	-15	-100	mA
		"	26	3/ 4/	"	"		"				GND	"				GND	Q_B	"	"	"
		"	27	3/ 5/	"	"		"			GND		"				"	Q_C	"	"	"
		"	28	3/ 6/	"	"		"					"	GND			"	Q_D	"	"	"
	I_{CC}	3005	29	GND				"					"					V_{CC}		15	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 B	2 3 Ro(1)	3 4 Ro(2)	4 6 NC	5 8 V_{CC}	6 9 NC	7 10 NC	8 12 Q_C	9 13 Q_B	10 14 GND	11 16 / Q_D	12 18 Q_A	13 19 NC	14 20 A	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 8/	3014	30	B 9/	A 9/	A		4.5 V			L	L	GND	L	L		B 9/	See 10/			
			31	A																	
			32	B																	
			33	B		B															
			34	A																	
			35	B								H									
			36			A						L									
			37		B																
			38	A																	
			39	A		B															
			40	B								H									
			41	A								H									
			42	B							H	L									
			43	A								L									
			44	B								H									
			45	A								H									
			46	B							L	L		H							
			47	A								L									
			48	B								H									
			49	A								H									
			50	B							H	L									
			51	A								L									
			52	B								H									
			53	A								H									
			54	B							L	L		L							
			55	A								L									
			56	B								H									
			57	A								H									
			58	B							H	L									
			59	A								L									
			60	B								H									
			61	A								H									
			62	B							L	L		H							
			63	A								L									
			64	B								H									
			65	B	A																
			66	A																	
			67	B							H	L									
			68	A								L									
			69	B								H									
			70	A								H									
			71	A		A					L	L		L							
			72	B		A															
			73		B	B											A				
			74												H		B				
			75												H		A				
			76												L		B				
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																				
9 T_C = +25°C	F_{MAX}	3003 (Fig. 5)	77		GND			5.0 V					GND		OUT		IN 11/	A to Q_A	29		MHz
	t_{PLH1}		78		12/	A 9/					OUT						IN	A to Q_C	3	53	ns
	t_{PHL1}		79		GND						OUT						IN	A to Q_C		58	
	t_{PLH2}		80	IN	12/	A								OUT				B to Q_D		56	
	t_{PHL2}		81	IN	GND									OUT				B to Q_D		56	

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 B	2 3 Ro(1)	3 4 Ro(2)	4 6 NC	5 8 V_{CC}	6 9 NC	7 10 NC	8 12 Q_C	9 13 Q_B	10 14 GND	11 16 Q_D	12 18 Q_A	13 19 NC	14 20 A	Measured terminal	Limits Min	Limits Max	Unit
10 T_C = +125°C	F_{MAX}	3003 (Fig. 5)	82		GND			5.0 V					GND		OUT		IN 11/	A to Q_A	29		MHz
	t_{PLH1}	"	83		12/ GND	A		"			OUT		"				IN	A to Q_C	3	74	ns
	t_{PHL1}	"	84					"			OUT		"				IN	A to Q_C	"	81	"
	t_{PLH2}	"	85	IN	12/ GND	A		"					"	OUT				B to Q_D	"	78	"
	t_{PHL2}	"	86	IN				"					"	OUT				B to Q_D	"	78	"
11	Same tests, terminal conditions, and limits as for subgroup 10 except, T_C = -55°C.																				

1/ Case 2 pins not referenced are N/C.

2/ For test 1, 4 mA $^{+}I_{IL3}$ (max).

3/ Apply 4.5 V pulse, then ground prior to taking measurements to set device in the desired state. Maintain ground for measurement.

4/ Input pulse must be applied one time after Ro pulse.

5/ Input pulse must be applied twice after Ro pulse.

6/ Input pulse must be applied four times after Ro pulse.

7/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Terminals	Circuits A	B	C	D	E	F	G
I_{IL1}	Ro(1) Ro(2)	-.12/-.36 "	-.03/-.40 "	-120/-360 "	-.03/-.40 "	-.12/-.36 "	-.12/-.36 "	
I_{IL2}	A	-0.5/-2.0	-1.0/-2.4	-0.5/-2.0	-1.0/-2.4	-1.0/-2.4	-0.5/-2.0	
I_{IL3}	B	-0.4/-1.6	-0.4/-1.6	-0.4/-1.6	-0.4/-1.6	-.65/-1.6	-0.4/-1.6	

8/ Only a summary of attributes data is required.

9/ A = 3.0 V minimum; B = 0.0 V or GND.

10/ H > 1.5 V; L < 1.5 V.

11/ F_{MAX} min limit specified is the frequency of the input pulse. The output frequency shall be one-half the input frequency.

12/ Momentary 3.0 V (min), then ground. Maintain ground for measurement.

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TABLE III. Group A inspection for device types 03, 04, 11, and 12.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 1/ 2 / Test no.	1 / 2 / Clear	2 / 3 / Clock	3 / 4 / A	4 / 5 / B	5 / 7 / C	6 / 8 / D	7 / 9 / EnP	8 / 10 / GND	9 / 12 / Load	10 / 13 / EnT	11 / 14 / Q_D	12 / 15 / Q_C	13 / 17 / Q_B	14 / 18 / Q_A	15 / 19 / Ripple carry	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1 2 3 4 5	4.5 V " " "	2/ " " "	0.7 V	0.7 V	0.7 V	0.7 V		GND " " " "	GND " "	0.7 V	4 mA	4 mA	4 mA	4 mA	4 mA	4.5 V " " " "	Q_D Q_C Q_B Q_A Ripple carry		0.4 " " " "	V " " " "
	V_{OH}	3006	6 7 8 9 10	" " " " "	2/ " " " "	2.0 V 2.0 V	2.0 V 3/	2.0 V 3/	2.0 V 2.0 V		" " " " "	GND " " "	2.0 V	-.4 mA	-.4 mA	-.4 mA	-.4 mA	-.4 mA	" " " " "	Q_D Q_C Q_B Q_A Ripple carry	2.5 " " " "		" " " " "
	V_{IC}		11 12 13 14 15 16 17 18 19	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	" " " " " " " " "	-18 mA	-18 mA						" " " " " " " " "	Clear Clock A B C D EnP Load EnT		-1.5 " " " " " " " "	" " " " " " " " "
	I_{IL4}	3009	20	0.4 V							"								5.5 V	Clear	6/	4/	μA
	I_{IL6}	"	21		0.4 V						"								"	Clock	"	"	"
	I_{IL4}	" " " " "	22 23 24 25 26			0.4 V	0.4 V	0.4 V	0.4 V	0.4 V	" " " " "	GND " " 4.5 V	4.5 V						" " " " "	A B C D EnP	" " " " "	" " " " "	" " " " "
	I_{IL5} I_{IL5}	" "	27 28 5/							4.5 V 4.5 V	" "	0.4 V 4.5 V	4.5 V 0.4 V						" "	Load EnT	" "	" "	" "
	I_{IH13}	3010	29 13/	2.7 V							"								"	Clear		20	"
	I_{IH9}	"	30		2.7 V						"								"	Clock		40	"
	I_{IH11}	" " " " "	31 32 33 34 35			2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	" " " " "	GND	GND						" " " " "	A B C D EnP		20 " " " "	" " " " "
	I_{IH9} I_{IH9}	" "	36 37							GND GND	" "	2.7 V GND	GND 2.7 V						" "	Load EnT		40 40	" "

See footnotes at end of device types 03, 04, 11, and 12.

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TABLE III. Group A inspection for device types 03, 04, 11, and 12 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	Clear	Clock	A	B	C	D	EnP	GND	Load	EnT	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
1 T_C = +25°C	I_{IH14}	3010	38 13/	5.5 V							GND								5.5 V	Clear		100	µA
	I_{IH10}	"	39		5.5 V						"								"	Clock		200	"
	I_{IH12}	"	40			5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	GND	GND						"	A		100	"
		"	41								"								"	B		"	"
		"	42								"								"	C		"	"
		"	43								"								"	D		"	"
		"	44								"								"	EnP		"	"
	I_{IH10}	"	45							GND	"	5.5 V	GND						"	Load		200	"
	I_{IH10}	"	46							GND	"	GND	5.5 V						"	EnT		200	"
	I_{OS}	3011	47	4.5 V	2/			4.5 V	4.5 V		"	GND		GND	GND	GND	GND		"	Q_D	-15	-100	mA
		"	48	"	"						"	"							"	Q_C	"	"	"
		"	49	"	"		4.5 V				"	"							"	Q_B	"	"	"
		"	50	"	"	4.5 V		6/	4.5 V		"	"							"	Q_A	"	"	"
		"	51	"	"	4.5 V	6/		4.5 V		"	"	4.5 V					GND	"	Ripple carry	"	"	"
	I_{CCH}	3005	52	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	5.5 V	5.5 V						"	V_{CC}		31	"
	I_{CCH}	"	53	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	GND	5.5 V						"			31	"
	I_{CCL}	"	54	GND	5.5 V	GND	GND	GND	GND	GND	"	GND	GND						"	"		32	"
	I_{CCL}	"	55	GND	GND	GND	GND	GND	GND	GND	"	GND	GND						"	"		32	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device types 03, 04, 11, and 12.

MIL-M-38510/315C

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	Clear	Clock	A	B	C	D	EnP	GND	Load	EnT	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
T_C = +25°C 7	Functional tests 7/	3014	56	B 8/	A 8/	A 8/	A 8/	A	A	A	GND	A	A	L	L	L	L	L	4.5 V	See 9/			
			57	A	B	B	B	B	B	=	=	=	B	=	=	=	=	=	=				
			58	=	A	A	A	A	A	=	=	=	A	=	=	=	H	=	=				
			59	=	A	B	B	A	A	=	=	=	A	=	=	=	=	=	=				
			60	=	B	A	A	A	B	=	=	=	B	=	=	=	=	=	=				
			61	=	B	A	A	A	A	=	=	=	A	=	=	=	L	=	=				
			62	=	A	A	A	A	A	=	=	=	A	=	=	=	=	=	=				
			63	=	A	B	B	A	B	=	=	=	B	=	=	=	=	=	=				
			64	=	B	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			65	=	A	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			66	=	A	B	B	A	B	=	=	=	B	=	=	=	=	=	=				
			67	=	B	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			68	=	A	A	B	A	A	=	=	=	A	=	H	=	L	=	=				
			69	=	A	B	B	A	B	=	=	=	B	=	=	=	=	=	=				
			70	=	B	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			71	=	A	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			72	=	A	B	B	A	B	=	=	=	B	=	=	=	=	=	=				
			73	=	B	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			74	=	A	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			75	=	A	B	B	A	B	=	=	=	B	=	=	=	=	=	=				
			76	=	B	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			77	=	A	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			78	=	A	B	B	A	B	=	=	=	B	=	=	=	=	=	=				
			79	=	B	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			80	=	A	A	B	A	A	=	=	=	A	=	H	=	=	=	=				
			81	=	A	B	B	A	B	=	=	=	B	=	=	=	=	=	=				
			82	=	B	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			83	=	A	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			84	=	A	B	B	A	B	=	=	=	B	=	=	=	=	=	=				
			85	=	B	A	B	A	A	=	=	=	A	=	=	=	=	=	=				
			86	=	A	A	B	A	A	=	=	=	A	=	L	=	=	=	=				
			87	B	A	B	A	B	B	=	=	=	B	=	=	=	=	=	=				
			88	A	B	B	A	B	B	=	=	=	B	=	=	=	=	=	=				
			89	=	A	B	A	B	B	=	=	=	B	=	=	=	=	=	=				
			90	=	B	B	A	B	B	=	=	=	B	=	=	=	=	=	=				
			91	=	A	B	A	B	B	=	=	=	B	=	=	=	=	=	=				
			92	=	A	A	A	B	B	=	=	=	B	=	=	=	=	=	=				
			93	=	B	A	A	B	B	=	=	=	B	=	=	=	=	=	=				
			94	=	A	B	A	B	B	=	=	=	B	=	=	=	=	=	=				
			95	=	A	A	A	B	B	=	=	=	B	=	=	=	=	=	=				
			96	=	B	A	A	B	B	=	=	=	B	=	=	=	=	=	=				
			97	=	A	B	A	B	B	=	=	=	B	=	=	=	=	=	=				
			98	=	A	A	A	B	B	=	=	=	B	=	=	=	=	=	=				
			99	=	B	A	A	B	B	=	=	=	B	=	=	=	=	=	=				
			100	=	A	B	A	B	B	=	=	=	B	=	H	=	=	=	=				
			101	=	A	A	A	B	B	=	=	=	A	=	=	=	=	=	=				
			102	=	B	A	A	B	B	=	=	=	B	=	=	=	=	=	=				
			103	=	A	B	A	B	B	=	=	=	B	=	=	=	=	=	=				
			104	=	A	A	A	B	A	=	=	=	A	=	=	=	=	=	=				
			105	=	B	A	A	B	=	=	=	=	B	=	=	=	=	=	=				
			106	=	A	B	A	B	=	=	=	=	A	=	L	=	=	=	=				
			107	=	A	A	A	B	=	=	=	=	B	=	=	=	=	=	=				
			108	=	B	A	A	B	=	=	=	=	A	=	=	=	=	=	=				
			109	=	A	B	A	B	=	=	=	=	B	=	=	=	=	=	=				
			110	B	A	A	A	A	=	=	=	=	A	=	L	=	=	=	=				
			111	A	A	B	A	=	=	=	=	=	=	=	=	=	=	=	=				
			112	A	A	A	B	=	=	=	=	=	=	=	=	=	=	=	=				
			113	A	B	A	A	=	=	=	=	=	=	H	H	H	H	X	=				
			114	B	A	A	A	=	=	=	=	=	A	L	L	L	L	L	=				

See footnotes at end of device types 03, 04, 11, and 12.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	Clear	Clock	A	B	C	D	EnP	GND	Load	EnT	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
7 T_C = +25°C	Functional tests 7/	3014	115	A 8/	A 8/	A 8/	B 8/	B	B	B	GND	B	B	L	L	L	L	L	4.5 V	See 9/			
			116	"	B	"	"	"	"	"	"	"	"	"	"	"	L	"	"	"				
			117	"	A	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"				
			118	"	A	B	"	"	A	A	"	"	A	"	"	"	"	"	"	"				
			119	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			120	"	A	"	"	"	"	"	"	"	"	H	"	"	L	"	"	"				
			121	"	A	"	A	"	B	B	"	"	"	"	"	"	"	"	"	"				
			122	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			123	"	A	"	"	"	"	"	"	"	"	L	"	"	H	"	"	"				
			124	"	A	"	B	A	A	"	"	"	"	"	"	"	"	"	"	"				
			125	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			126	"	A	"	"	"	"	"	"	"	"	H	H	L	"	"	"	"				
			127	"	A	"	A	"	"	"	"	"	B	"	"	"	"	"	"				
			128	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			129	"	A	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"				
			130	"	A	A	B	B	"	A	"	"	A	"	"	"	"	"	"	"				
			131	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			132	"	A	"	"	"	"	"	"	"	"	"	L	L	"	H	H	"				
			133	"	A	"	A	A	B	B	"	"	B	"	"	"	"	"	L	"				
			134	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			135	"	A	"	"	"	"	"	"	"	"	L	H	H	"	"	"	"				
			136	"	A	B	"	B	A	A	"	"	"	"	"	"	"	"	"	"				
			137	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			138	"	A	"	"	"	"	"	"	"	"	H	L	"	"	L	"	"				
			139	"	A	A	"	"	B	"	"	"	A	"	"	"	"	"	"	"				
			140	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			141	"	A	"	"	"	"	"	"	"	"	L	"	"	"	H	"	"				
			142	"	A	B	B	A	"	"	"	"	"	"	"	"	"	"	"	"				
			143	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			144	"	A	"	"	"	"	"	"	"	"	"	H	L	"	L	"	"				
			145	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			146	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			147	"	A	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"				
			148	"	A	B	A	"	"	"	"	"	"	"	"	"	"	"	"	"				
			149	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			150	"	A	"	"	"	"	"	"	"	"	"	"	H	"	L	"	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						

See footnotes at end of device types 03, 04, 11, and 12.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 Clear	2 3 Clock	3 4 A	4 5 B	5 7 C	6 8 D	7 9 EnP	8 10 GND	9 12 Load	10 13 EnT	11 14 Q_D	12 15 Q_C	13 17 Q_B	14 18 Q_A	15 19 Ripple carry	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 7/	3014	56	B 8/	A 8/	A 8/	A	A	A	A 8/	GND	A	A	L	L	L	L	L	4.5 V	See 9/			
			57	A	A	B	B	B	B				B										
			58		B	A	A	A	A				A				H						
			59		A	B	A	A	A				A										
			60		B	B	B	B	B				B										
			61		B	A	A	A	A				A										
			62		A	B	A	A	A				A			H	L						
			63		B	B	B	B	B				B										
			64		B	A	A	A	A				A										
			65		A	B	A	A	A				A				H						
			66		B	B	B	B	B				B										
			67		B	A	A	A	A				A										
			68		A	B	A	A	A				A		H	L	L						
			69		B	B	B	B	B				B										
			70		B	A	A	A	A				A										
			71		A	B	A	A	A				A				H						
			72		B	B	B	B	B				B										
			73		B	A	A	A	A				A										
			74		A	B	A	A	A				A			H	L						
			75		B	B	B	B	B				B										
			76		B	A	A	A	A				A										
			77		A	B	A	A	A				A				H						
			78		B	B	B	B	B				B										
			79		B	A	A	A	A				A										
			80		A	B	A	A	A				A	H	L	L	L						
			81		B	A	B	B	B				A										
			82		B	B	A	A	A				A				H						
			83		A	B	A	A	A				A										
			84		B	A	B	B	B				B										
			85		B	B	A	A	A				A			H	L						
			86		A	B	A	A	A				A										
			87		B	A	B	B	B				B										
			88		B	B	A	A	A				A				H						
			89		A	B	A	A	A				A										
			90		A	B	B	B	B				B										
			91		B	B	A	A	A				A				L						
			92		A	A	A	A	A				A		H	L							
			93		A	B	A	A	A				B										
			94		B	B	A	A	A				A				H						
			95		A	A	A	A	A				A										
			96		A	B	A	A	A				B										
			97		B	A	A	A	A				A				L						
			98		A	A	A	A	A				A			H							
			99		A	B	A	A	A				B										
			100		B	A	A	A	A				A				H						
			101		A	A	A	A	A				A					H					
			102		A	B	A	A	A				B				L	L					
			103		B	A	A	A	A				A					H					
			104		A	A	A	A	A				A	L	L	L	L	L					
			105	B	A	A	A	A	A	B			A										
			106	A	A	A	A	A	A	B			A										
			107		B	-	-	-	-	B			B										
			108		A	-	-	-	-	A			-										
			109		B	-	B	B	B	-			-										
			110		A	-	-	-	-	-			-										
			111		B	-	-	-	-	-		B	-										
			112		A	-	-	-	-	-		-	-				H						
			113		A	-	A	A	-	-		-	-										
			114		B	-	A	A	-	-		-	-										

See footnotes at end of device types 03, 04, 11, and 12.

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TABLE III. Group A inspection for device type 04 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 Clear	2 3 Clock	3 4 A	4 5 B	5 7 C	6 8 D	7 9 EnP	8 10 GND	9 12 Load	10 13 EnT	11 14 Q_D	12 15 Q_C	13 17 Q_B	14 18 Q_A	15 19 Ripple carry	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 1/	3014	115	A 8/	A 8/	A	A	A	B	A	GND	B	B	L	H	H	H	L	4.5 V	See 9/			
		"	116	"	A	"	"	"	"	"	"	A	"	"	"	"	"	"	"				
		"	117	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	118	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	119	"	B	"	B	B	A	"	"	B	"	"	"	"	"	"	"				
		"	120	"	A	"	B	B	"	"	"	"	"	H	L	L	"	"	"				
		"	121	B	A	"	A	A	"	"	"	"	A	L	"	"	L	"	"				
		"	122	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	123	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	124	"	A	"	"	"	"	"	"	"	"	H	H	H	H	H	"				
		"	125	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	126	B	A	"	"	"	"	"	"	A	"	L	L	L	L	L	"				
		"	127	A	A	B	"	"	"	"	"	B	"	"	"	"	"	"	"				
		"	128	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	129	"	A	"	"	"	"	"	"	"	"	H	H	H	"	"	"				
		"	130	"	A	"	"	B	B	"	"	"	"	"	"	"	"	"	"				
		"	131	"	B	"	"	"	"	"	"	"	"	L	L	"	"	"	"				
		"	132	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	133	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	134	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	135	"	A	"	"	"	"	"	"	"	"	"	"	"	H	"	"				
		"	136	"	A	B	B	A	"	"	"	"	"	"	"	"	"	"	"				
		"	137	"	B	"	"	"	"	"	"	"	"	"	"	"	L	"	"				
		"	138	"	A	"	"	"	"	"	"	"	"	"	H	"	"	"	"				
		"	139	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	140	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	141	"	A	"	"	"	"	"	"	"	"	"	"	"	H	"	"				
		"	142	"	A	B	A	"	"	"	"	"	"	"	"	"	"	"	"				
		"	143	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	144	"	A	"	"	"	"	"	"	"	"	"	"	"	L	"	"				
		"	145	"	A	B	B	B	A	"	"	"	"	"	"	"	"	"	"				
		"	146	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	147	"	A	"	"	"	"	"	"	"	"	H	L	L	"	"	"				
		"	148	"	A	A	A	"	"	"	"	"	"	"	"	"	"	"	"				
		"	149	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	150	"	A	"	"	"	"	"	"	"	"	"	"	"	H	"	"				
		"	151	"	A	B	B	A	"	"	"	"	"	"	"	"	"	"	"				
		"	152	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	153	"	A	"	"	"	"	"	"	"	"	"	H	"	L	"	"				
		"	154	"	A	A	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	155	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
		"	156	"	A	"	"	"	"	"	"	"	"	"	"	"	H	"	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						

See footnotes at end of device types 03, 04, 11, and 12.

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	Clear	Clock	A	B	C	D	EnP	GND	Load	EnT	Q_D	Q_C	ϕ_B	Q_A	Ripple carry	V_{CC}				
7 T_C = +25°C	Functional tests 7/	3014	56	B 8/	B 8/	B 8/	B	B	B 8/	A 8/	GND	B	A	X	X	X	X	X	4.5 V	See 9/			
			57	B	A	"	"	"	"	"	"	"	"	L	L	L	L	L	"				
			58	A	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			59	A	A	"	"	"	A	"	"	A	"	"	"	"	"	"	"				
			60	B	A	A	A	A	A	"	"	"	"	"	"	"	H	"	"				
			61	B	B	A	A	A	A	"	"	"	"	"	"	"	"	"	"				
			62	B	A	A	A	A	A	"	"	"	"	"	"	"	L	"	"				
			63	A	B	A	B	B	B	"	"	"	B	"	"	"	"	"	"				
			64	"	A	A	A	A	A	"	"	"	A	"	"	"	H	"	"				
			65	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			66	"	A	B	B	B	B	"	"	"	B	"	"	"	L	"	"				
			67	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			68	"	A	B	B	B	B	"	"	"	B	"	"	"	H	"	"				
			69	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			70	"	A	B	B	B	B	"	"	"	B	"	"	"	L	"	"				
			71	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			72	"	A	B	B	B	B	"	"	"	B	"	"	"	H	"	"				
			73	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			74	"	A	B	B	B	B	"	"	"	B	"	H	"	L	"	"				
			75	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			76	"	A	B	B	B	B	"	"	"	B	"	"	"	H	"	"				
			77	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			78	"	A	B	B	B	B	"	"	"	B	"	"	"	L	"	"				
			79	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			80	"	A	B	B	B	B	"	"	"	B	"	"	"	H	"	"				
			81	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			82	"	A	B	B	B	B	"	"	"	B	"	"	"	L	"	"				
			83	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			84	"	A	B	B	B	B	"	"	"	B	"	"	"	H	"	"				
			85	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			86	"	A	B	B	B	B	"	"	"	B	"	H	"	L	"	"				
			87	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			88	"	A	B	B	B	B	"	"	"	B	"	"	"	H	"	"				
			89	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			90	"	A	B	B	B	B	"	"	"	B	"	"	"	H	"	H				
			91	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	L				
			92	"	A	B	B	B	B	"	"	"	B	"	"	"	L	"	H				
			93	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	L				
			94	"	A	B	B	B	B	B	"	"	B	"	"	"	H	"	"				
			95	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			96	"	A	B	B	B	B	"	"	"	B	"	"	"	H	"	"				
			97	"	B	A	A	A	A	A	"	"	B	"	"	"	"	"	"				
			98	"	A	B	B	B	B	"	"	"	A	"	"	"	H	"	"				
			99	"	B	A	A	A	A	"	"	B	"	"	"	"	"	"	"				
			100	"	A	B	B	B	B	"	"	"	A	"	"	"	H	"	"				
			101	"	B	A	A	A	A	"	"	"	B	"	"	"	"	"	"				
			102	"	A	B	B	B	B	"	"	"	A	"	"	"	H	"	"				
			103	"	B	A	A	A	A	"	"	"	B	"	"	"	"	"	"				
			104	"	A	B	B	B	B	"	"	"	A	"	"	"	H	"	"				
			105	"	B	A	A	A	A	"	"	"	B	"	"	"	"	"	"				
			106	A	A	B	B	B	B	A	"	"	B	"	"	"	L	"	"				
			107	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			108	"	A	B	B	B	B	"	"	"	B	"	H	"	L	"	"				
			109	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			110	"	A	B	B	B	B	"	"	"	B	"	"	"	"	"	"				
			111	"	B	A	A	A	A	"	"	"	A	"	"	"	"	"	"				
			112	B	A	B	B	B	B	"	"	"	B	"	"	"	"	"	H				
			113	B	B	A	A	A	A	"	"	"	B	"	"	"	"	"	L				
			114	A	A	"	"	"	"	"	"	"	"	L	"	"	L	L	"				

See footnotes at end of device types 03, 04, 11, and 12.

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TABLE III. Group A inspection for device type 11 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	Clear	Clock	A	B	C	D	EnP	GND	Load	EnT	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
7 T_C = +25°C	Functional tests 7/	3014	115	A 8/	B 8/	A	B	B	A	A	GND	B	A	L	L	L	L	L	4.5 V	See 9/			
			116		A									H			H	H					
			117		B																		
			118		A																		
			119	B	B																		
			120	B	A									L			L	L					
			121	A	A	B			B				B										
			122		B		A																
			123		A		A									H							
			124		A		B		A														
			125		B																		
			126		A									H		L							
			127		A				B				A										
			128		B																		
			129		A									L									
			130		A		A	A	A														
			131		B																		
			132		A									H	N	H							
			133		A	A		B	B														
			134		B																		
			135		A									L	L		H						
			136		A	B	B	A															
			137		B																		
			138		A										H	L	L						
			139		A	A																	
			140		B																		
			141		A												H						
			142		A	B	A																
			143		B																		
			144		A											H	L						
			145		A		B	B	A														
			146		B																		
			147		A									H	L	L							
			148		A	A																	
			149		B																		
			150		A												H	H					
			151		A		A	A				A											
			152		B																		
			153		A									L			L	L					
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						

See footnotes at end of device types 03, 04, 11, and 12.

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TABLE III. Group A inspection for device type 12 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 Clear	2 3 Clock	3 4 A	4 5 B	5 7 C	6 8 D	7 9 EnP	8 10 GND	9 11 Load	10 13 EnT	11 14 Q_D	12 15 Q_C	13 17 Q_B	14 18 Q_A	15 19 Ripple carry	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 7/	3014	56	B 8/	B 8/	B 8/	B	B	B	A 8/	GND	B	A	X	X	X	X	X	4.5 V	See 9/			
			57	B	A									L	L	L	L	L					
			58	A	B												H						
			59	A	A												L						
			60	B	A	A			A			A											
			61	B	B																		
			62	B	A																		
			63	A	A	B	B	B	B				B										
			64		B	A	A	A	A				A										
			65		A	A	A	A	A				A				H						
			66		A	B	A	A	A				B										
			67		B	A	A	A	A				A			H							
			68		A	A	A	A	A				A				L						
			69		A	B	A	A	A				B										
			70		B	A	A	A	A				A				H						
			71		A	A	A	A	A				A										
			72		A	B	A	A	A				B				L						
			73		B	A	A	A	A				A										
			74		A	A	A	A	A				A		H		H						
			75		A	B	A	A	A				B										
			76		B	A	A	A	A				A				L						
			77		A	A	A	A	A				A										
			78		A	B	A	A	A				B				H						
			79		B	A	A	A	A				A										
			80		A	A	A	A	A				A			H							
			81		A	B	A	A	A				B				L						
			82		B	A	A	A	A				A										
			83		A	A	A	A	A				A		L		H						
			84		A	B	A	A	A				B										
			85		B	A	A	A	A				A				L						
			86		A	A	A	A	A				A	H									
			87		A	B	A	A	A				B		L		H						
			88		B	A	A	A	A				A										
			89		A	A	A	A	A				A				L						
			90		A	B	A	A	A				B										
			91		B	A	A	A	A				A				H						
			92		A	A	A	A	A				A										
			93		A	B	A	A	A				B		L		L						
			94		B	A	A	A	A				A										
			95		A	A	A	A	A				A				H						
			96		A	B	A	A	A				B										
			97		B	A	A	A	A				A				L						
			98		A	A	A	A	A				A		H								
			99		A	B	A	A	A				B										
			100		B	A	A	A	A				A				H						
			101		A	A	A	A	A				A										
			102		A	B	A	A	A				B		L		L						
			103		B	A	A	A	A				A										
			104		A	A	A	A	A				A				H						
			105		A	B	A	A	A				B										
			106		B	A	A	A	A				A				L						
			107		A	A	A	A	A				A										
			108		A	B	A	A	A				B		L		H						
			109		B	A	A	A	A				A										
			110		A	A	A	A	A				A	L			L						
			111		A	B	A	A	A	B			B		L								
			112		B	A	A	A	A	A			A				H						
			113		B	A	A	A	A	A			A				L						
			114		A	B	A	B	A	A			A		L		L						

See footnotes at end of device types 03, 04, 11, and 12.

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TABLE III. Group A inspection for device type 12 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	16	18	19	20		Min	Max	
			Test no.	Clear	Clock	A	B	C	D	EnP	GND	Load	EnT	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
7 T_C = +25°C	Functional tests 7/	3014	115	A 8/	B	A	B 8/	B 8/	B	A	GND	8	B	L	L	L	L	L	4.5 V	See 9/			
			116		A		B	B									H						
			117		A		A	A															
			118		B																		
			119		A										H	H							
			120		A							A											
			121		B																		
			122		A		B	B	A														
			123		B							B											
			124		A									H	L	L							
			125		A		A	A					A										
			126		B																		
			127		A										H	H		H					
			128		A	B																	
			129		B	A																	
			130		B	A	B	B	B														
			131		A	A								L	L	L		L					
			132			B																	
			133			A		A	A														
			134			B			A														
			135			A			B														
			136			B																	
			137			B	A	B															
			138			A																	
			139		B																		
			140		A											H							
			141		A	B	B	A															
			142		B																		
			143		A										H	L	L						
			144		A	A																	
			145		B																		
			146		A												H						
			147		A	B	A																
			148		B																		
			149		A											H	L						
			150		A		B	B	A														
			151		B																		
			152		A									H	L	L							
			153		A		A																
			154		B																		
			155		A											H							
			156		A	A																	
			157		B																		
			158		A												H						
			159		A	B	B	A															
			160		B																		
			161		A										H	L	L						
			162		A	A																	
			163		B																		
			164		A												M						
			165		A							A											
			166		B																		
			167		A											H	L						
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						

See footnotes at end of device types 03, 04, 11, and 12.

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TABLE III. Group A inspection for device types 03, 04, 11, and 12 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 1/ 2 / Test no. (Device types) 03 04 11 12	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				Clear	Clock	A	B	C	D	EnP	GND	Load	EnT	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
9 T_C = +25°C	F_{MAX} 10/	3003 (Fig 6)	151 157 154 168	4.5 V	IN					4.5 V	GND	4.5 V	4.5 V				OUT		5.0 V	Q_A	22		MHz
	t_{PLH4}	"	152 158 155 169	"	"					"	"	"	"					OUT	"	Clk to carry	3	40	ns
	t_{PHL4}	"	153 159 156 170	"	"					"	"	"	"					OUT	"	Clk to carry	"	40	"
	t_{PLH5}	"	154 160 157 171	"	"					"	"	"	"				OUT		"	Clk to Q_A	"	29	"
		"	155 161 158 172	"	"					"	"	"	"			OUT			"	Clk to Q_B	"	"	"
		"	156 162 159 173	"	"					"	"	"	"		OUT				"	Clk to Q_C	"	"	"
		"	157 163 160 174	"	"					"	"	"	"	OUT					"	Clk to Q_D	"	"	"
	t_{PHL5}	"	158 164 161 175	"	"					"	"	"	"				OUT		"	Clk to Q_A	"	32	"
		"	159 165 162 176	"	"					"	"	"	"			OUT			"	Clk to Q_B	"	"	"
		"	160 166 163 177	"	"					"	"	"	"		OUT				"	Clk to Q_C	"	"	"
		"	161 167 164 178	"	"					"	"	"	"	OUT					"	Clk to Q_D	"	"	"
	t_{PLH6}	"	162 168 165 179	"	"	IN					"	GND					OUT		"	Clk to Q_A	"	29	"
	t_{PHL6}	"	163 169 166 180	"	"	IN					"	"					OUT		"	Clk to Q_A	"	32	"
	t_{PLH6}	"	164 170 167 181	"	"		IN				"	"				OUT			"	Clk to Q_B	"	29	"
	t_{PHL6}	"	165 171 168 182	"	"		IN				"	"				OUT			"	Clk to Q_B	"	32	"
	t_{PLH6}	"	166 172 169 183	"	"			IN			"	"			OUT				"	Clk to Q_C	"	29	"
	t_{PHL6}	"	167 173 170 184	"	"			IN			"	"			OUT				"	Clk to Q_C	"	32	"
	t_{PLH6}	"	168 174 171 185	"	"				IN		"	"		OUT					"	Clk to Q_D	"	29	"
	t_{PHL6}	"	169 175 172 186	"	"				IN		"	"		OUT						Clk to Q_D		32	"
	t_{PLH7}	"	170 176 173 187	"	"					4.5 V	"	4.5 V	IN					OUT	"	EnT to Cy	"	19	"
	t_{PHL7}	"	171 177 174 188	"	"					4.5 V	"	4.5 V	IN					OUT	"	EnT to Cy	"	11/	
	t_{PHL8}	"	172 178 175 189	IN	12/	4.5 V					"	GND					OUT		"	Clr to Q_A	"	33	"
		"	173 179 176 190	"	"		4.5 V				"	"				OUT			"	Clr to Q_B	"	"	"
		"	174 180 177 191	"	"			4.5 V			"	"			OUT				"	Clr to Q_C	"	"	"
		"	175 181 178 192	"	"				4.5 V		"	"		OUT					"	Clr to Q_D			

See footnotes at end of device types 03, 04, 11, and 12.

MIL-M-38510/315C

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TABLE III. Group A inspection for device types 03, 04, 11, and 12 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 1/ 2 / Test no. (Device types) 03 04 11 12	1 Clear	2 Clock	3 A	4 B	5 C	6 D	7 EnP	8 GND	9 Load	10 EnT	11 Q_D	12 Q_C	13 Q_B	14 Q_A	15 Ripple carry	16 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
10 T_C = +125°C	F_{MAX} 10/	3003 (Fig 6)	176 182 179 193																	Clk to Q_A	22		MHz
	t_{PLH4}	"	177 183 180 194																	Clk to carry	3	56	ns
	t_{PHL4}	"	178 184 181 195																	Clk to carry	"	56	"
	t_{PLH5}	"	179 185 182 196																	Clk to Q_A	"	41	"
		"	180 186 183 197																	Clk to Q_B	"	"	"
		"	181 187 184 198																	Clk to Q_C	"	"	"
		"	182 188 185 199																	Clk to Q_D	"	"	"
	t_{PHL5}	"	183 189 186 200																	Clk to Q_A	"	45	"
		"	184 190 187 201																	Clk to Q_B	"	"	"
		"	185 191 188 202																	Clk to Q_C	"	"	"
		"	186 192 189 203																	Clk to Q_D	"	"	"
	t_{PLH6}	"	187 193 190 204	Same terminal conditions as for subgroup 9.																Clk to Q_A	"	42	"
	t_{PHL6}	"	188 194 191 205																	Clk to Q_A	"	48	"
	t_{PLH6}	"	189 195 192 206																	Clk to Q_B	"	42	"
	t_{PHL6}	"	190 196 193 207																	Clk to Q_B	"	48	"
	t_{PLH6}	"	191 197 194 208																	Clk to Q_C	"	42	"
	t_{PHL6}	"	192 198 195 209																	Clk to Q_C	"	48	"
	t_{PLH6}	"	193 199 196 210																	Clk to Q_D	"	42	"
	t_{PHL6}	"	194 200 197 211																	Clk to Q_D	"	48	"
	t_{PLH7}	"	195 201 198 212																	EnT to carry	"	28	"
	t_{PHL7}	"	196 202 199 213																	EnT to carry	"	28	"
	t_{PHL8}	"	197 203 200 214																	Clr to Q_A	"	46	"
	t_{PHL8}	"	198 204 201 215																	Clr to Q_B	"	46	"
	t_{PHL8}	"	199 205 202 216																	Clr to Q_C	"	46	"
	t_{PHL8}	"	200 206 203 217																	Clr to Q_D	"	46	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

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1/ For case 2, pins not referenced are NC.

2/ Apply one pulse prior to measurement as follows:



3/ Apply 0.7 V for types 03 and 11; apply 2.0 V for types 04 and 12.

4/ I_{IL} limits (µA) min/max values for circuits shown:

Parameter	Terminals	Circuits						
		A	B	C	D	E	F	G
I_{IL4}	Clear 03	-160/-400	-30/-300	-120/-360	-160/-400	-120/-360	-150/-380	-160/-400
	Clear 04	"	"	"	"	"	"	"
	Clear 11	"	"		"	"	-300/-760	"
	Clear 12	"	"	-290/-630	"			"
	EnP	"	"	-120/-360	"	-120/-360	-150/-380	"
	A, B, C, D	"	"	-160/-400	"	-150/-380	-160/-400	"
I_{IL5}	Load	-320/-800	-30/-300	-290/-630	-320/-800	-120/-360	-300/-760	-320/-800
	EnT			-340/-860		-240/-720		
I_{IL6}	Clock	-160/-400	-0/-100	-290/-630	-160/-400	-180/-420	-120/-360	-160/-400

5/ For types 03 and 11, set outputs to 9th count (Q_A = 1, Q_D = 1, Q_B and Q_C = 0) prior to measurement.

For types 04 and 12, set outputs to 15th count (Q_A, Q_B, Q_C and Q_D = 1) prior to measurement.

6/ Apply GND for types 03 and 11; apply 4.5 V for types 04 and 12.

7/ Only a summary of attributes data is required.

8/ A = 3.0 V minimum; B = 0.0 V or GND.

9/ H > 1.5 V; L < 1.5 V; X = don't care.

10/ The F_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency. For type 04, circuit C, 20 MHz minimum.

11/ The limit for circuit B shall be 23 ns.

12/ For types 03 and 04, apply one clock pulse prior to test. For types 11 and 12 apply one clock pulse prior to test and another pulse during test.

13/ I_{IH13} limit for types 11 and 12; 40 µA maximum.

I_{IH14} limit for types 11 and 12; 200 µA maximum.

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TABLE III. Group A inspection for device types 05 and 06.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 1/ 2 / Test no.	1 / 2 / U/$\overline{D}$	2 / 3 / CP	3 / 4 / A	4 / 5 / B	5 / 7 / C	6 / 8 / D	7 / 9 / EP	8 / 10 / GND	9 / 12 / L	10 / 13 / ET	11 / 14 / Q_D	12 / 15 / Q_C	13 / 17 / Q_B	14 / 18 / Q_A	15 / 19 / Ripple carry	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1	4.5 V	2/	0.7 V	0.7 V	0.7 V	0.7 V	4.5 V	GND	0.7 V	4.5 V	4 mA	4 mA				4.5 V	Q_D		0.4	V
			2																	Q_C			
			3													4 mA				Q_B			
			4														4 mA			Q_A			
			5	0.7 V						0.7 V			0.7 V					4 mA		Ripple carry			
	V_{OH}	3006	6	4.5 V		2.0 V	2.0 V	2.0 V	2.0 V					-4 mA	-4 mA					Q_D	2.5		
			7													-4 mA				Q_C			
			8														-4 mA			Q_B			
			9															-4 mA		Q_A			
			10	0.7 V																Ripple carry			
	V_{IC}		11	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA		-18 mA	-18 mA							U/$\overline{D}$		-1.5	
			12																	CP			
			13																	A			
			14																	B			
			15																	C			
			16																	D			
			17																	EP			
			18																	L			
			19																	ET			
	I_{IL12}	3009	20			0.4 V	0.4 V	0.4 V	0.4 V			GND							5.5 V	A	3/	3/	μA
			21																	B			
			22																	C			
			23																	D			
	I_{IL13}		24	0.4 V	0.4 V							0.4 V								U/$\overline{D}$	3/	3/	
			25																	CP			
			26																	L			
	I_{IL14}		27							0.4 V										EP	3/	3/	
	I_{IL15}		28										0.4 V							ET	3/	3/	
	I_{IH17}	3010	29	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V		2.7 V								U/$\overline{D}$		20	
			30																	CP			
			31																	A			
			32																	B			
			33																	C			
			34																	D			
			35																	EP			
			36																	L			
	I_{IH19}		37										2.7 V							ET		40	

See footnotes at end of device types 05 and 06.

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TABLE III. Group A inspection for device types 05 and 06.

Terminal conditions (pins not designated may be H $\geq$ 2.0 V, or L $\leq$ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	U/$\overline{D}$	CP	A	B	C	D	EP	GND	L	ET	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
1 T_C = +25°C	I_{IH1B}	3010	38	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	5.5 V							5.5 V	U/$\overline{D}$		100	µA
		"	39								"								"	CP		"	"
		"	40								"								"	A		"	"
		"	41								"								"	B		"	"
		"	42								"								"	C		"	"
		"	43								"								"	D		"	"
		"	44								"								"	EP		"	"
		"	45								"								"	L		"	"
	I_{IH2O}	"	46								"		5.5 V						"	ET		200	"
	I_{OS}	3011	47	5.5 V	2/	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	GND	5.5 V	GND	GND	GND	GND	GND	"	Q_D	-15	-100	mA
		"	48		"						"								"	Q_C	"	"	"
		"	49		"						"								"	Q_B	"	"	"
		"	50		"						"								"	Q_A	"	"	"
		"	51		"						"								"	Ripple carry	"	"	"
	I_{CC}	3005	52	GND	"	GND	GND	GND	GND	GND	"	5.5 V	GND						"	V_{CC}		34	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device types 05 and 06.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 U/D	2 3 CP	3 4 A	4 5 B	5 7 C	6 8 D	7 9 EP	8 10 GND	9 12 L	10 13 ET	11 14 Q_D	12 15 Q_C	13 17 Q_B	14 18 Q_A	15 19 Ripple carry	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 4/	3014	53	A 5/	B 5/	B	B	B	B	B	GND	B	B	X	X	X	X	X	4.5 V	See 6/			
			54		A							B		L 5/	L	L	L	H 5/					
			55		B							A				L	L						
			56		A											H	H						
			57		B											H	L						
			58		A											H	L						
			59		B											L	L						
			60		A											H	L						
			61		B											H	L						
			62		A										H	L	L						
			63		B											L	L						
			64		A											H	L						
			65		B											H	L						
			66		A											H	L						
			67		B											L	L						
			68		A											L	L						
			69		B											L	L						
			70		A									H		L	L						
			71		B											L	L						
			72		A											L	L						
			73		B					A						L	L						
			74		A					A						L	L						
			75		B					B						L	L						
			76		A								A			L	L						
			77		B								A			L	L						
			78		A								B			L	L						
			79		B									L		L	L						
			80		A	A			A			B		L		L	L						
			81		B							B		H		L	L						
			82	B	A							A				L	L						
			83		B											L	L						
			84		A											L	L						
			85		B										H	L	L						
			86		A											L	L						
			87		B											L	L						
			88		A											L	L						
			89		B											L	L						
			90		A											L	L						
			91		B										H	L	L						
			92		A											L	L						
			93		B											L	L						
			94		A											L	L						
			95		B											L	L						
			96		A											L	L						
			97		B											L	L						
			98		A											L	L						
			99		B											L	L						
			100		A					A						L	L						
			101		B					A						L	L						
			102		A					B						L	L						
			103		B								A			L	L						
			104		A								A			L	L						
			105		B								B			L	L						
			106		B	B	A	A				B				L	L						
			107		A	B	A	A				B			H	H	H						
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C																						

See footnotes at end of device types 05 and 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	U/$\overline{D}$	CP	A	B	C	D	EP	GND	L	ET	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
T_C = +25°C	Functional tests 4/	3014	53	A 5/	B 5/	B	B	B	B	B	GND	B	B	X	X	X	X	X	4.5 V	See 6/			
			54		A							B		L 5/	L	L	L	H 5/					
			55		B							A					L						
			56		A												H						
			57		B												L						
			58		A											H							
			59		B												L						
			60		A												H						
			61		B												H						
			62		A											L							
			63		B										H		L						
			64		A												H						
			65		B												L						
			66		A											H							
			67		B												L						
			68		A												H						
			69		B												L						
			70		A										H		L						
			71		B												L						
			72		A												H						
			73		B												L						
			74		A											H							
			75		B												L						
			76		A												H						
			77		B												L						
			78		A										H		L						
			79		B												L						
			80		A											H							
			81		B												L						
			82		A												H						
			83		B												L						
			84		A											H							
			85		B					A								L					
			86		A					A													
			87		B					B													
			88		B								A					H					
			89		A								A					H					
			90		B								B					L					
			91		A									L	L		L	H					
			92		B	A	A	A	A			B		L	L		L	H					
			93		A							B		H	H		H	L					
			94	B	B							A					H						
			95		A												L						
			96		B												H						
			97		A												L						
			98		B												H						
			99		A												L						
			100		B												H						
			101		A												L						
			102		B												H						
			103		A												L						
			104		B												H						
			105		A												L						
			106		B												H						
			107		A												L						
			108		B																		

See footnotes at end of device types 05 and 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	U/D	CP	A	B	C	D	EP	GND	-	ET	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
7 T_C = +25°C	Functional tests 4/	3014	109	B 5/	A 5/	A	A	A	A	B	GND	\	B	L	H	H	H	H	4.5 V	See 6/			
			110		B												H						
			111		A												L						
			112		B												L						
			113		A											L	H						
			114		B												H						
			115		A												L						
			116		B											H	L						
			117		B										L		H						
			118		B												H						
			119		A												L						
			120		B											L	L						
			121		A												H						
			122		B											L	L						
			123		A												H						
			124		B					A						H	L						
			125		A								A				L						
			126		B					B						H	L						
			127		A								A				L						
			128		B								B				L						
8	Repeat subgroup 7 at T_C = -55°C and T_C = +125°C.																						

See footnotes at end of device types 05 and 06.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	U/D	CP	A	B	C	D	EP	GND	L	ET	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
9 T_C = +25°C	t_{PLH5}	See fig. 7	108 109 110 111	5.0 V " " "	IN 7/ IN IN 2/ IN 8/	GND " " "	GND " " "	GND " " "	GND " " "	GND " " "	GND " " "	IN 7/ 5.0 V " "	GND	OUT	OUT	OUT	OUT		5.0 V " " "	CP to Q_A CP to Q_B CP to Q_C CP to Q_D	3 " " "	22 " " "	ns " " "
	t_{PHL5}	" " " "	112 113 114 115	" " " "	IN 7/ " " "	5.0 V " " "	5.0 V " " "	5.0 V " " "	5.0 V " " "	" " " "	" " " "	IN " " "		OUT	OUT	OUT	OUT		" " " "	CP to Q_A CP to Q_B CP to Q_C CP to Q_D	" " " "	" " " "	" " " "
	t_{PHL15}	"	116	"	"	"	"	"	"	"	"	"						OUT	"	CP to RC	"	32	"
	t_{PLH12}	"	117	"	IN	"	"	"	"	"	"	5.0 V						OUT	"	CP to RC	"	30	"
	t_{PLH5} t_{PHL5} t_{PLH5} t_{PHL5} t_{PLH5} t_{PHL5} t_{PLH5} t_{PHL5}	" " " " " " " "	118 119 120 121 122 123 124 125	GND " " " " " " "	IN 9/ IN " " " " " "	IN GND " " " " " "	GND GND 5.0 V GND " " " "	GND " " 5.0 V GND " " "	GND " " " " 5.0 V GND "	" " " " " " " "	" " " " " " " "	GND		OUT OUT	OUT OUT	OUT OUT	OUT OUT		" " " " " " " "	CP to Q_A CP to Q_A CP to Q_B CP to Q_B CP to Q_C CP to Q_C CP to Q_D CP to Q_D	" " " " " " " "	22 " " " " " " "	" " " " " " " "
	t_{PHL7} t_{PLH7}	" "	126 127	5.0 V 5.0 V	IN 7/ GND	5.0 V "	" "	" "	5.0 V "	" "	" "	IN 5.0 V	IN IN					OUT OUT	" "	ET to RC ET to RC	" "	24 15	" "
	t_{PHL11}	"	128	IN	IN 7/	"	"	"	"	"	"	IN	GND					OUT	"	U/D to RC	"	28	"
	t_{PLH9}	"	129	IN	GND	"	"	"	"	"	"	5.0 V	GND						"	U/D to RC	"	22	"
	t_{PHL7} t_{PLH7}	" "	130 131	GND GND	IN 7/ GND	GND "	" "	" "	GND "	" "	" "	IN 5.0 V	IN IN						" "	ET to RC ET to RC	" "	24 15	" "
	t_{PHL11}	"	132	IN	IN 7/	"	"	"	"	"	"	IN	GND						"	U/D to RC	"	28	"
	t_{PLH9}	"	133	IN	GND	"	"	"	"	"	"	5.0 V							"	U/D to RC	"	22	"
	F_{MAX} 10/	"	134	5.0 V	IN					"	"	5.0 V		OUT	OUT	OUT	OUT		"	CP to Q_n	25		MHz
	F_{MAX} 10/	"	135	GND	IN					"	"	5.0 V		OUT	OUT	OUT	OUT		"	CP to Q_n	25		MHz

See footnotes at end of device types 05 and 06.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 U/$\overline{D}$	2 CP	3 A	4 B	5 C	6 D	7 EP	8 GND	9 L	10 ET	11 Q_D	12 Q_C	13 Q_B	14 Q_A	15 Ripple carry	16 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
			Cases E, F Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
10 T_C = +125°C	t_{PLH5}	See fig. 7	136																	CP to Q_A	3	26	ns
		"	137																	CP to Q_B	"	"	"
		"	138																	CP to Q_C	"	"	"
		"	139																	CP to Q_D	"	"	"
	t_{PHL5}	"	140																	CP to Q_A	"	"	"
		"	141																	CP to Q_B	"	"	"
		"	142																	CP to Q_C	"	"	"
		"	143																	CP to Q_D	"	"	"
	t_{PHL15}	"	144																	CP to RC	"	37	"
	t_{PLH12}	"	145																	CP to RC	"	35	"
	t_{PLH5}	"	146																	CP to Q_A	"	26	"
	t_{PHL5}	"	147																	CP to Q_A	"	"	"
	t_{PLH5}	"	148																	CP to Q_B	"	"	"
	t_{PHL5}	"	149																	CP to Q_B	"	"	"
	t_{PLH5}	"	150																	CP to Q_C	"	"	"
	t_{PHL5}	"	151																	CP to Q_C	"	"	"
	t_{PLH5}	"	152																	CP to Q_D	"	"	"
	t_{PHL5}	"	153																	CP to Q_D	"	"	"
	t_{PHL7}	"	154																	ET to RC	"	28	"
	t_{PLH7}	"	155																	ET to RC	"	18	"
	t_{PHL11}	"	156																	U/$\overline{D}$ to RC	"	33	"
	t_{PLH9}	"	157																	U/$\overline{D}$ to RC	"	26	"
	t_{PHL7}	"	158																	ET to RC	"	28	"
	t_{PLH7}	"	159																	ET to RC	"	18	"
	t_{PHL11}	"	160																	U/$\overline{D}$ to RC	"	33	"
	t_{PLH9}	"	161																	U/$\overline{D}$ to RC	"	26	"
	F_{MAX}	"	162																	CP to Q_n	25		MHz
	F_{MAX}	"	163																	CP to Q_n	25		MHz

Same conditions as for subgroup 9.

11	Same tests, conditions and limits as for subgroup 10 except T_C = -55°C and V_{CC} = 4.5 for F_{MAX}.

See footnotes at end of device types 05 and 06.

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TABLE III. Group A inspection for device type 06.

Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	11	13	14	15	17	18	19	20		Min	Max	
			Test no.	U/D	CP	A	B	C	D	EP	GND	L	ET	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
9 T_C = +25°C	t_{PLH5}	See fig. 7	129	5.0 V	IN	GND	GND	GND	GND	GND	GND	IN	GND						5.0 V	CP to Q_A	3	22	ns
			130	"	"	"	"	"	"	"	"	"	"						"	CP to Q_B	"	"	"
			131	"	"	"	"	"	"	"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_C	"	"	"
			132	"	"	"	"	"	"	"	"	"	"						"	CP to Q_D	"	"	"
	t_{PHL5}	"	133	"	"	5.0 V	"	"	"	"	"	IN	IN						"	CP to Q_A	"	"	"
			134	"	"	"	5.0 V	"	"	"	"	"	"						"	CP to Q_B	"	"	"
			135	"	"	"	"	5.0 V	"	"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_C	"	"	"
			136	"	"	"	"	"	5.0 V	"	"	"	"						"	CP to Q_D	"	"	"
	t_{PHL15}	"	137	"	"	GND	"	"	"	"	"	"						OUT	"	CP to RC	"	32	"
	t_{PLH12}	"	138	"	"	GND	"	"	"	"	"	5.0 V						OUT	"	CP to RC	"	30	"
	t_{PLH5}	"	139	GND	"	5.0 V	GND	GND	GND	"	"	GND					OUT		"	CP to Q_A	"	22	"
	t_{PHL5}	"	140	"	"	GND	GND	"	"	"	"	"					OUT		"	CP to Q_A	"	"	"
	t_{PLH5}	"	141	"	"	"	5.0 V	"	"	"	"	"					OUT		"	CP to Q_B	"	"	"
	t_{PHL5}	"	142	"	"	"	GND	5.0 V	"	"	"	"					OUT		"	CP to Q_B	"	"	"
	t_{PLH5}	"	143	"	"	"	"	GND	"	"	"	"					OUT		"	CP to Q_C	"	"	"
	t_{PHL5}	"	144	"	"	"	"	"	"	"	"	"					OUT		"	CP to Q_C	"	"	"
	t_{PLH5}	"	145	"	"	"	"	"	5.0 V	"	"	"					OUT		"	CP to Q_D	"	"	"
	t_{PHL5}	"	146	"	"	"	"	"	GND	"	"	"		OUT OUT	OUT OUT	OUT OUT	OUT OUT		"	CP to Q_D	"	"	"
	t_{PHL7}	"	147	5.0 V	"	5.0 V	5.0 V	5.0 V	5.0 V	"	"	IN 2/ 5.0 V	IN					OUT	"	ET to RC	"	24	"
	t_{PLH7}	"	148	5.0 V	GND	"	"	"	"	"	"	"	IN					OUT	"	ET to RC	"	15	"
	t_{PHL11}	"	149	IN	IN	"	"	"	"	"	"	IN 2/	GND					OUT	"	U/$\overline{D}$ to RC	"	28	"
	t_{PLH9}	"	150	IN	GND	"	"	"	"	"	"	5.0 V	GND					"	"	U/$\overline{D}$ to RC	"	22	"
	t_{PHL7}	"	151	GND	IN	GND	GND	GND	GND	"	"	IN 2/ 5.0 V	IN					"	"	ET to RC	"	24	"
	t_{PLH7}	"	152	GND	GND	"	"	"	"	"	"	"	IN					"	"	ET to RC	"	15	"
	t_{PHL11}	"	153	IN	IN	"	"	"	"	"	"	IN 2/	GND					"	"	U/$\overline{D}$ to RC	"	28	"
	t_{PLH9}	"	154	IN	GND	"	"	"	"	"	"	5.0 V	"					"	"	U/$\overline{D}$ to RC	"	22	"
	F_{MAX}	"	155	5.0 V	IN	"	"	"	"	"	"	5.0 V	"	OUT	OUT	OUT	OUT		"	CP to Q_n	25		MHz
	F_{MAX}	"	156	GND	IN	5.0 V	5.0 V	5.0 V	5.0 V	"	"	5.0 V	"	OUT	OUT	OUT	OUT		"	CP to Q_n	25		MHz

See footnotes at end of device types 05 and 06.

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TABLE III. Group A inspection for device type 06 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

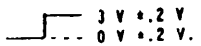
Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	U/D	CP	A	B	C	D	EP	GND	L	ET	Q_D	Q_C	Q_B	Q_A	Ripple carry	V_{CC}				
10 T_C = +125°C	t_{PLH5}	See fig. 7	157	Same conditions as for subgroup 9.																CP to Q_A	3	26	ns
		"	158																	CP to Q_B	"	"	"
		"	159																	CP to Q_C	"	"	"
		"	160																	CP to Q_D	"	"	"
	t_{PHL5}	"	161																	CP to Q_A	"	"	"
		"	162																	CP to Q_B	"	"	"
		"	163																	CP to Q_C	"	"	"
		"	164																	CP to Q_D	"	"	"
	t_{PHL15}	"	165																	CP to RC	"	37	"
	t_{PLH12}	"	166																	CP to RC	"	35	"
	t_{PLH5}	"	167																	CP to Q_A	"	26	"
	t_{PHL5}	"	168																	CP to Q_A	"	"	"
	t_{PLH5}	"	169																	CP to Q_B	"	"	"
	t_{PHL5}	"	170																	CP to Q_B	"	"	"
	t_{PLH5}	"	171																	CP to Q_C	"	"	"
	t_{PHL5}	"	172																	CP to Q_C	"	"	"
	t_{PLH5}	"	173																	CP to Q_D	"	"	"
	t_{PHL5}	"	174																	CP to Q_D	"	"	"
	t_{PHL7}	"	175																	ET to RC	"	28	"
	t_{PLH7}	"	176																	ET to RC	"	18	"
	t_{PHL11}	"	177																	$U/\overline{D}$ to RC	"	33	"
	t_{PLH9}	"	178																	$U/\overline{D}$ to RC	"	26	"
	t_{PHL7}	"	179																	ET to RC	"	28	"
	t_{PLH7}	"	180																	ET to RC	"	18	"
	t_{PHL11}	"	181																	$U/\overline{D}$ to RC	"	33	"
	t_{PLH9}	"	182																	$U/\overline{D}$ to RC	"	26	"
	F_{MAX} -	"	183																	CP to Q_n	25		MHz
	F_{MAX}	"	184																	CP to Q_n	25		MHz
11	Same tests, conditions and limits as for subgroup 10 except T_C = -55°C and V_{CC} = 4.5 for F_{MAX}.																						

See footnotes at end of device types 05 and 06.

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1/ Case 2 pins not referenced are N/C.

2/ Apply one clock pulse prior to test as follows:



3/ I_{IL} limits (μA) min/max values for circuits shown:

Parameter	Terminals	Circuits						
		A	B	C	D	E	F	G
I_{IL12}	A, B, C, D			-160/-400		-3.0/-400		
I_{IL13}	U/$\overline{D}$, CP, L			-160/-400		-135/-370		
I_{IL14}	EP			-160/-400		-150/-385		
I_{IL15}	ET			-290/-720		-280/-760		

4/ Only a summary of attributes data is required.

5/ A = 3.0 V minimum; B = 0.0 V or GND.

6/ H > 1.5 V; L < 1.5 V; X = don't care.

7/ Apply one clock pulse with "L" low prior to test.

8/ Apply three clock pulses prior to test.

9/ Apply one clock pulse with "A" low prior to test.

10/ On (Q_A, Q_B, Q_C, and Q_D) shall respond as specified in the truth table with the minimum F_{MAX} frequency input to "CP".

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TABLE III. Group A inspection for device types 07 and 08.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				B	Q_B	Q_A	Count down	Count up	Q_C	Q_D	GND	D	C	Load	Ripple carry	Borrow f	Clear	A	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1	0.7 V	4 mA	4 mA			4 mA	4 mA	GND		0.7 V	0.7 V			0.7 V	0.7 V	4.5 V	Q_A		0.4	V
			2											"			"		"	Q_B		"	
			3											"			"		"	Q_C		"	
			4									0.7 V		"			"		"	Q_D		"	
			5	2/				0.7 V				2.0 V	2/	"	4 mA		"	2.0 V	"	Ripple carry		"	
			6				0.7 V									4 mA	2.0 V			Borrow			
	V_{OH}	3006	7	2.0 V	-0.4 mA	-0.4 mA			-0.4 mA	-0.4 mA			2.0 V	0.7 V			0.7 V	2.0 V	"	Q_A	2.5		
			8											"			"		"	Q_B	"		
			9											"			"		"	Q_C	"		
			10									2.0 V		"			"		"	Q_D	"		
			11					2.0 V						"	-0.4 mA		"		"	Ripple carry	"		
			12				2.0 V									-0.4 mA			"	Borrow	"		
	V_{IC}		13	-18 mA									2.0 V					-18 mA	"	A		-1.5	
			14																"	B		"	
			15																"	C		"	
			16									-18 mA							"	D		"	
			17										-18 mA	-18 mA					"	Load		"	
			18														-18 mA		"	Clear		"	
			19					-18 mA											"	Count up		"	
			20				-18 mA												"	Count down		"	
	I_{IL9}	3009	21	0.4 V									0.4 V	GND			GND	0.4 V	5.5 V	A	3/	3/	μA
			22											"			"		"	B	"	"	
			23											"			"		"	C	"	"	
			24									0.4 V	0.4 V	"			"		"	D	"	"	
	I_{IL10}	"	25								"			0.4 V					"	Load	"	"	"
	I_{IL11}	"	26					0.4 V			"						0.4 V		"	Clear	"	"	"
			27																"	Count up	"	"	
			28				0.4 V												"	Count down	"	"	
	I_{IH17}	3010	29	2.7 V									2.7 V	5.5 V			5.5 V	2.7 V	"	A		20	
			30											"			"		"	B		"	
			31											"			"		"	C		"	
			32									2.7 V		"			"		"	D		"	
			33											2.7 V			"		"	Load		"	
			34														2.7 V		"	Clear		"	
			35					2.7 V											"	Count up		"	
			36				2.7 V												"	Count down		"	
	I_{IH18}	"	37	5.5 V									5.5 V	5.5 V			5.5 V	5.5 V	"	A		100	
			38											"			"		"	B		"	
			39											"			"		"	C		"	
			40									5.5 V	5.5 V	"			"		"	D		"	
			41											"			"		"	Load		"	
			42														5.5 V		"	Clear		"	
			43					5.5 V											"	Count up		"	
			44				5.5 V												"	Count down		"	

See footnotes at end of device types 07 and 08.

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TABLE III. Group A inspection for device types 07 and 08 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	B	Q_B	Q_A	Count down	Count up	Q_C	Q_D	GND	D	C	Load	Ripple carry	Borrow	Clear	A	V_{CC}				
1 T_C = +25°C	I_{OS}	3011	45	5.5 V	GND	GND			GND	GND	GND	5.5 V	5.5 V	GND	GND		GND	5.5 V	5.5 V	Q_A	4/	4/	mA
		"	46								"			"			"		"	Q_B	"	"	"
		"	47								"			"			"		"	Q_C	"	"	"
		"	48					5.5 V			"			"			"		"	Q_D	"	"	"
		"	49								"			"			"		"	Ripple carry	"	"	"
		"	50				5.5 V				"			"		GND			"	Borrow			
	I_{CC}	3005	51								"			GND			GND		"	V_{CC}		34	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device types 07 and 08.

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 B	2 3 Q_B	3 4 Q_A	4 5 Count down	5 7 Count up	6 8 Q_C	7 9 Q_D	8 10 GND	9 12 D	10 13 C	11 14 Load	12 15 Ripple carry	13 17 Borrow	14 18 Clear	15 19 A	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 5/	3014	52	A 6/	L	L	A	A	L	L	GND	A	A	A	H	H	A	A	4.5 V	See 7/			
			53			L		A									B 6/						
			54			L		B															
			55			H		A															
			56			H		B															
			57		H	L		A															
			58			L		B															
			59			H		A															
			60			H		B															
			61		L	L		A	H														
			62			L		B															
			63			H		A															
			64			H		B															
			65		H	L		A															
			66			L		B															
			67			H		A															
			68			H		B															
			69		L	L		A	L	H													
			70			L		B															
			71			H		A															
			72			H		B															
			73			L		A		L						H							
			74			L	B																
			75			H	A			H						L							
			76			H	B									H							
			77			L	A			L													
			78			L	B																
			79		H	H	A		H	L													
			80			H	B																
			81			L	A																
			82			L	B																
			83			H	A																
			84			H	B																
			85			L	A																
			86			L	B																
			87		H	H	A		L	H													
			88			H	B																
			89			L	A																
			90			L	B																
			91		L	H	A			H						L							
			92			H	B									H							
			93			L	A																
			94			L	B																
			95			H	A			H						L							
			96		H	H	B		H	L				B									
			97		L	L			L	H							A						
			98		L	H			H	H							B						
			99		H	H			H	L				A			B						
			100		L	L			L	L							A						
			101		L	L			L	L							B						

See footnotes at end of device types 07 and 08.

MIL-M-38510/315C

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TABLE III. Group A inspection for device type 07 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	B	Q_B	Q_A	Count down	Count up	Q_C	Q_D	GND	D	C	Load	Ripple carry	Borrow	Clear	A	V_{CC}				
7 T_C = +25°C	Functional tests 5/	3014	102	A 6/	H	H	A	A	H	H	GND	A	A	B 6/	H	H	B	A	4.5 V	See 7/			
			103	A								A	A	A				A					
			104	B								B	B	B				B					
			105		L	L	B		L	L						L							
			106				A							A		H							
			107					B									A						
			108					A															
			109				B									L							
			110				A									H							
			111					B									A						
			112				B																
			113				A																
			114					B															
			115				A																
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						

See footnotes at end of device types 07 and 08.

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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 B	2 3 Q_B	3 4 Q_A	4 5 Count down	5 7 Count up	6 8 Q_C	7 9 Q_D	8 10 GND	9 12 D	10 13 C	11 14 Load	12 15 Ripple carry	13 17 Borrow	14 18 Clear	15 19 A	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 5/	3014	52	B 6/	L	L	A 6/	A	L	L	GND	9	B	B	H	H	B	B	4.5 V	See 7/			
			53	"	"	L	"	"	"	"	"	"	"	A	"	"	"	A	"				
			54	"	"	H	"	"	"	"	"	"	"	B	"	"	"	A	"				
			55	"	"	H	"	"	"	"	"	"	"	A	"	"	"	A	"				
			56	A	H	L	"	"	"	"	"	"	"	B	"	"	"	B	"				
			57	"	"	L	"	"	"	"	"	"	"	A	"	"	"	B	"				
			58	"	"	H	"	"	"	"	"	"	"	B	"	"	"	A	"				
			59	"	"	H	"	"	"	"	"	"	"	A	"	"	"	A	"				
			60	B	L	L	"	"	H	"	"	"	A	B	"	"	"	B	"				
			61	"	"	L	"	"	"	"	"	"	"	A	"	"	"	B	"				
			62	"	"	H	"	"	"	"	"	"	"	B	"	"	"	A	"				
			63	"	"	H	"	"	"	"	"	"	"	A	"	"	"	A	"				
			64	A	H	L	"	"	"	"	"	"	"	B	"	"	"	B	"				
			65	"	"	L	"	"	"	"	"	"	"	A	"	"	"	B	"				
			66	"	"	H	"	"	"	"	"	"	"	B	"	"	"	A	"				
			67	"	"	H	"	"	"	"	"	"	"	A	"	"	"	A	"				
			68	B	L	L	"	"	L	H	"	A	B	B	"	"	"	B	"				
			69	"	"	L	"	"	"	"	"	"	"	A	"	"	"	B	"				
			70	"	"	H	"	"	"	"	"	"	"	B	"	"	"	A	"				
			71	"	"	H	"	"	"	"	"	"	"	A	"	"	"	A	"				
			72	A	H	L	"	"	"	"	"	"	"	B	"	"	"	B	"				
			73	"	"	L	"	"	"	"	"	"	"	A	"	"	"	B	"				
			74	"	"	H	"	"	"	"	"	"	"	B	"	"	"	A	"				
			75	"	"	H	"	"	"	"	"	"	"	A	"	"	"	A	"				
			76	B	L	L	"	"	H	"	"	"	A	B	"	"	"	B	"				
			77	"	"	L	"	"	"	"	"	"	"	A	"	"	"	B	"				
			78	"	"	H	"	"	"	"	"	"	"	B	"	"	"	A	"				
			79	"	"	H	"	"	"	"	"	"	"	A	"	"	"	A	"				
			80	A	H	L	"	"	"	"	"	"	"	B	"	"	"	B	"				
			81	"	"	L	"	"	"	"	"	"	"	A	"	"	"	B	"				
			82	"	"	H	"	"	"	"	"	"	"	B	"	"	"	A	"				
			83	"	"	L	"	"	L	L	"	"	"	B	"	"	"	"	"				
			84	"	"	L	"	"	"	"	"	"	"	A	"	"	"	A	"				
			85	"	"	L	"	"	"	"	"	"	"	"	"	"	"	B	"				
			86	"	"	L	B	"	"	"	"	"	"	"	"	"	L	"	"				
			87	"	H	H	A	"	H	H	"	"	"	"	"	H	"	"	"				
			88	"	"	H	A	"	"	"	"	"	"	"	"	"	"	"	"				
			89	"	"	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			90	"	"	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			91	"	L	H	A	"	"	"	"	"	"	"	"	"	"	"	"				
			92	"	"	H	A	"	"	"	"	"	"	"	"	"	"	"	"				
			93	"	"	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			94	"	"	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			95	"	H	H	A	"	L	"	"	"	"	"	"	"	"	"	"				
			96	"	"	H	A	"	"	"	"	"	"	"	"	"	"	"	"				
			97	"	"	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			98	"	"	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			99	"	L	H	A	"	"	"	"	"	"	"	"	"	"	"	"				
			100	"	"	H	A	"	"	"	"	"	"	"	"	"	"	"	"				
			101	"	"	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			102	"	"	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			103	"	H	H	A	"	H	L	"	"	"	"	"	"	"	"	"				
			104	"	"	H	A	"	"	"	"	"	"	"	"	"	"	"	"				
			105	"	"	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			106	"	"	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			107	"	L	H	A	"	"	"	"	"	"	"	"	"	"	"	"				

See footnotes at end of device types 07 and 08.

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TABLE III. Group A inspection for device type 08 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 1/ 2 / Test no.	1 / 2 / B	2 / 3 / Q_B	3 / 4 / Q_A	4 / 5 / Count down	5 / 7 / Count up	6 / 8 / Q_C	7 / 9 / Q_D	8 / 10 / GND	9 / 12 / D	10 / 13 / C	11 / 14 / Load	12 / 15 / Ripple carry	13 / 17 / Borrow	14 / 18 / Clear	15 / 19 / A	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 5/	3014	108	A 6/	L	H	B 6/	A	H	L	GND	A	A	A	H	H	B 8/	A	4.5 V	See 7/			
			109			L	A																
			110			L	B																
			111		H	H	A		L														
			112			H	B																
			113			L	A																
			114			L	B																
			115		L	H	A																
			116			H	B																
			117			L	A																
			118			L		B															
			119			H		A															
			120			H		B															
			121		H	L		A															
			122			L		B															
			123			H		A															
			124			H		B															
			125		L	L		A	H														
			126			L		B															
			127			H		A															
			128			H		B															
			129		H	L		A															
			130			L		B															
			131			H		A															
			132			H		B															
			133		L	L		A	L	H													
			134			L		B															
			135			H		A															
			136			H		B															
			137		H	L		A															
			138			L		B															
			139			H		A															
			140			H		B															
			141		L	L		A	H														
			142			L		B															
			143			H		A															
			144			H		B															
			145		H	L		A															
			146			L		B															
			147			H		A															
			148			H		B								L							
			149		L	L		A	L	L						H							
			150			L		B										A					
			151			L		A															
			152			L		B															
			153			L	B										L						
			154			L	A										H						
			155			L		B										B					
			156		H	H		A	H	H													
			157			H		B								L							
			158			H		A								H							
			159			H	B	A															
			160			H	A	A															
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						

See footnotes at end of device types 07 and 08.

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TABLE III. Group A inspection for device types 07 and 08 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 1/ 2 / (Device type) 07 / 08	1 / 2 / B	2 / 3 / Q_B	3 / 4 / Q_A	4 / 5 / Count down	5 / 7 / Count up	6 / 8 / Q_C	7 / 9 / Q_D	8 / 10 / GND	9 / 12 / D	10 / 13 / C	11 / 14 / Load	12 / 15 / Ripple carry	13 / 17 / Borrow	14 / 18 / Clear	15 / 19 / A	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	F_{MAX} 8/	3003 9/	116 161			OUT	5.0 V	IN			GND			5.0 V			GND		5.0 V	Count up to Q_A	22		MHz
	F_{MAX} 8/	"	117 162			OUT	IN	5.0 V			"			5.0 V			GND		"	Count down to Q_A	22		MHz
	t_{PLH8}	"	118 163	5.0 V	OUT	OUT					"			IN			GND	5.0 V	"	Load to Q_A	3	45	ns
		"	119 164								"			"			"		"	Load to Q_B	"	"	"
		"	120 165						OUT		"		5.0 V	"			"		"	Load to Q_C	"	"	"
		"	121 166							OUT	"	5.0 V		"			"		"	Load to Q_D	"	"	"
	t_{PHL10}	"	122 167	GND	OUT	OUT					"			"			GND	GND	"	Load to Q_A	"	"	"
		"	123 168								"			"			"		"	Load to Q_B	"	"	"
		"	124 169						OUT		"		GND	"			"		"	Load to Q_C	"	"	"
		"	125 170							OUT	"	GND		"			"		"	Load to Q_D	"	"	"
	t_{PLH9}	"	126 171			OUT	5.0 V	IN			"			5.0 V			"		"	Count up to Q_A	"	43	"
		"	127 172		OUT		"	"			"			"			"		"	Count up to Q_B	"	"	"
		"	128 173				"	"	OUT		"			"			"		"	Count up to Q_C	"	"	"
		"	129 174							OUT	"			"			"		"	Count up to Q_D	"	"	"
		"	130 175				IN	5.0 V		OUT	"			"			"		"	Count down to Q_D	"	"	"
		"	131 176				"	"	OUT		"			"			"		"	Count down to Q_C	"	"	"
		"	132 177		OUT		"	"			"			"			"		"	Count down to Q_B	"	"	"
		"	133 178			OUT	"	"			"			"			"		"	Count down to Q_A	"	"	"
	t_{PHL11}	"	134 179		OUT	OUT	5.0 V	IN			"			"			"		"	Count up to Q_A	"	52	"
		"	135 180				"	"	OUT		"			"			"		"	Count up to Q_B	"	"	"
		"	136 181							OUT	"			"			"		"	Count up to Q_C	"	"	"
		"	137 182							OUT	"			"			"		"	Count up to Q_D	"	"	"
		"	138 183				IN	5.0 V		OUT	"			"			"		"	Count down to Q_D	"	"	"
		"	139 184				"	"	OUT		"			"			"		"	Count down to Q_C	"	"	"
		"	140 185		OUT		"	"			"			"			"		"	Count down to Q_B	"	"	"
		"	141 186			OUT	"	"			"			"			"		"	Count down to Q_A	"	"	"
	t_{PHL12}	"	142 187	5.0 V	OUT	OUT					"			10/			IN	5.0 V	"	Clear to Q_A	"	40	"
		"	143 188								"			"			"		"	Clear to Q_B	"	"	"
		"	144 189						OUT		"		5.0 V	"			"		"	Clear to Q_C	"	"	"
		"	145 190							OUT	"	5.0 V		"			"		"	Clear to Q_D	"	"	"

See footnotes at end of device types 07 and 08.

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TABLE III. Group A inspection for device types 07 and 08 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 (Device type) 07	08	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
					2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
					B	Q_B	Q_A	Count down	Count up	Q_C	Q_D	GND	D	C	Load	Ripple carry	Borrow	Clear	A	V_{CC}				
10 T_C = +125°C	F_{MAX} 8/	3003 9/	146	191																	Count up to Q_A	22		MHz
	F_{MAX} 8/	"	147	192																	Count down to Q_A	22		MHz
	t_{PLH8}	"	148	193																	Load to Q_A	3	63	ns
		"	149	194																	Load to Q_B	"	"	"
		"	150	195																	Load to Q_C	"	"	"
		"	151	196																	Load to Q_D	"	"	"
	t_{PHL10}	"	152	197																	Load to Q_A	"	"	"
		"	153	198																	Load to Q_B	"	"	"
		"	154	199																	Load to Q_C	"	"	"
		"	155	200																	Load to Q_D	"	"	"
	t_{PLH9}	"	156	201																	Count up to Q_A	"	60	"
		"	157	202																	Count up to Q_B	"	"	"
		"	158	203																	Count up to Q_C	"	"	"
		"	159	204																	Count up to Q_D	"	"	"
		"	160	205																	Count down to Q_D	"	"	"
		"	161	206																	Count down to Q_C	"	"	"
		"	162	207																	Count down to Q_B	"	"	"
		"	163	208																	Count down to Q_A	"	"	"
	t_{PHL11}	"	164	209																	Count up to Q_A	"	73	"
		"	165	210																	Count up to Q_B	"	"	"
		"	166	211																	Count up to Q_C	"	"	"
		"	167	212																	Count up to Q_D	"	"	"
		"	168	213																	Count down to Q_D	"	"	"
		"	169	214																	Count down to Q_C	"	"	"
		"	170	215																	Count down to Q_B	"	"	"
		"	171	216																	Count down to Q_A	"	"	"
	t_{PHL12}	"	172	217																	Clear to Q_A	"	56	"
		"	173	218																	Clear to Q_B	"	"	"
		"	174	219																	Clear to Q_C	"	"	"
		"	175	220																	Clear to Q_D	"	"	"
	Same terminal conditions as for subgroup 9.																							
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																							

See footnotes at end of device types 07 and 08.

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1/ Case 2 pins not referenced are N/C.

2/ Apply 0.7 V for device type 07; apply 2.0 V for device type 08.

3/ I_{IL} limits (μA) min/max values for circuits shown:

Parameter	Terminal	Circuits						
		A	B (08 only)	C	D	E	F	G
		-160/-400	-160/-400	-160/-400	-100/-340	-100/-340	-120/-360	-135/-370
I_{IL9}	A	"	"	"	"	"	"	"
	B	"	"	"	"	"	"	"
	C	"	"	"	"	"	"	"
	D	"	"	"	"	"	"	"
I_{IL10}	Load	-100/-340	"	-150/-380	-120/-360	-120/-360	"	-100/-340
I_{IL11}	Clear	-160/-400	"	-150/-380	"	"	"	-135/-370
	Count up	"	"	"	"	"	"	"
	Count down	"	"	"	"	"	"	"

4/ I_{OS} limits (mA) min/max values for circuits shown: -15/-100 for circuits A, B, C, D, E, F, and G.

5/ Only a summary of attributes data is required.

6/ A = 3.0 V minimum; B = 0.0 V or GND.

7/ H > 1.5 V; L < 1.5 V; X = don't care.

8/ F_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

9/ See figure 8 for device type 07 and figure 9 for device type 08.

10/ Apply momentary ground, then 4.5 V prior to input pulses. Maintain 4.5 V for measurement.

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TABLE III. Group A inspection for device types 09 and 13 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Case 1/ 2 / Test no.	1 B	2 Q_B	3 Q_A	4 Enable G	5 Down/up	6 Q_C	7 Q_D	8 GND	9 D	10 C	11 Load	12 Max/Min	13 Ripple carry	14 Clock	15 A	16 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1	0.7 V	4 mA	4 mA			4 mA	4 mA	GND		0.7 V	0.7 V				0.7 V	4.5 V	Q_B		0.4	V
			2									0.7 V								Q_A			
			3																	Q_C			
			4					2.0 V							4 mA					Q_D			
			5	2.0 V 2/			0.7 V	0.7 V				2.0 V	2/			4 mA	0.7 V	2.0 V		Max/Min			
			6																	Ripple carry			
	V_{OH}	3006	7	2.0 V	-0.4 mA	-0.4 mA			-0.4 mA	-0.4 mA			2.0 V					2.0 V		Q_B	2.5		
			8																	Q_A			
			9									2.0 V								Q_C			
			10					2.0 V				0.7 V	0.7 V		-0.4 mA	-0.4 mA		0.7 V		Q_D			
			11	0.7 V			2.0 V													Max/Min			
			12																	Ripple carry			
	V_{IC}		13	-18 mA			-18 mA	-18 mA				-18 mA	-18 mA	-18 mA			-18 mA	-18 mA		B		-1.5 V	
			14																	Enable G			
			15																	Down/up			
			16																	D			
			17																	C			
			18														-18 mA			Load			
			19															-18 mA		Clock			
			20																	A			
	I_{IL7}	3009	21				0.4 V	5.5 V											5.5 V	Enable G	3/	3/	μA
	I_{IL8}		22	0.4 V				0.4 V				0.4 V	0.4 V	GND			0.4 V	0.4 V		B			
			23																	Down/up			
			24																	D			
			25																	C			
			26											0.4 V						Load			
			27																	Clock			
			28											GND						A			
	I_{IH15}	3010	29				2.7 V													Enable G		60	μA
	I_{IH16}		30				5.5 V													Enable G		300	
	I_{IH17}		31	2.7 V				2.7 V				2.7 V	2.7 V	5.5 V			2.7 V	2.7 V		B		20	μA
			32											5.5 V						Down/up			
			33											5.5 V						D			
			34											2.7 V						C			
			35																	Load			
			36											5.5 V						Clock			
			37																	A			

See footnotes at end of device types 09 and 13.

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TABLE III. Group A inspection for device types 09 and 13 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	B	Q_B	Q_A	Enable G	Down/ up	Q_C	Q_D	GND	D	C	Load	Max/ Min	Ripple carry	Clock	A	V_{CC}				
1 T_C = +25°C	I_{IH18}	3010	38	5.5 V				5.5 V			GND	5.5 V	5.5 V	5.5 V			5.5 V	5.5 V	5.5 V	B		100	µA
		"	39								"			5.5 V					"	Down/up		"	"
		"	40								"			"					"	D		"	"
		"	41								"			"					"	C		"	"
		"	42								"			"					"	Load		"	"
		"	43								"			"					"	Clock		"	"
		"	44								"			5.5 V					"	A		"	"
	I_{OS}	3011	45	5.5 V	GND	GND			GND	GND	"		5.5 V	GND				5.5 V	"	Q_B	4/	4/	mA
		"	46								"			"					"	Q_A	"	"	"
		"	47								"			"					"	Q_C	"	"	"
		"	48								"			"					"	Q_D	"	"	"
		"	49	GND			5.5 V	5.5 V			"	5.5 V GND	GND	"	GND	GND		GND	"	Max/Min	"	"	"
		"	50								"								"	Ripple carry	"	"	"
	I_{CC}	3005	51	GND			GND	GND			"	GND	GND	GND			GND	GND	"	V_{CC}		35	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of device types 09 and 13.

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TABLE III. Group A inspection for device type 09 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 B	2 3 Q_B	3 4 Q_A	4 5 Enable G	5 7 Down/ up	6 8 Q_C	7 9 Q_D	8 10 GND	9 12 D	10 13 C	11 14 Load	12 15 Max/Min	13 17 Ripple carry	14 18 Clock	15 19 A	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 5/	3014	52	A 6/	H	H	B 6/	B	H	H	GND	A	A	B	H	H	A	A	4.5 V	See 7/			
			53				A							B									
			54											A									
			55	B								B	B					B					
			56	B														A					
			57	B			B											B					
			58	A								A	A			L		A					
			59		L	L		L	L	L					L	H		B					
			60			L																	
			61			H												B					
			62			H																	
			63	B	H	L						B	B					A					
			64	A		L						A	A										
			65			H												A					
			66			H						B	B					B					
			67		L	L			H									B					
			68			L						B	A					A					
			69			H												B					
			70			H						A	B					B					
			71		H	L						A	A					B					
			72			L						B	B					A					
			73			H												B					
			74	B		H						A	B					B					
			75	B	L	L			L	H		A	A					B					
			76	A		L												A					
			77			H												A					
			78			H												B					
			79	B	H	L						A	B					A					
			80			L												B					
			81			H												A					
			82			L												B					
			83		L	L			H									A					
			84			L						B	B					B					
			85	A		H						A	A					B					
			86	B		H						B	A					A					
			87	A	H	L						A	A					B					
			88	B		L						B	A					A					
			89	A		L						A	A					B					
			90	B		L						B	A					A					
			91	A	L	L			L	L		A	A		H	L		B					
			92	B		L		A				A	A		L	H		A					
			93	A		L	A					A	A					B					
			94	B		L						B	A					A					
			95	A		L						A	A					B					
			96	B		L	B					A	B					A					
			97	A		L						B	A			L		B					
			98	A	H	L			H	H		B	B		L	H		A					
			99	B		H						B	B					B					
			100	B		H						A	A					A					
			101	A		L																	

See footnotes at end of device types 09 and 13.

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TABLE III. Group A inspection for device type 09 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 B	2 3 Q_B	3 4 Q_A	4 5 Enable G	5 7 Down/ up	6 8 Q_C	7 9 Q_D	8 10 GND	9 12 D	10 13 C	11 14 Load	12 15 Max/Min	13 17 Ripple carry	14 18 Clock	15 19 A	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 5/	3014	102	A 6/	L	H	B 6/	A	H	H	GND	B	A	A	L	H	A	B	4.5 V	See 7/			
			103	B	H	H						A	A				B	B					
			104	A		L						B	B				A	A					
			105	A		L						B	B				B	A					
			106	A	H	H			L			A	A				A	B					
			107	B		H						A	B				B	A					
			108	B		L						B	A				A	A					
			109	A		L						A	B				B	A					
			110	A	L	H						B	A				A	B					
			111	A		H						B	B				B	A					
			112	B		L						A	B				A	B					
			113	B		L				L		B	B				B	A					
			114	B	H	H			H			B	A				A	A					
			115	A		H						A	B				B	A					
			116	B		L						A	A				B	B					
			117	A		L						B	A				B	B					
			118	B	L	H						A	B				A	B					
			119	A		H						B	A				B	A					
			120	A		L						B	B				A	B					
			121	A		L						B	B				B	A					
			122	B	H	H			L			A	A				A	B					
			123	B		H						B	A				B	A					
			124	B		L						A	B				B	A					
			125	A		L						B	B				A	B					
			126	A		H						A	B				B	A					
			127	A		H						B	A				A	B					
			128	B		L						A	B		H		B	A					
			129	A		L						B	A		H	L	A	B					
			130	B	H	H			H	H		A	B		L	H	B	A					
			131	A		H						B	A	B	L	H	A	B					
			132	B		H		B				B	A		H	L	B	B					
			133	B		H						A	B				A	B					
			134	B		H	A					B	A				B	B					
			135	B	L	L			L	L		B	B		L		A	B					
			136	B	L	H			H	L		A	B				B	B					
			137	A	H	L			L	H		B	A				A	B					
			138	B	L	H		A	H	L		A	B				B	B					
			139	A	H	L	B		L	H		B	A				A	B					
			140	B	L	H			H	L		A	B				B	B					
			141	A	H	L			L	H		B	A				A	B					
			142	A		L						A	B	A			B	B					
			143	B		L						B	A				A	B					
			144					B															
			145				A																
			146				B																
			147					A															
			148															B					
			149	A														A					
			150						H	L				B			B	B					
			151						H	L				A			B	B					
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						

See footnotes at end of device types 09 and 13.

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TABLE III. Group A inspection for device type 13 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	B	Q_B	Q_A	Enable G	Down/ up	Q_C	Q_D	GND	D	C	Load	Max/Min	Ripple carry	Clock	A	V_{CC}				
7 T_C = +25°C	Functional tests 5/	3014	52	B 6/	L	H	A 6/	B	L	H	GND	A	B	B 6/	H	H	B	A	4.5 V	See 7/			
			53											A			B						
			54														A						
			55				B									L	B						
			56														A						
			57			L				L					L	H	B						
			58	A		L							A				A						
			59	A		H											B						
			60	B		H							B				A						
			61	B	H	L						B					B						
			62	A		L						A	B				A						
			63	B		H							B				A						
			64			L							A				A						
			65		L				H				A				A						
			66	A		H							A				A						
			67	B		L						B					A						
			68	A		H							B				A						
			69		H								B				A						
			70			L							A				A						
			71			H							A				A						
			72	B		L			L				A				A						
			73	A		L				H			A				A						
			74	A		H							B				A						
			75	B		L							B				A						
			76	B		H				L			B				A						
			77	B		L							A				A						
			78	A		H							A				A						
			79	B		L							A				A						
			80	A		H		A					B				A						
			81	A		L							A				A						
			82	B		H							B				A						
			83			L							A				A						
			84			H	B						B				A						
			85			L							A				A						
			86			H							B				A						
			87	A		L				H			A				A						
			88			H							B				A						
			89			L							A				A						
			90			H							B				A						
			91			H			H	L			A				A						
			92			L							B				A						
			93			H							A				A						
			94			L							B				A						
			95			H							A				A						
			96	B		L							A				A						
			97	B		H							A				A						
			98	A		L							A				A						
			99			H			L				A				A						
			100			L							A				A						
			101			H							A				A						

See footnotes at end of device types 09 and 13.

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TABLE III. Group A inspection for device type 13 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 Test no.	1 2 B	2 3 Q_B	3 4 Q_A	4 5 Enable G	5 7 Down/ up	6 8 Q_C	7 9 Q_D	8 10 GND	9 12 D	10 13 C	11 14 Load	12 15 Max/Min	13 17 Ripple carry	14 18 Clock	15 19 A	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 5/	3014	102	B 6/	H	L	B 6/	A 6/	L	L	GND	3	B 6/	A	L	H	B	B	4.5 V	See 7/			
		"	103	"	L	H	"	"	"	"	"	"	"	"	"	"	A	B	"				
		"	104	"	"	H	"	"	"	"	"	"	"	"	"	"	B	A	"				
		"	105	"	"	L	"	"	"	"	"	"	"	"	H	"	A	A	"				
		"	106	A	"	L	A	"	"	"	"	A	A	"	"	"	A	B	"				
		"	107	A	"	"	"	"	"	"	"	A	A	"	"	"	B	A	"				
		"	108	B	"	"	"	"	"	"	"	B	B	"	"	"	A	A	"				
		"	109	"	"	"	B	"	"	"	"	"	"	"	"	"	A	B	"				
		"	110	"	"	"	"	"	"	"	"	"	"	"	"	L	B	A	"				
		"	111	A	"	H	"	"	"	H	"	A	A	"	L	H	A	B	"				
		"	112	"	"	H	"	"	"	"	"	"	"	"	"	"	B	A	"				
		"	113	"	"	L	"	"	"	"	"	"	"	"	"	"	A	B	"				
		"	114	B	"	H	"	"	"	"	"	B	B	A	"	"	B	A	"				
		"	115	"	"	"	"	"	"	"	"	"	"	"	"	"	A	B	"				
		"	116	"	"	"	"	"	"	"	"	B	"	"	"	"	B	B	"				
		"	117	"	"	"	A	"	"	"	"	"	"	"	"	"	B	A	"				
		"	118	"	"	"	"	"	"	"	"	"	"	"	"	"	A	B	"				
		"	119	"	"	"	"	"	"	"	"	"	"	"	"	"	B	A	"				
		"	120	"	"	"	"	B	"	"	"	"	"	"	H	"	A	B	"				
		"	121	"	"	"	"	"	"	"	"	"	"	"	"	"	B	A	"				
		"	122	"	"	"	"	"	"	"	"	"	"	"	"	"	A	B	"				
		"	123	A	"	"	B	"	"	"	"	A	A	"	"	"	B	A	"				
		"	124	"	"	"	"	"	"	"	"	"	"	"	"	"	A	B	"				
		"	125	"	"	L	"	"	L	L	"	"	"	"	L	H	B	A	"				
		"	126	B	"	H	"	"	H	"	"	B	"	B	"	"	A	B	"				
		"	127	"	"	"	"	"	"	"	"	"	"	"	"	"	B	A	"				
		"	128	"	"	"	"	"	"	"	"	"	"	"	"	"	A	B	"				
		"	129	"	"	"	"	"	"	"	"	"	"	A	"	"	B	A	"				
		"	130	A	"	"	"	"	"	"	"	A	B	"	"	"	A	B	"				
		"	131	A	"	"	"	"	"	"	"	"	"	B	"	"	B	A	"				
		"	132	B	"	L	"	"	L	M	"	"	"	A	"	"	B	A	"				
		"	133	"	"	"	"	"	"	"	"	"	"	B	"	"	A	B	"				
		"	134	"	"	"	A	A	"	"	"	"	"	A	"	"	B	A	"				
		"	135	"	"	"	"	"	"	"	"	B	A	"	"	"	A	B	"				
		"	136	"	"	"	"	"	"	"	"	B	A	"	"	"	B	A	"				
		"	137	"	"	H	"	"	"	"	"	A	B	B	"	"	"	B	"				
		"	138	A	H	L	"	"	H	L	"	B	A	"	"	"	"	B	"				
		"	139	A	H	H	"	B	H	"	"	A	A	"	"	"	"	A	"				
		"	140	B	L	L	"	"	L	"	"	"	B	"	"	"	"	B	"				
		"	141	B	"	"	B	"	"	"	"	A	B	A	"	"	"	B	"				
		"	142	A	"	"	"	A	"	"	"	"	A	"	"	"	"	A	"				
		"	143	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"				
		"	144	"	"	"	"	"	"	"	"	"	"	"	H	L	"	"	"				
		"	145	"	"	H	"	"	"	H	"	B	B	"	L	H	"	B	"				
		"	146	"	"	"	"	B	"	"	"	"	A	"	H	L	"	B	"				
		"	147	"	"	"	"	"	"	"	"	"	"	"	H	H	"	A	"				
		"	148	"	"	L	"	"	"	L	"	"	"	"	L	H	"	A	"				
		"	149	"	H	H	"	"	H	H	"	"	"	B	H	H	A	"	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						

See footnotes at end of device types 09 and 13.

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TABLE III. Group A inspection for device types 09 and 13 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			(Device type) 09 | 13	B	Q_B	Q_A	Enable G	Down/ up	Q_C	Q_D	GND	D	C	Load	Max/Min	Ripple carry	Clock	A	V_{CC}				
9 T_C = +25°C	F_{MAX} 8/	3003 9/	152 | 150			OUT	GND	GND			GND			5.0 V			IN		5.0 V	Ck to Q_A	18		MHz
	t_{PLH10}	"	153 | 151 154 | 152 155 | 153 156 | 154	5.0 V	OUT	OUT			OUT	OUT	"	5.0 V	5.0 V	IN " " "			GND " " "	5.0 V	" " " "	Load to Q_A Load to Q_B Load to Q_C Load to Q_D	3 " " "	38 " " "	ns " " "
	t_{PHL13}	"	157 | 155 158 | 156 159 | 157 160 | 158	GND	OUT	OUT			OUT	OUT	"	GND	GND	" " " "			" " " "	GND	" " " "	Load to Q_A Load to Q_B Load to Q_C Load to Q_D	" " " "	55 " " "	" " " "
	t_{PLH11}	"	161 | 159 162 | 160 163 | 161 164 | 162		OUT	OUT	GND " " "	GND " " "	OUT	OUT	"			5.0 V " " "			IN " " "		" " " "	Ck to Q_A Ck to Q_B Ck to Q_C Ck to Q_D	" " " "	29 " " "	" " " "
	t_{PHL14}	"	165 | 163 166 | 164 167 | 165 168 | 166		OUT	OUT	" " " "	" " " "	OUT	OUT	"			" " " "			" " " "		" " " "	Ck to Q_A Ck to Q_B Ck to Q_C Ck to Q_D	" " " "	41 " " "	" " " "
	t_{PLH12}	"	169 | 167				"	"			"			"	OUT		"		"	Ck to Max/Min	"	47	"
	t_{PHL15}	"	170 | 168				"	"			"			"	OUT		"		"	Ck to Max/Min	"	57	"

See footnotes at end of device types 09 and 13.

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TABLE III. Group A inspection for device types 09 and 13 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Case 1/ 2 (Device type) 09	13	1 B	2 Q_B	3 Q_A	4 Enable G	5 Down/ up	6 Q_C	7 Q_D	8 GND	9 D	10 C	11 Load	12 Max/Min	13 Ripple carry	14 Clock	15 A	16 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
			2		3	4	5	7	8	9	10	12	13	14	15	17	18	19	20					
10 T_C = +125°C	f_{MAX} 8/	3003 9/	171	169	Same terminal conditions as for subgroup 9.																Ck to Q_A	18		MHz
	t_{PLH10}	"	172	170																	Load to Q_A	3	53	ns
		"	173	171																	Load to Q_B	"	"	"
		"	174	172																	Load to Q_C	"	"	"
		"	175	173																	Load to Q_D	"	"	"
	t_{PHL13}	"	176	174																	Load to Q_A	"	77	"
		"	177	175																	Load to Q_B	"	"	"
		"	178	176																	Load to Q_C	"	"	"
		"	179	177																	Load to Q_D	"	"	"
	t_{PLH11}	"	180	178																	Ck to Q_A	"	41	"
		"	181	179																	Ck to Q_B	"	"	"
		"	182	180																	Ck to Q_C	"	"	"
		"	183	181																	Ck to Q_D	"	"	"
	t_{PHL14}	"	184	182																	Ck to Q_A	"	57	"
		"	185	183																	Ck to Q_B	"	"	"
		"	186	184																	Ck to Q_C	"	"	"
		"	187	185																	Ck to Q_D	"	"	"
	t_{PLH12}	"	188	186																	Ck to Max/Min	"	66	"
	t_{PHL15}	"	189	187																	Ck to Max/Min	"	80	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																							

See footnotes at end of device types 09 and 13.

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1/ Case 2 pins not referenced are N/C.

2/ Apply 2.0 for device type 09; apply 0.7 V for device type 13.

3/ I_{IL} limits (μA) min/max values for circuits shown:

Parameter	Terminal	Circuits						
		A	B	C	D	E	F	G
I_{IL7}	Enable G	-360/-1080	-160/-400	-360/1080	-360/-1080	-360/-1080	-360/-1080	-360/-1080
I_{IL8}	A, B, C, D	-160/-400	-160/-400	-160/-400	-160/-400	-120/-360	-120/-360	-120/-360
	Down/up	"	"	-150/-380	"	"	"	"
	Clock	"	"	"	"	"	"	"
	Load	-100/-340	"	"	-100/-340	"	"	"

4/ I_{OS} limits (mA) min/max values for circuits shown: -15/-100 for circuits A, B, C, D, E, F, and G.

5/ Only a summary of attributes data is required.

6/ A = 3.0 V minimum; B = 0.0 V or GND.

7/ H > 1.5 V; L < 1.5 V; X = don't care.

8/ F_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

9/ See figure 10 for device type 09 and figure 12 for device type 13.

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TABLE III. Group A inspection for device type 10.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 B	2 3 NC	3 4 NC	4 6 NC	5 8 V_{CC}	6 9 Ro(1)	7 10 Ro(2)	8 12 Q_D	9 13 Q_C	10 14 GND	11 16 Q_B	12 18 Q_A	13 19 NC	14 20 A	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OL}	3007	1	GND				4.5 V	2.0 V	2.0 V			GND		4 mA		2.0 V	Q_A		0.4	V
		"	2	2.0 V				"	"	"			"	4 mA	$*I_{IL3}$ (max)		GND	Q_B		"	"
		"	3	"				"	"	"		4 mA	"				"	Q_C		"	"
		"	4	"				"	"	"	4 mA		"				"	Q_D		"	"
	V_{OH}	3006	5	GND				"	2/	2/			"				2/ 3/ GND	Q_A	2.5		"
		"	6	2/ 3/				"	"	"			"	-0.4 mA	-0.4 mA		"	Q_B	"		"
		"	7	2/ 4/				"	"	"		-0.4 mA	"				"	Q_C	"		"
		"	8	2/ 5/				"	"	"	-0.4 mA		"				"	Q_D	"		"
	V_{IC}		9					"					"				-18 mA	A		-1.5	"
			10	-18 mA				"					"					B		"	"
			11					"	-18 mA				"					Ro(1)		"	"
			12					"		-18 mA			"					Ro(2)		"	"
	I_{IL1}	3009	13					5.5 V	0.4 V	5.5 V			"					Ro(1)	6/	6/	mA
		"	14					"	5.5 V	0.4 V			"					Ro(2)	"	"	"
	I_{IL2}	"	15					"	2/	2/			"				0.4 V	A	"	"	"
	I_{IL3}	"	16	0.4 V				"	"	"			"					B	"	"	"
	I_{IH1}	3010	17					"	2.7 V	GND			"					Ro(1)		20	μA
	I_{IH1}	"	18					"	GND	2.7 V			"					Ro(2)		20	"
	I_{IH2}	"	19					"	5.5 V	GND			"					Ro(1)		100	"
	I_{IH2}	"	20					"	GND	5.5 V			"					Ro(2)		100	"
	I_{IH3}	"	21					"	5.5 V	5.5 V			"				2.7 V	A		80	"
	I_{IH4}	"	22					"	"	"			"				5.5 V	A		400	"
	I_{IH5}	"	23	2.7 V				"	"	"			"					B		80	"
	I_{IH6}	"	24	5.5 V				"	"	"			"					B		400	"
	I_{OS}	3011	25	GND				"	2/	2/			"		GND		2/ 3/ GND	Q_A	7/	7/	mA
		"	26	2/ 3/				"	"	"			"	GND			"	Q_B	"	"	"
		"	27	2/ 4/				"	"	"		GND	"				"	Q_C	"	"	"
		"	28	2/ 5/				"	"	"	GND		"				"	Q_D	"	"	"
	I_{CC}	3005	29	GND		↓		"					"					V_{CC}		15	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				

See footnotes at end of device type 10.

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TABLE III. Group A inspection for device type 10 - Continued.
Terminal conditions (pins not designated may be H ≥ 2.0 V, or L ≤ 0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 B	2 3 NC	3 4 NC	4 5 NC	5 8 V_{CC}	6 9 Ro(1)	7 10 Ro(2)	8 12 Q_D	9 13 Q_C	10 14 GND	11 16 Q_B	12 18 Q_A	13 19 NC	14 20 A	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional tests 8/	3014	30	B 9/				4.5 V	A 9/	A 9/	L	L	GND	L	L		B 9/	See 10/			
			31	A																	
			32	B					B												
			33	B										H							
			34	A					A					L							
			35	B						X				H							
			36	B						B				H							
			37	B								H		L							
			38	A							H	L		H							
			39	B										H							
			40	A								H		L							
			41	B								L		H							
			42	A							H	L		L							
			43	B										H							
			44	A								H		H							
			45	B										L							
			46	A								L		L							
			47	B						A	L			H							
			48	B					B					H							
			49	B										H							
			50	A										L							
			51	B								H		H							
			52	A							H	L		L							
			53	B										H							
			54	A								H		H							
			55	B										L							
			56	A								L		L							
			57	B						A	L			H							
			58	B					B					H							
			59	A										H							
			60	B								H		H							
			61	A							H	L		L							
			62	B										H							
			63	B								H		H							
			64	A										L							
			65	B																	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																				
9 T_C = +25°C	F_{MAX}	3003	66					5.0 V	GND				GND		OUT		IN 12/	A to Q_A	29		MHz
	t_{PLH1}	(Fig 11)	67						11/ GND	A 9/		OUT					IN	A to Q_C	3	53	ns
	t_{PHL1}	"	68									OUT					IN	A to Q_C	"	58	"
	t_{PLH2}	"	69	IN					11/ GND	A 9/	OUT							B to Q_B	"	37	"
	t_{PHL2}	"	70	IN							OUT							B to Q_D	"	40	"
10 T_C = +125°C	F_{MAX}	"	71					"	GND				"		OUT		IN 12/	A to Q_A	29		MHz
	t_{PLH1}	"	72						11/ GND	A 9/		OUT					IN	A to Q_C	3	74	ns
	t_{PHL1}	"	73									OUT					IN	A to Q_C	"	81	"
	t_{PLH2}	"	74	IN					11/ GND	A 9/	OUT							B to Q_B	"	52	"
	t_{PHL2}	"	75	IN							OUT							B to Q_D	"	56	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																				

See footnotes at end of device type 10.

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1/ Case 2 pins not referenced are N/C.

2/ Apply 4.5 volts pulse, then ground prior to taking measurements to set device in the desired state. Maintain ground for measurement.

3/ Input pulse must be applied one time after Ro pulse.

4/ Input pulse must be applied twice after Ro pulse.

5/ Input pulse must be applied four times after Ro pulse.

6/ I_{IL} limits (mA) min/max values for circuits shown:

Parameter	Terminals	Circuits						
		A	B	C	D	E	F	G
I_{IL1}	Ro(1) Ro(2)	-.12/-.36	-.03/-.40	-.03/-.40	-.03/-.40		-.12/-.36	
I_{IL2}	A	-0.5/-2.0	-1.0/-2.4	-1.0/-2.4	-1.0/-2.4		-0.5/-2.0	
I_{IL3}	B	-0.7/-3.2	-0.7/-3.2	-0.7/-3.2	-0.4/-1.6		-0.7/-3.2	

7/ I_{OS} limits (mA) min/max values for circuits shown:

Parameter	Measured terminals	Circuits						
		A	B	C	D	E	F	G
I_{OS}	Q_A, Q_B, Q_C, Q_D	-15/-100	-15/-100	-30/-130	-15/-100		-15/-100	

8/ Only a summary of attributes data is required.

9/ A = 3.0 V minimum; B = 0.0 V or GND.

10/ H > 1.5 V; L < 1.5 V; X = don't care.

11/ Momentary 3.0 V (min), then ground. Maintain ground for measurement.

12/ F_{MAX} min limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

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b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions, and limits specified for subgroups 10 and 11 of group A.

c. Steady-state life-test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D using the circuit shown on figure 13, or equivalent.

(2) T_A = +125°C, minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the reference terminal.

4.6 Inclusion with other detail specifications. For qualification and quality conformance inspection purposes, devices covered by this specification may be treated as though they were included on the same detail specification as devices covered by MIL-M-38510/320. In addition, if a manufacturer is already part I qualified for types /32003 and /32004, and if the respective devices on this specification (MIL-M-38510/315C) are designed and manufactured identically (same die, same process, same screening) in all respects except electrical testing, then device types /31501 and /31502 may be part I qualified by conducting only group A electrical tests, with approval of qualifying activity, including subgroups A-10 and A-11, and submitting data in accordance with MIL-M-38510, appendix D (i.e., groups B, C, and D tests are not required).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
V_{IN}	Voltage level at an input terminal.
I_{IN}	Current flowing into an input terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

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6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS90
02	54LS93
03	54LS160
04	54LS161
05	54LS168
06	54LS169
07	54LS192
08	54LS193
09	54LS191
10	54LS92
11	54LS162
12	54LS163
13	54LS190

6.7 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers designation.

Device type	Circuit	A	B	C	D	E	F	G
	Manufacturer Commercial Type	Texas Instrument, Incorporated	Signetics Corpo-ration	National Semi-conductor Corpo-ration	Raytheon Company	Fairchild Semi-conductor	Motorola Incorpo-rated	Advanced Micro Devices, Incorpo-rated
01	54LS90	X	X		X	X	X	
02	54LS93	X	X	X	X	X	X	
03	54LS160A	X	X	X	X	X	X	X
04	54LS161A	X	X	X	X	X	X	X
05	54LS168			X		X		
06	54LS169A			X		X		
07	54LS192	X	X	X	X	X	X	X
08	54LS193	X	X	X	X	X	X	X
09	54LS191	X	X	X	X	X	X	X
10	54LS92	X		X	X		X	
11	54LS162A	X	X	X	X	X	X	X
12	54LS163A	X	X	X	X	X	X	X
13	54LS190	X	X	X	X	X	X	X

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/315C	Microcircuits, Digital, Low-Power Schottky TTL, Counters,

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
Monolithic Silicon	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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